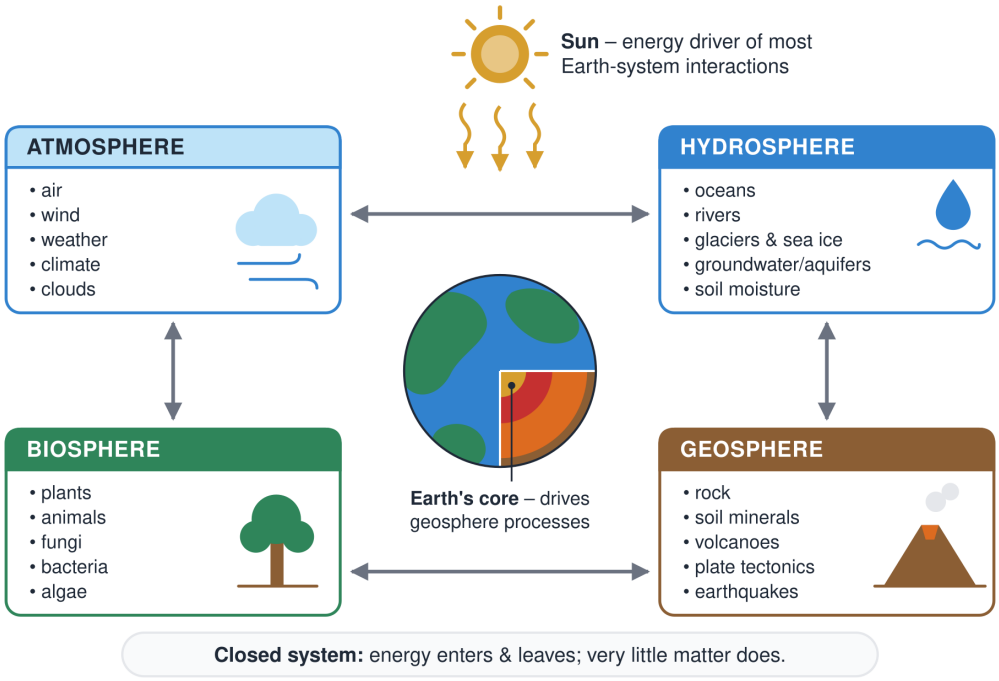
ECOSYSTEMS UNIT

Study Guide & Practice Questions

Ms. Hero's assessment: 25 multiple-choice questions · 30 minutes · one handwritten 3" × 5" index card allowed



What's inside	Page section
Start here — test format, 5-day plan, topic map, gaps from your handouts	Start Here
Study guide — 9 topics with diagrams, tables, handout callouts, traps, model answers	Topics 1–9
Common traps (30 confused pairs) · Index-card model	Review tools
101 practice questions: 76 topic drills + a 25-question mock test	Practice
Hints (peek only if stuck) · Answer key with full explanations	Answers
Weakness tracker — map every missed question to the section to re-read	Tracker

The interactive version (**Ecosystems_Study_Guide.html**) has the same content plus instant feedback, hints, retries, and a results tracker. Question IDs (e.g., T4-03, M-12) and answer letters are identical in both versions. Graphs labeled "practice data — illustrative" are not real records.

Start Here

Your test at a glance

Format	25 multiple-choice questions · 30 minutes · on your school Chromebook (charge it, bring the charger)
Allowed	One pale-blue 3" × 5" index card from Ms. Hero — **handwritten only**, front and back, with your first & last name and period. You turn it in at the end.
What Ms. Hero wants	For every topic: know the **definition**, its **importance**, and be able to **identify it in context** (in a scenario, table, or graph).
Absent?	Contact Ms. Hero right away to make it up.

How to use this guide

1. **Read** a topic in the **Study Guide** tab. Each topic opens with a green **"Know it cold"** box, then has diagrams, tables, **"From your handouts"** boxes (blue), **traps** (red), and **model answers** (purple).
2. **Practice** that topic right away with the button at the end of each topic, or in the **Practice** tab. Stuck? Click **Hint**. Wrong on the first try? You'll see *why* that choice is wrong and get one more try. Then you'll get the full explanation and a link back to the exact study-guide section.
3. **Check your Results** — the mastery table shows which topics need more work, and you can **retry only the questions you missed**.
4. **Take the Mock Test** (25 questions, 30-minute timer, no hints) once you've finished the drills — just like the real thing.
5. **Make your index card** using the **Index Card** tab as a model. Writing it by hand is studying!

A 5-day study plan

Day	Do this	Time
1	Study Guide Topics 1–3 → drills for Topics 1–3	~40 min
2	Study Guide Topics 4–5 → drills for Topics 4–5 (the biggest topics)	~45 min
3	Study Guide Topics 6–9 → drills for Topics 6–9 → **Common Traps** tab	~45 min
4	**Mock Test** (timed) → review every missed question → **retry missed**	~45 min
5	Re-read weak topics (Results tab) → write your **index card** → retry missed once more	~30 min

Topic map: what's on the test, and where it came from

#	Topic (from the Review Guide)	Checklist items	Class handouts to re-read
1	Topic 1 · Earth Systems Science	Earth Systems Science · atmosphere · hydrosphere · biosphere · geosphere	Earth Systems Science Graphic Organizer
2	Topic 2 · Ecological Relationships	Symbiosis · predator/prey · mutualism · parasitism · commensalism · inter- & intraspecific competition · keystone species	Ecological Relationships; Gizmos Ecosystems STEM Case (Atlas NP)
3	Topic 3 · Food Chains, Webs & Cycling of Energy	Trophic levels (autotroph, heterotroph) · producers, consumers, decomposers · metabolism · flow of energy (10%)	Creating Chains & Webs (Redwood Forest); Gizmos Handbook
4	Topic 4 · Limiting Factors & Carrying Capacity	Limiting factors · density-dependent vs. -independent · exponential growth · overshoot · decline · recovery · carrying capacity · cyclical/periodic	Limiting Factors & Carrying Capacity; Carrying Capacity Simulation & Explanatory Model
5	Topic 5 · Succession, Habitat Fragmentation & Fire	Habitat fragmentation · primary & secondary succession · pioneer species, climax community · crown & surface fires · fire severity · disturbances	Habitat Fragmentation Simulation; Life from Stone (Yosemite); CZU / Big Basin; Wildfires: What is happening in California?
6	Topic 6 · Biodiversity	Biodiversity · relative abundance · richness / evenness	Sampling Diversity
7	Topic 7 · What is an Ecosystem?	Ecosystem · equilibrium · biotic / abiotic factors	Gizmos Handbook ("What is an Ecosystem")
8	Topic 8 · Graphs	X & Y axes · IV & DV · DRY MIX	DRY MIX notes; Carrying Capacity Simulation graph; wildfire graph analysis
9	Topic 9 · Scientific Argumentation (CER)	Claim · evidence · reasoning	Sampling Diversity; Habitat Fragmentation recommendations; Gizmo case study

Spots to double-check from your own worksheets

These came straight from your marked-up handouts. Each one is a likely test question.

WATCH OUT

1. **Sampling Diversity Q1–Q3** were left incomplete (which community is most/least even; richness vs. evenness). → Relative abundance (with your class data)
2. **Crown fire vs. surface fire** comparison table (Life from Stone Q12) was blank, as were Q13 ("Does fire always mean destruction?") and Q14 ("How does biodiversity change during recovery?"). → Fire: surface vs. crown, severity & disturbances and Secondary & tertiary succession
3. **Primary vs. secondary succession "Both" column** (Q11) was blank. → Secondary & tertiary succession
4. **Bean simulation graph** was marked **"Wrong type of graph"** — population over time needs a **line** graph. → Choosing the right graph
5. **What increases vs. decreases carrying capacity** — the two answers were swapped ("swap"). → Population growth & carrying capacity
6. **Omnivores and top predators at more than one trophic level** (black bear, puma) in your food web. → Trophic levels: who eats whom
7. **Density-dependent vs. density-independent** was only covered briefly on the worksheet — expect scenario questions. → Density-dependent vs. density-independent

Topic 1 · Earth Systems Science

KNOW IT COLD

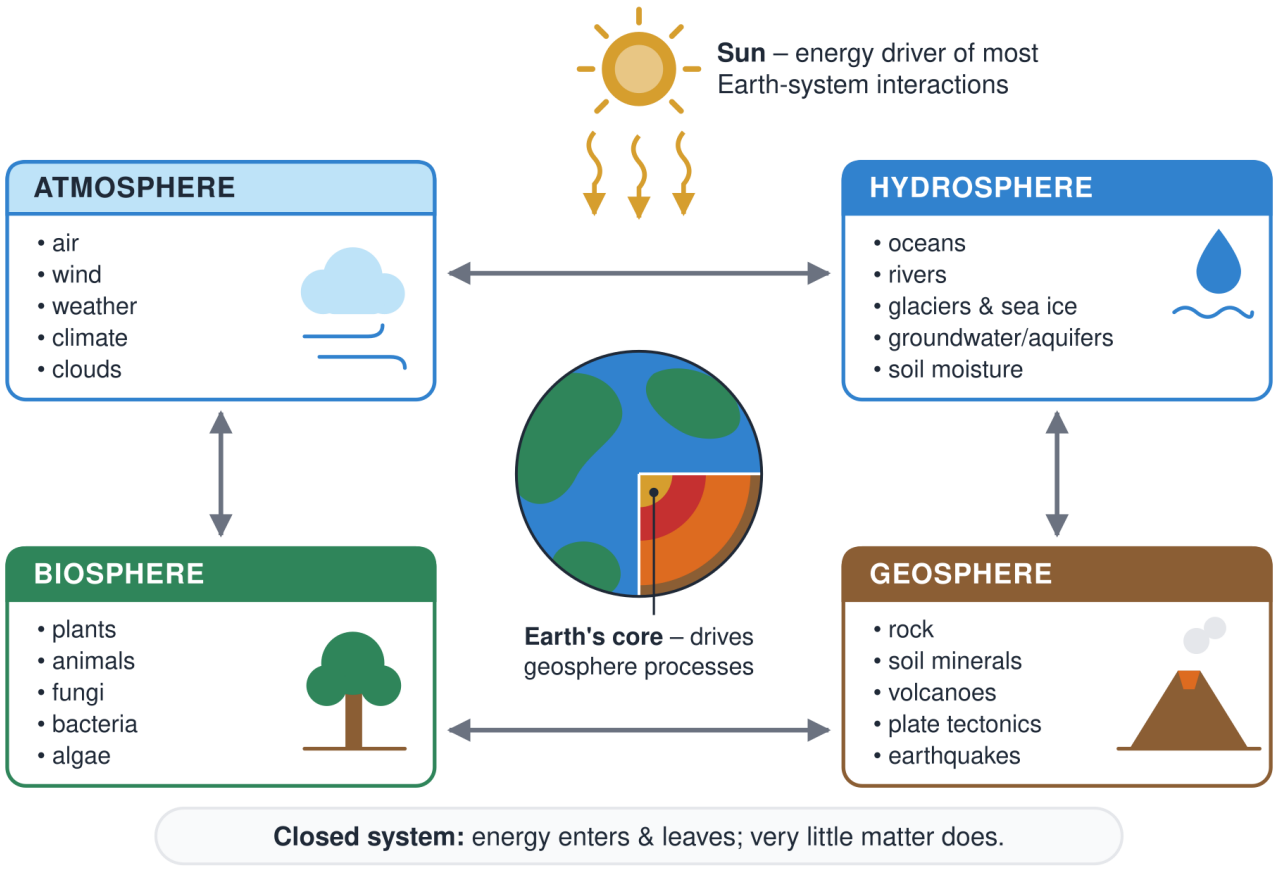
Know it cold: Earth system science is the study of the **relationships** between air, land, life, and water on our planet. You must be able to (1) put any item into the right sphere and (2) describe how an event in one sphere causes changes in another.

What is Earth system science?

Earth system scientists treat Earth as **four connected systems ("spheres")**. A change in one sphere ripples into the others. That is why a lightning storm (air) can burn a forest (life), wash soil into creeks (land → water), and change the climate (air) — all from one event.

Big idea	What it means	Why it matters
Closed system	Earth receives **energy** from the Sun (and loses heat to space), but **very little matter** enters or leaves.	Matter must be **recycled** — water cycle, carbon cycle, and nutrients returned by decomposers.
The Sun	The energy driver of **most** Earth-system interactions.	Powers weather, the water cycle, and photosynthesis (the start of every food chain).
Earth's core	Internal heat; the energy driver of many **geosphere** processes.	Powers plate tectonics, volcanoes, earthquakes, and the rock cycle.
The Earth	Third planet from the Sun with unique properties that allow life.	Liquid water + a protective atmosphere make the biosphere possible.

The four spheres



The four spheres from your Earth System Science Graphic Organizer. Arrows show that every sphere interacts with every other sphere.

Sphere	Word clue	What belongs here (from your class organizer)	Watch out!
Atmosphere	*atmo* = vapor/air	air, winds, weather, climate, clouds, jet stream, solar-radiation buffer, CO_2/O_2 mixing, heat mixing	Anything that is a **gas** or **weather** goes here.
Hydrosphere	*hydro* = water	ALL water in every form — liquid, solid, gas: oceans, lakes & rivers, the water cycle, **sea ice, glaciers, snow cover, soil moisture, aquifers** (groundwater)	**Frozen water and underground water are still hydrosphere.** Glaciers are *not* geosphere.
Biosphere	*bio* = life	plants, animals, fungi, bacteria, algae, **coral reefs**, producers/consumers/decomposers, evolution & adaptation	Anything **alive** — including tiny bacteria and the process of adaptation.
Geosphere	*geo* = earth	land, rock, soil minerals, plate tectonics, volcanoes, the rock cycle, hot spots, sedimentation, **fossilization**, earthquakes, **magnetism**	Fossils, sediment, and Earth's magnetic field are geosphere (per your organizer).

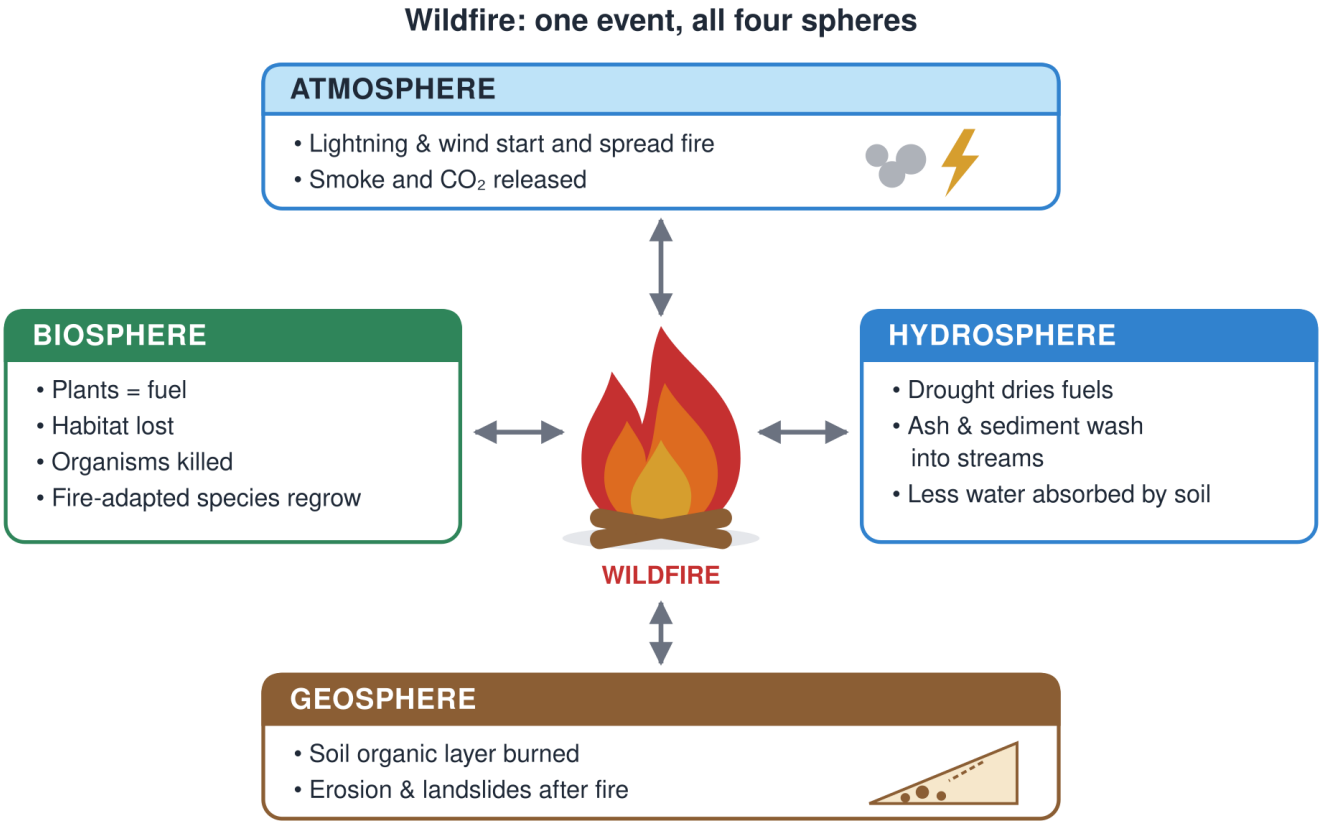
WATCH OUT

Common trap — soil: the *minerals* in soil are **geosphere**, the *water* in soil (soil moisture) is **hydrosphere**, and the *worms and bacteria* in soil are **biosphere**. Read the question to see which part it is asking about.

How the spheres interact

A strong answer names **the cause sphere → the effect sphere**: "Lightning in the **atmosphere** ignited dry plants in the **biosphere**."

Event	Sphere interactions
Photosynthesis	Plants (bio) take in CO_2 (atmo) + water (hydro) + minerals (geo) → release O_2 (atmo)
Volcanic eruption	Magma (geo) → ash & gases into the air (atmo) → acid rain (hydro) → harms fish and plants (bio)
Hurricane	Warm ocean water (hydro) fuels the storm (atmo) → erodes coastline (geo) → destroys habitat (bio)
Lichen on bare rock	Lichens (bio) break rock into tiny pieces → first soil (geo) — the start of **primary succession**
Glaciers carving Yosemite	Ice (hydro) scrapes and carves granite (geo) → bare rock where life (bio) must start over
Rain after a fire	Rain (hydro) washes burned, bare soil (geo) into creeks (hydro) → muddy water harms fish (bio)



One wildfire touches all four spheres. Your CZU Lightning Complex notes are a perfect example.

FROM YOUR CLASS HANDOUTS

From your handouts — CZU fire as an Earth-systems event: dry **lightning** (atmosphere) ignited vegetation (biosphere) that had dried out during the hot, dry summer of Big Basin's **Mediterranean climate**; **wind** (atmosphere) and **steep slopes** (geosphere) spread the fire; smoke and CO_2 went into the atmosphere; after the fire, bare soil (geosphere) could wash into streams (hydrosphere).

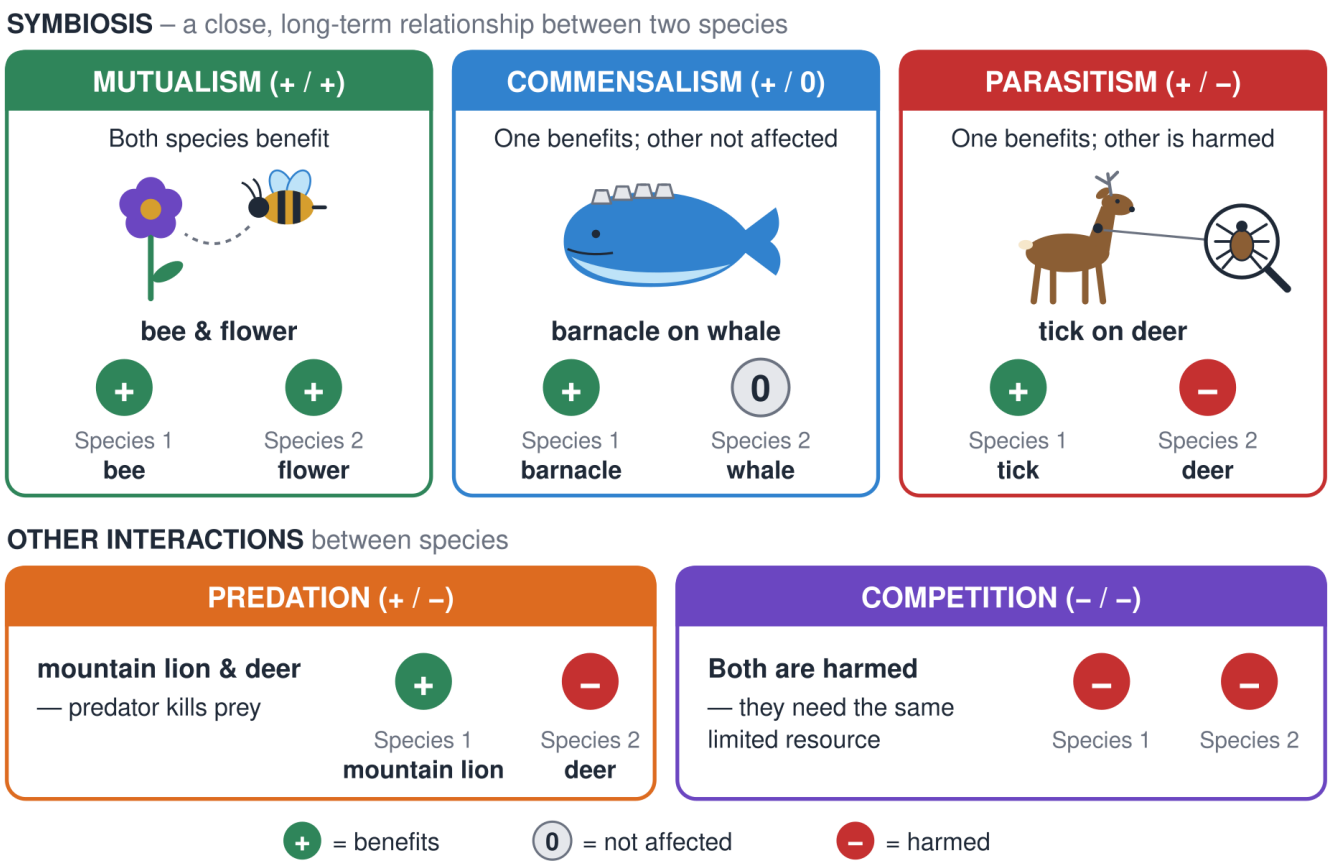
Practice: Topic 1 drills (questions T1-xx in the Practice Questions section).

Topic 2 · Ecological Relationships

KNOW IT COLD
Know it cold: Every relationship can be scored from the point of view of **both** species: **+** (helped), **0** (not affected), **−** (harmed). Score both species and the name follows.

Symbiosis: mutualism, commensalism, parasitism

Symbiosis = a **close, long-term relationship** between two **different** species. There are three types.



The "scoreboard" for each relationship. Predation and competition are interactions too, but they are not usually called symbiosis.

Relationship	Score	Definition	Classic examples	Clue words in a question
Mutualism	+ / +	Both species benefit	bees & flowers (food ↔ pollination); **lichen** (fungus gives shelter & water, alga makes food); clownfish & sea anemone; coral & algae	"both," "trade," "in exchange," "each gets"
Commensalism	+ / 0	One benefits; the other is **neither helped nor harmed**	barnacles on a whale; remora riding a shark; bird nesting in a tree; cattle egret eating insects stirred up by cattle	"not affected," "neither helped nor harmed," "doesn't notice"
Parasitism	+ / –	One benefits; the other (the **host**) is harmed	tick on a mule deer; flea on a dog; tapeworm in an intestine; mistletoe on a tree	"host," "feeds on," "weakens," "lives on/in"

WATCH OUT

Common traps

- **Parasite vs. predator** — both are + / −. A **predator kills and eats** its prey (usually quickly). A **parasite lives on or in its host** and feeds on it over time, usually without killing it right away.
- **Commensalism vs. mutualism** — ask: "Does the *second* species get anything?" If yes → mutualism. If nothing changes for it → commensalism.

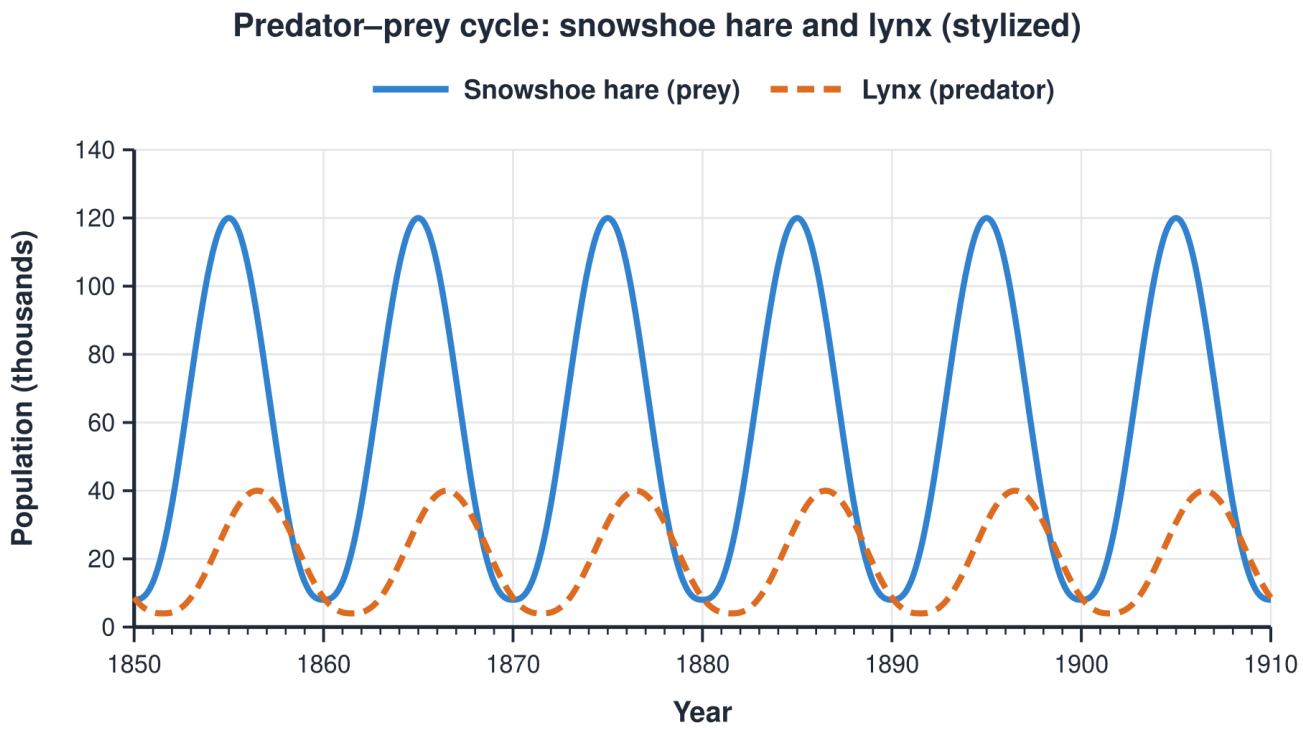
Predator and prey

Predation (+ / −): the predator hunts, kills, and eats the prey — puma & mule deer, bobcat & brush rabbit, lynx & snowshoe hare, wolves & deer.

Why it matters: predators keep prey populations from growing past what their food supply can handle (predation is a **density-dependent limiting factor**). Prey, in turn, limit predators: no prey, no food.

Predator-prey cycles — the populations rise and fall in a repeating pattern:

1. Prey increase → more food for predators.
2. Predators increase (a little **later** — there is a **lag**).
3. More predators eat more prey → prey decrease.
4. Less food → predators decrease (again, later).
5. Fewer predators → prey recover → the cycle repeats (snowshoe hare and lynx: about every 10 years).



Stylized after Hudson's Bay Company fur records: ~10-year cycle; predator peaks lag prey peaks.
(Illustrative pattern, not real data.)
Predator peaks come AFTER prey peaks, and the predator line is lower (fewer predators — remember the 10% rule).

FROM YOUR CLASS HANDOUTS

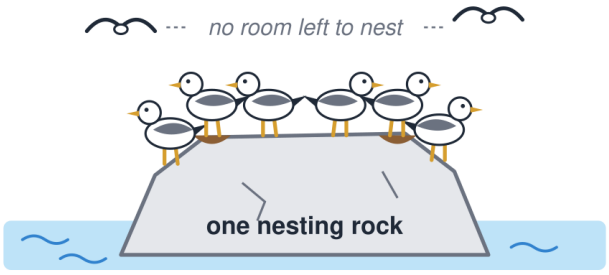
From your handouts: rabbits introduced to Australia in the mid-1800s had no predators, so they exploded, ate the land bare, and crashed. The worksheet notes that a predator (the lynx) helped **stabilize** the rabbit population — predators can keep prey near carrying capacity.

Competition: intraspecific vs. interspecific

Competition (− / −): two organisms need the **same limited resource** — food, water, space, sunlight, mates, or nesting sites. Both lose something.

INTRASPECIFIC COMPETITION

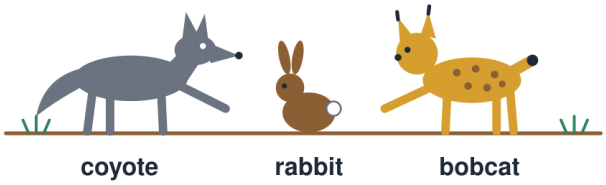
within the **SAME** species



Gulls compete with other gulls for limited **nesting space** on one rock.

INTERSPECIFIC COMPETITION

between **DIFFERENT** species



Coyote and bobcat compete for the same limited **food**: one rabbit.

Intra = within ONE species. Inter = between DIFFERENT species.

Type	Word clue	Definition	Examples
Intraspecific	*intra* = within (like *intramural* sports inside one school)	Competition between members of the **same species**	**Seagulls competing for nesting space on a rock** (class example); two mule deer bucks competing for mates; redwood seedlings competing for light
Interspecific	*inter* = between (like *interstate* highways between states)	Competition between **different species**	**A lion and a tiger competing for a water buffalo** (class example); a coyote and a bobcat competing for brush rabbits

Why it matters: competition gets **stronger as a population gets more crowded**, so it is a **density-dependent** limiting factor. Intraspecific competition is often the most intense, because members of the same species need exactly the same things.

Keystone species

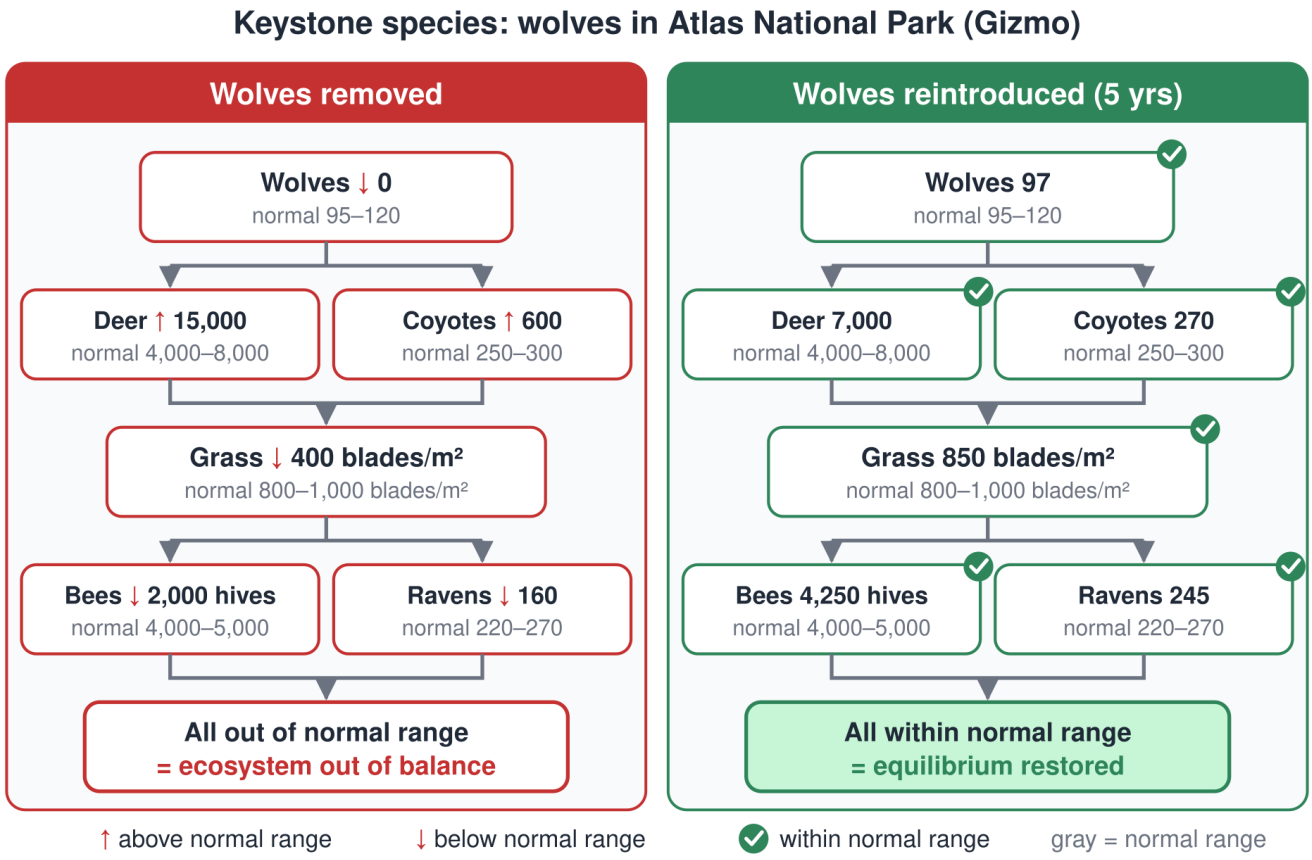
Keystone species = a species that has a **much bigger effect on its ecosystem than its numbers suggest**. If it is removed, the ecosystem's balance (equilibrium) is destroyed and the ecosystem can collapse. **There can be more than one keystone species** in an ecosystem.

WATCH OUT

A keystone species is NOT necessarily the most abundant species, the biggest species, or always the top predator. Bees (pollinators) and beavers (dam builders) can be keystone species too.

Keystone species	What it does	What happens without it
Bees (class example)	Pollinate plants	Plants decline → primary consumers lose food → the decline cascades upward
Wolves (Yellowstone; Atlas NP Gizmo)	Keep deer & coyote numbers in check	Deer overpopulate and overgraze → grass, bees, ravens decline
Sea otters	Eat sea urchins	Urchins explode and eat the kelp forest
Pisaster sea stars	Eat mussels in tide pools	Mussels crowd out most other tide-pool species

Trophic cascade: a change at the top of a food web "cascades" down through several levels (wolves ↓ → deer ↑ → grass ↓ → bees ↓).



Your Gizmo case study. The keystone is the species whose loss started the imbalance — and whose return restored equilibrium.

FROM YOUR CLASS HANDOUTS

From your Gizmo case study (Atlas National Park):

Species	Normal range	Now	5 yrs after adding bees	5 yrs after adding wolves
Grass	800–1,000 blades/m²	400	350	**850**
Coyotes	250–300	600	650	**270**
Ravens	220–270	160	145	**245**
Deer	4,000–8,000	15,000	17,000	**7,000**
Bees	4,000–5,000 hives	2,000	3,000	**4,250**
Wolves	95–120	0	0	**97**

People might guess **bees** are the keystone (pollinators are important!), but adding bees made things *worse*. Adding **wolves** brought every population back into its normal range → **wolves are the keystone species** in Atlas National Park.

Practice: Topic 2 drills (questions T2-xx in the Practice Questions section).

Topic 3 · Food Chains, Webs & Cycling of Energy

KNOW IT COLD

Know it cold: Energy enters as **sunlight**, producers turn it into **chemical energy**, and only about **10%** passes up each trophic level. The other ~90% is used for **metabolism** and leaves as **heat**. Arrows point **from the food to the eater**.

Trophic levels: who eats whom

A **trophic level** ("food level") is an organism's position in a food chain.

Level	Name	Also called	Gets its energy from	Big Basin example
1	**Producer**	**Autotroph** (*auto* = self, *troph* = food)	Sunlight, by **photosynthesis** (light energy → chemical energy in sugar)	redwood sorrel, California poppy, coast redwood
2	**Primary consumer**	Herbivore (a heterotroph)	Producers	mule deer, California brush rabbit
3	**Secondary consumer**	Carnivore or omnivore	Primary consumers	gray fox, bobcat, coyote
4	**Tertiary consumer**	Often the apex (top) predator	Secondary consumers (may also eat primary consumers)	puma (mountain lion)
5	Quaternary consumer	Apex predator	Tertiary consumers	(rare — few chains are this long)

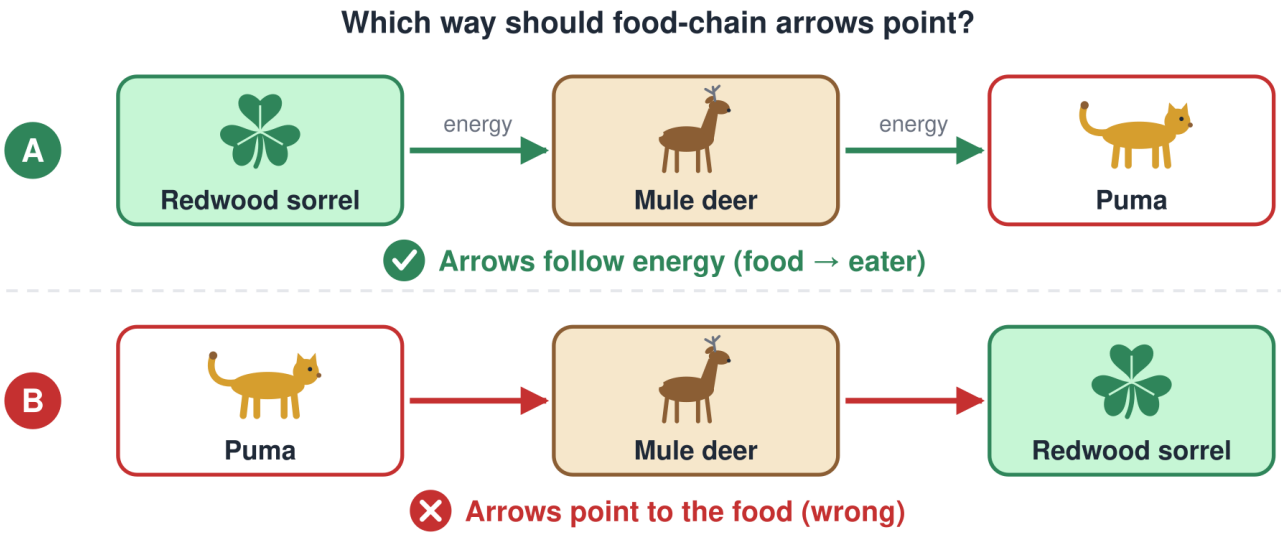
- **Heterotroph** (*hetero* = other) = any organism that gets food by eating **other organisms** — every consumer and decomposer.
- **Herbivore** eats only producers (deer). **Carnivore** eats only animals (puma). **Omnivore** eats both (black bear) — so an omnivore can sit at **more than one** trophic level.

TEST TIP

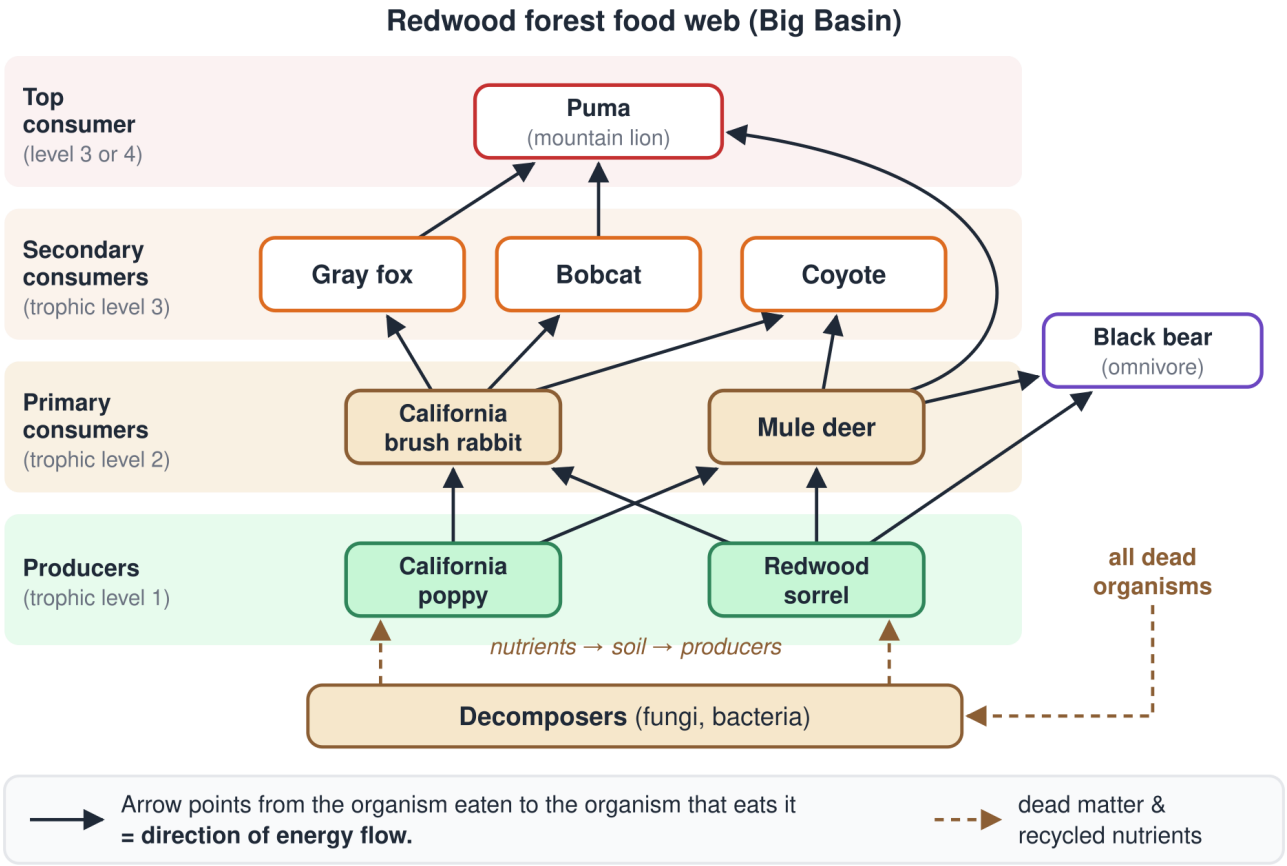
Tip — trophic level depends on the meal: a puma eating a **mule deer** is a *secondary* consumer (level 3). The same puma eating a **gray fox** is a *tertiary* consumer (level 4). A black bear eating **redwood sorrel** is a *primary* consumer; eating a **deer**, it is a *secondary* consumer.

Food chains vs. food webs: reading the arrows

The arrow means **"is eaten by"** or **"gives energy to."** It points **from the food → to the eater** — the direction energy flows.



Your Creating Chains & Webs worksheet asked this exact question: model B (producer → consumer) is correct.



The Big Basin food web built from your Redwood Forest Food Web Cards. Follow any arrow to see where energy goes.

Predicting ripple effects with a food web — if the **California brush rabbit** disappeared:

- **Gray fox** and **bobcat** lose their only food in this web → they decline.
- **Coyotes** depend more on mule deer → more pressure on deer.
- **Pumas** lose fox and bobcat prey and rely on deer.
- **Redwood sorrel and poppy** may increase (fewer rabbits eating them).

FROM YOUR CLASS HANDOUTS

From your handout — comparing models (Part 4: Model Evaluation):

Model	Useful for…	Lacks / could cause a misconception…
Food chain	One simple, step-by-step path of energy; who eats whom	Doesn't show the whole picture — real organisms eat (and are eaten by) many species
Energy pyramid	How much energy is at each level; why there are fewer organisms at the top	Doesn't show how the organisms are related to each other
Food web	The whole ecosystem and how species interact; predicting ripple effects	Hard to read with many arrows; doesn't show energy amounts

For a real ecosystem like Big Basin, the **food web** is the most complete model.

FROM YOUR CLASS HANDOUTS

From your handout — disturbance (fire) effects on trophic levels (Part 5): producers **+** (fire clears debris and small trees → less competition, more sunlight reaches the forest floor); primary consumers **neutral** (food regrows quickly, but fire and smoke are direct dangers); secondary and tertiary consumers **−** (habitat loss, fewer prey, poor air quality). Part 5 also defines the abiotic parts of the cards (drought, fire) as **ecological forces or disturbances**.

Decomposers, scavengers & detritivores

Role	What it does	Examples
Scavenger	Animal that eats **dead animals**	vultures eating a dead raccoon
Detritivore	Survives on **dead or decaying** organisms and wastes	earthworms, millipedes, banana slugs
Decomposer	Uses **enzymes** to break down dead organisms and wastes into simple nutrients	**fungi** and **bacteria**

Why decomposers matter: decomposition **returns nutrients to the soil**, where they are taken up again by **producers**. Without decomposers, dead material would pile up and producers would run out of nutrients — the whole food web would starve.

WATCH OUT

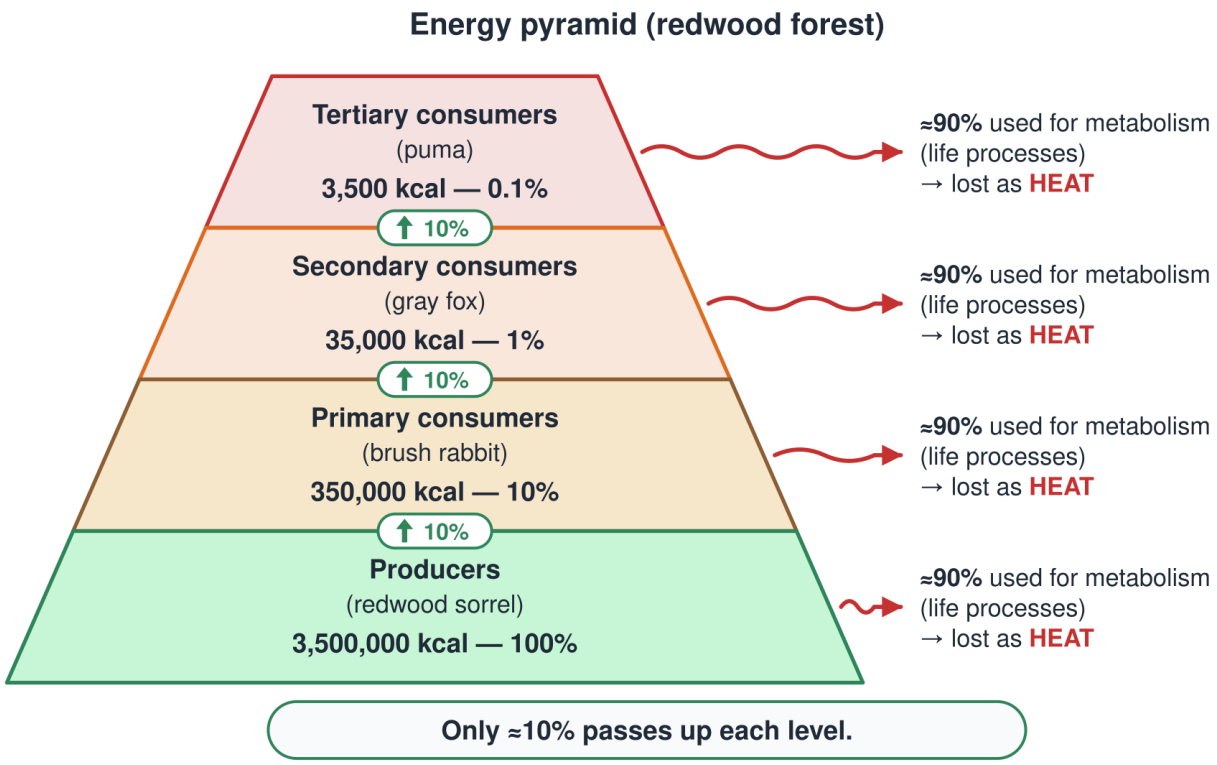
Common traps

- Decomposers are **not** primary consumers. They feed on dead material from **every** trophic level.
- **Energy flows; matter cycles.** Nutrients (matter) are recycled back to producers again and again. Energy enters as sunlight and eventually leaves as **heat** — it cannot be reused. (Your Gizmo handbook titles this section "Cycling of Energy"; on a test, remember that the *nutrients* are what get recycled.)

Energy flow, metabolism & the 10% rule

Only about **10%** of the energy at one trophic level is passed on to the next level. Where does the other ~90% go?

- **Metabolism** = all the chemical reactions that keep an organism alive (moving, growing, reproducing, staying warm). **During metabolism, some chemical energy is converted to heat energy. The heat leaves the organism, goes into the environment, and leaves the food chain.**
- Some energy is never passed on (parts not eaten — bones, fur, wood — and undigested waste go to decomposers).



Your handout's pyramid with the 3,500,000 kcal example: 100% → 10% → 1% → 0.1%.

How to calculate	Rule	Example
Up one level	multiply by 0.10 (÷ 10)	10,000 kcal producers → **1,000 kcal** primary consumers (Gizmo)
Up several levels	÷ 10 for each step	3,500,000 → 350,000 → 35,000 → **3,500 kcal** (Creating Chains, Q8)
Down (backward)	× 10 for each step	Tertiary has 50 kcal → secondary 500 → primary 5,000 → producers **50,000 kcal**
As a percent of producers	100% → 10% → 1% → 0.1%	Tertiary consumers get 0.1% of the producers' energy

Units: energy is measured in **calories**; **1,000 calories = 1 kilocalorie (kcal)**.

Why the 10% rule matters:

1. **Fewer organisms at higher levels** — lots of plants, fewer rabbits, very few pumas (Q9 on your worksheet).
2. **Food chains are short** — usually only 4–5 levels, because energy runs out.
3. **Top predators need huge territories** to find enough food, so they are hit hardest by **habitat fragmentation**.

> **FROM YOUR CLASS HANDOUTS**
> **From your handout:** "Three hundred trout are needed to support one man for a year. The trout, in turn, must consume 90,000 frogs, that must consume 27 million grasshoppers that live off of 1,000 tons of grass." — G. Tyler Miller, Jr. (1971). Every step up loses most of the energy.

Practice: Topic 3 drills (questions T3-xx in the Practice Questions section).

Topic 4 · Limiting Factors & Carrying Capacity

KNOW IT COLD

Know it cold: A population grows **exponentially** when resources are plentiful, **overshoots** its **carrying capacity (K)**, **declines**, **recovers**, and then rises and falls around K in a **cyclical/periodic** pattern. **Limiting factors** set K.

Limiting factors

A **limiting factor** is an environmental factor that **slows or stops a population's growth** (or causes it to decrease). Limiting factors can be **biotic** (food, predators, disease, competitors), **abiotic** (water, space, sunlight, temperature, weather), or both.

Limiting factor	Example from your worksheet	The lesson
Food & water	A bear must eat **10 fish a day**; the river provides about **100 fish a day** → the river supports at most **10 bears**.	Extra shelter or water won't help — **food** is the limiting factor. 100 ÷ 10 = 10 bears.
Living space	Gulls need nesting space on rocky shores; a pair with no room can't raise chicks. Crowded plants run out of root space, and taller plants shade smaller ones.	More nesting space → the gull population could increase.
Waste	Animal waste pollutes water and promotes dangerous **algae** growth.	Algae blooms block sunlight and use up oxygen → harm fish.
Plant growth	Photosynthesis is limited by sunlight (shading), CO_2, water, temperature, and soil nutrients.	The resource in shortest supply sets the limit.
Weather & random events	A late-spring cold front kills newborn offspring; a hurricane or flood washes away turtle nests and burrows.	These are **density-independent** (next section).

Density-dependent vs. density-independent

The one-question test: *Would this factor hurt a crowded population more (a bigger percentage) than a sparse one?* **Yes → density-dependent. No → density-independent.**

	Density-dependent	Density-independent
Meaning	Effect gets **stronger as the population gets more crowded**	Affects the population the **same way no matter its size or density** — "not affected by population size" (your worksheet)
Usually	**Biotic**	**Abiotic**
Examples	competition for food, water, space or mates; predation; **disease & parasites** (spread faster in crowds); waste buildup; nesting space; the bean-simulation rule "**5 or more rabbits in a square die**"	weather (cold front, heat wave), **drought**, **floods**, **hurricanes**, **wildfire**, volcanic eruptions, some human actions (chemical spill, clear-cutting)

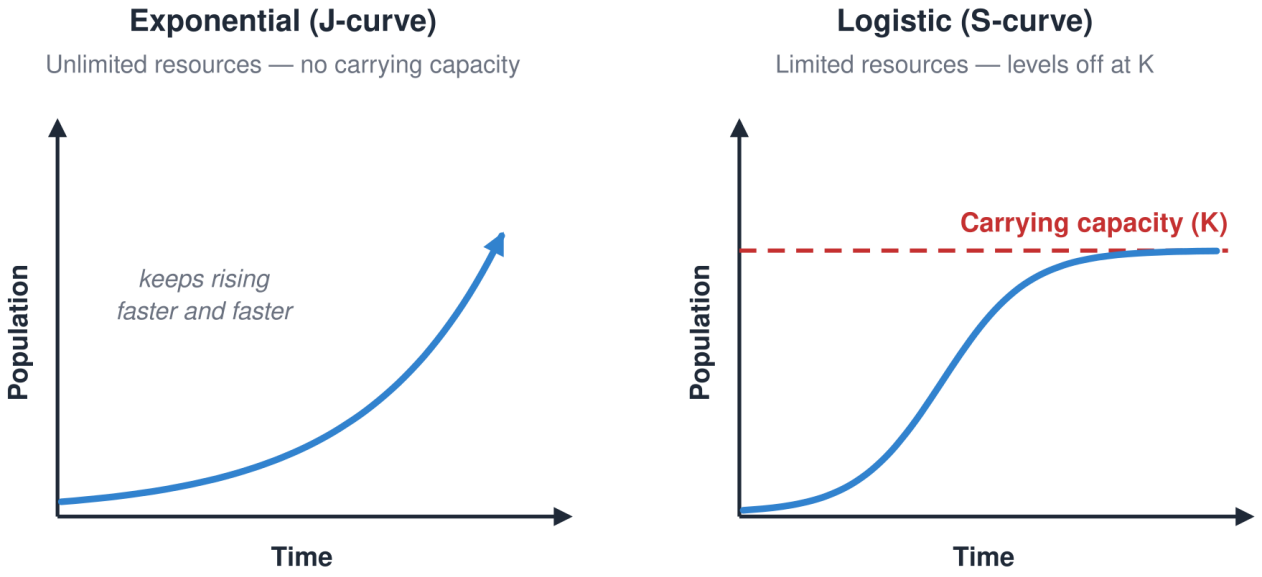
WATCH OUT

The #1 trap: "density-**independent**" does **NOT** mean "doesn't affect the population." A flood can kill 90% of a population! It means the **percent affected doesn't depend on how crowded** the population is.

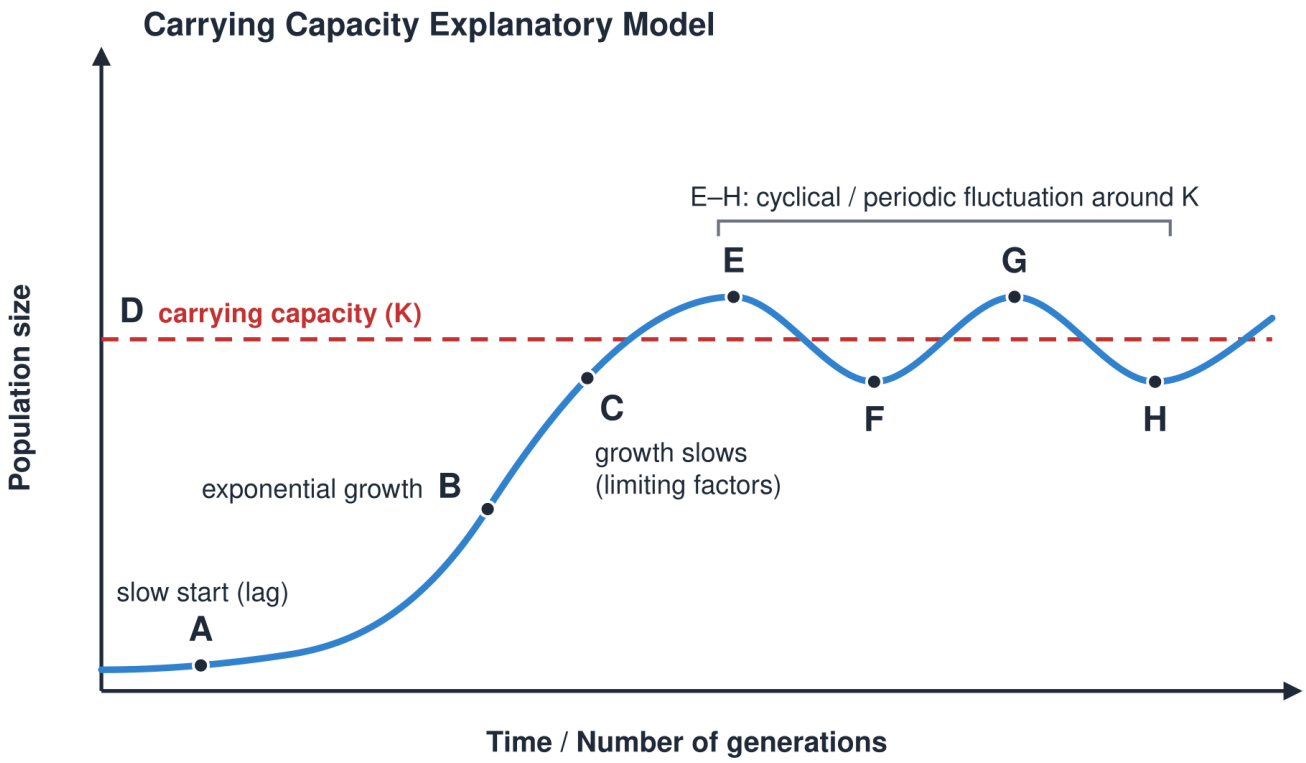
Reading a data table: if the % killed **rises** as density rises → density-dependent. If the % killed stays the **same** at every density → density-independent.

Population growth & carrying capacity

Carrying capacity (K) = the **largest population an area can support** over time with its resources (food, water, land/space) without damaging the environment. A population usually stays near its carrying capacity because of the habitat's **limiting factors**.



Exponential growth (J-curve) happens when nothing is limiting. Logistic growth (S-curve) levels off at K.



Your Carrying Capacity Explanatory Model with every letter labeled. Cover the labels and quiz yourself!

Letter (class model)	What it shows
A	Slow start — few individuals to reproduce
B	**Exponential growth** — plenty of resources; growth gets faster and faster
C	Growth slows as limiting factors kick in
D	**Carrying capacity (K)** — the dashed horizontal line
E, F, G, H	**Cyclical / periodic** rise and fall **around K** (peaks E & G above; dips F & H below)

Carrying capacity can change:

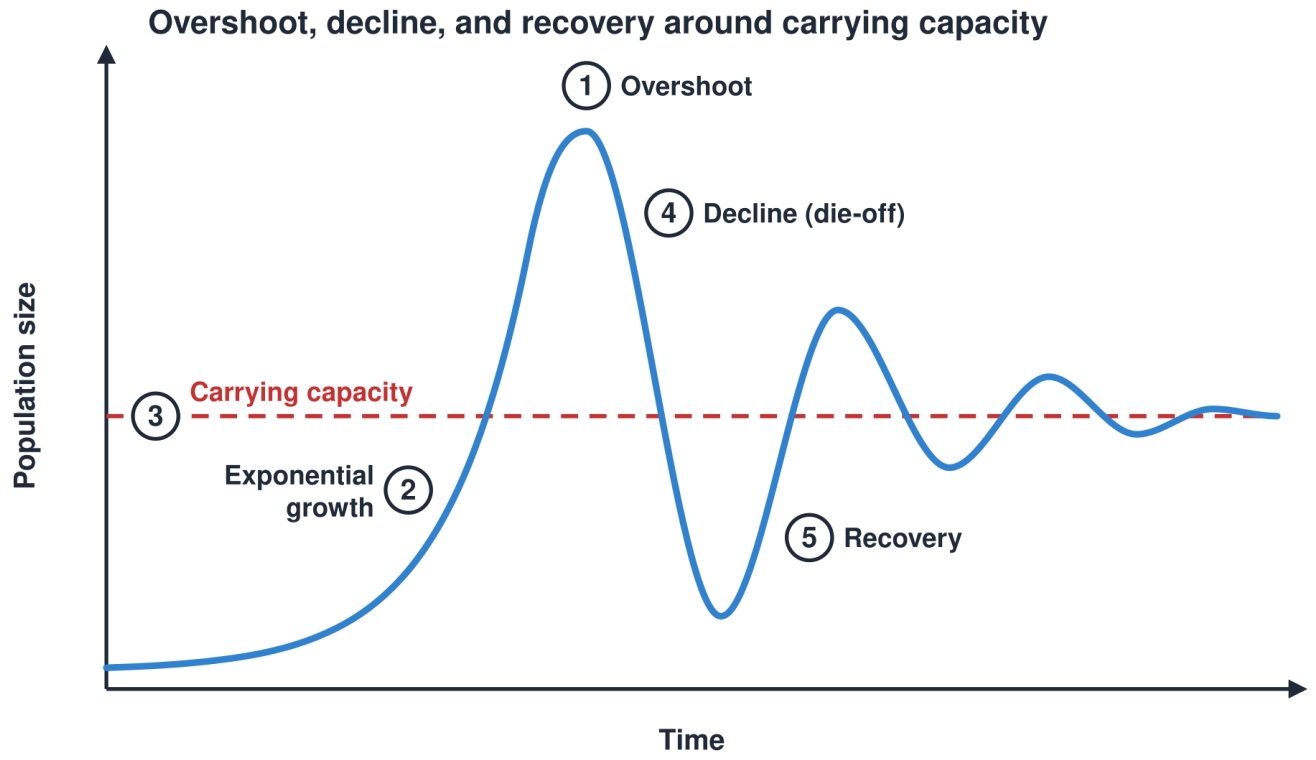
Raises K (more can survive)	Lowers K (fewer can survive)
more food, water, or shelter	habitat destruction or **fragmentation** (development, roads)
more space or restored habitat	drought, severe fire, pollution
fewer predators, competitors, or disease (e.g., humans hunting a rabbit's predators)	overgrazing that damages the food supply
wildlife corridors reconnecting habitat	new predators, competitors, invasive species, disease

WATCH OUT

Double-check this one: on your bean-simulation sheet, the "what could make K **increase**" and "what could make K **decrease**" answers were swapped (your teacher marked "swap"). Remember: **more resources or fewer threats → higher K; habitat loss → lower K.**

Overshoot, decline & recovery

From your worksheet: a growing population shows **exponential growth** and keeps growing until it **overshoots** the carrying capacity. It runs out of resources and **declines** rapidly until it can **recover** and stabilize around K.



The five stages from your Limiting Factors worksheet graph (Q14).

Real example	What happened
Australian rabbits (mid-1800s, worksheet)	Plenty of vegetation + no predators → exponential growth → ate the land bare → **crash** (starvation) → vegetation recovered → rabbits increased again. A predator helped stabilize them.
St. Matthew Island reindeer (Alaska)	29 reindeer released in 1944, no predators → 1,350 (1957) → about **6,000** (1963) = overshoot; they overgrazed the lichen, then a brutal winter → only **42** left by 1966. Overgrazing **lowered K** permanently.

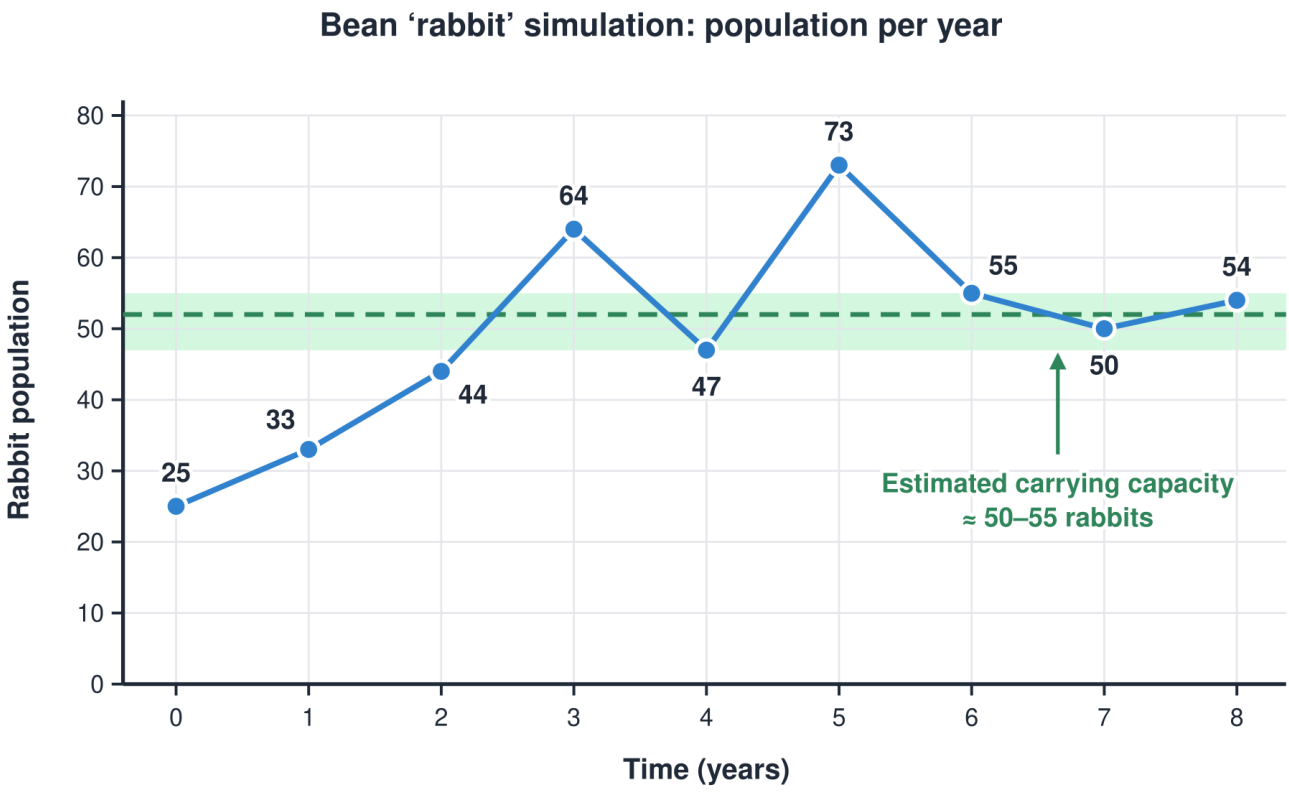
FROM YOUR CLASS HANDOUTS

From your Carrying Capacity Simulation (beans = rabbits, paper grid = habitat):

Rule	What it represents
5 or more rabbits in one square → remove all	Overpopulation — not enough resources → they die (**density-dependent**)
Exactly **1** rabbit in a square → survives, no babies	No mate → can't reproduce
2, 3, or 4 in a square → all become parents; add the same number of babies	Reproduction

Year	0	1	2	3	4	5	6	7	8
Rabbits	25	33	44	64	47	73	55	50	54

- Years 0–3: rapid growth (positive relationship) — 25 → 64.
- Years 3 and 5: **overshoot** (64, 73) followed by **declines** (47, 55).
- Years 6–8: the population settles between **50 and 55 → K ≈ 50 rabbits**.
- **Graph it as a LINE graph** (change over time) — your teacher marked the bar graph "Wrong type of graph."



Your class data as a line graph, with the estimated carrying capacity.

Cyclical / periodic patterns

In the class explanatory model, after the population reaches K it keeps **rising and falling above and below K** (E–F–G–H). Your class labeled this the **cyclical / periodic relationship**. Why does it happen? When the population is **above K**, limiting factors (food shortage, competition, predators) push it down. When it is **below K**, resources recover and the population grows again.

Term (Ms. Hero's review notes)	Meaning
Cyclical	The pattern repeats, returning to the same state over and over (recurring cycles)
Periodic	Changes happen at intervals, but may not return to exactly the same state

Where you'll see repeating patterns: populations fluctuating around K; **predator–prey cycles** (snowshoe hare and lynx, about every 10 years — see Topic 2); seasonal changes (populations rise in spring and summer, fall in winter).

> **TEST TIP**
> **On the test:** if a graph shows a **wavy, repeating up-and-down pattern**, it is the **cyclical/periodic** part. In a predator–prey graph, the **predator peaks come after** the prey peaks.

Practice: Topic 4 drills (questions T4-xx in the Practice Questions section).

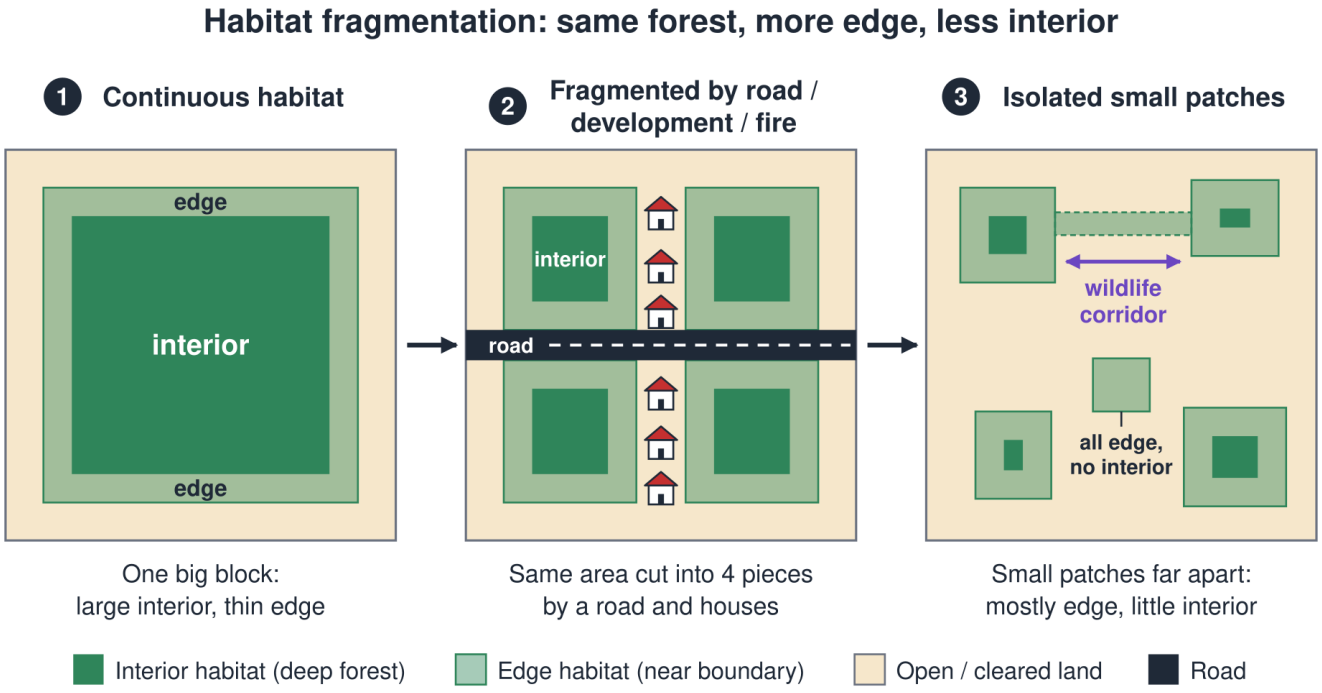
Topic 5 · Succession, Habitat Fragmentation & Fire

KNOW IT COLD

Know it cold: No soil → primary succession (starts with lichens). **Soil left → secondary** succession (starts with grasses and wildflowers; faster). **Soil lost + humans restore → tertiary** (Ms. Hero's term). **Surface fires** burn the forest floor; **crown fires** burn the treetops. **Fragmentation** breaks habitat into small, isolated patches.

Habitat fragmentation

Habitat fragmentation = breaking a large, continuous habitat into **smaller, isolated patches** (and often shrinking it), caused by human activities (roads, development, farming, logging) or natural causes like **fire**.



Fragmentation turns one big habitat into small patches with more edge and less interior. Corridors reconnect them.

Why it matters:

- Smaller, isolated populations → animals can't move to find food, mates, or new territory → less genetic variety.
- **Less interior habitat, more edge** (edges are brighter, hotter, windier, and easier for predators and invasive species to reach).
- Species that need **large areas** (pumas, interior-forest birds) decline first.
- It disrupts the ecosystem's **equilibrium** and lowers **carrying capacity**.

FROM YOUR CLASS HANDOUTS

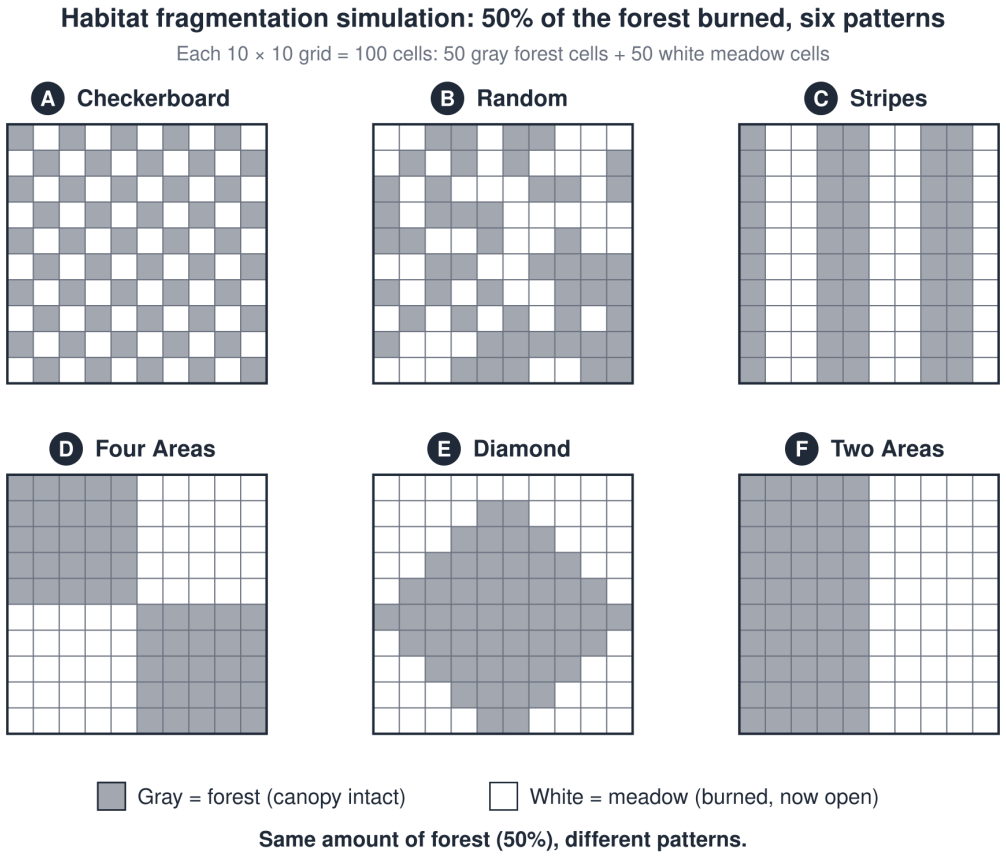
From your Habitat Fragmentation Simulation (10×10 grid; gray = forest with canopy intact; white = meadow, previously burned and now open):

Exercise 1 — Marbled Murrelet (your species):

% meadow (burned)	10	20	40	60	80
Forest cells left	90	80	60	40	20
Suitable cells	90	79	55	24	4
% habitat available (suitable ÷ forest × 100)	100%	98.8%	91.7%	60%	20%

Forest dropped from 90 to 20 cells (−78%), but suitable habitat dropped from 90 to 4 cells (−96%). **Habitat loss plus fragmentation hurts more than habitat loss alone**, because the remaining forest is broken into pieces too small to use.

Exercise 2 — same amount of forest (50%), six different patterns:



The six clearing patterns from Exercise 2. Every grid has exactly 50 forest cells.

Lesson from Exercise 2	Explanation
Pattern matters, not just amount	All six grids have 50% forest, yet each species found very different amounts of usable habitat.
Checkerboard = most fragmented	No two forest cells share a side → it is *all edge* and no interior. Worst for species that need big, connected forest.
Big blocks keep the most interior	Two Areas, Four Areas, and Diamond keep large connected patches → best for interior-forest species that need large areas for hunting and finding mates.
Different species, different needs	Open-habitat birds like **Say's Phoebe** use the meadow; closed-canopy birds need intact forest; some species like edges.
Best plan for ALL species	A mix: keep some **large connected forest blocks**, some open meadow, and **corridors** linking patches — support it with claim, evidence, and reasoning.

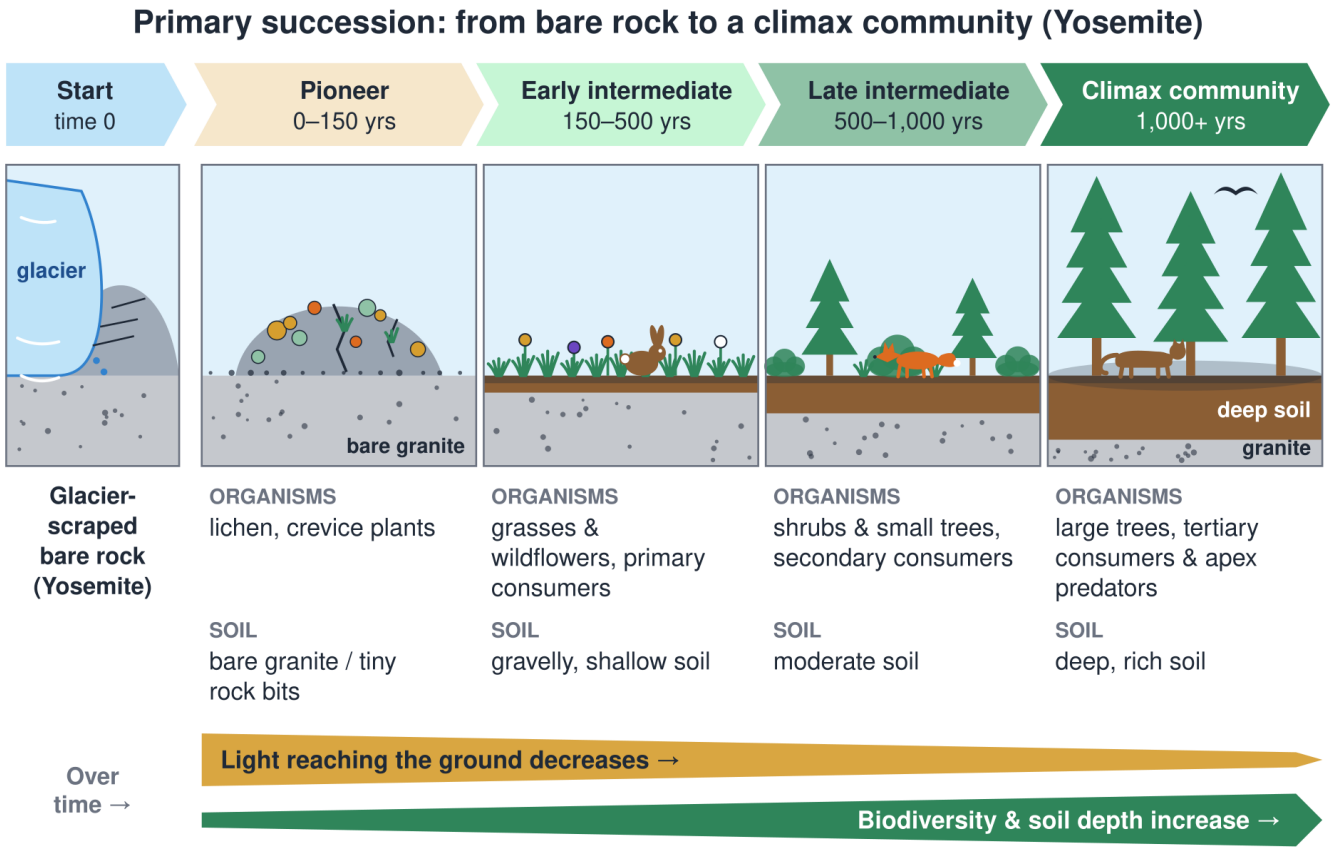
Real-world fix — wildlife crossings: the **Highway 17 Laurel Curve wildlife undercrossing** in the Santa Cruz Mountains (finished in 2023) lets mountain lions and other animals pass safely under a busy highway — reconnecting fragmented habitat.

Primary succession

Ecological succession = the gradual, fairly predictable change in the species living in an area over time.

Primary succession = building an ecosystem **where there is NO soil** — bare rock such as **glacier-scraped granite (Yosemite)**, new **lava flows**, new volcanic islands, or rock exposed by a landslide.

Pioneer species = the very first organisms to colonize an area. In primary succession they are **lichens** (a fungus + alga mutualism) and mosses: tough, and they don't need soil. Lichens **chemically and physically break rock into tiny pieces**; their remains add organic matter → **the first soil**.



From your "Life from Stone" timeline: soil deepens and biodiversity increases, while light reaching the ground decreases.

FROM YOUR CLASS HANDOUTS

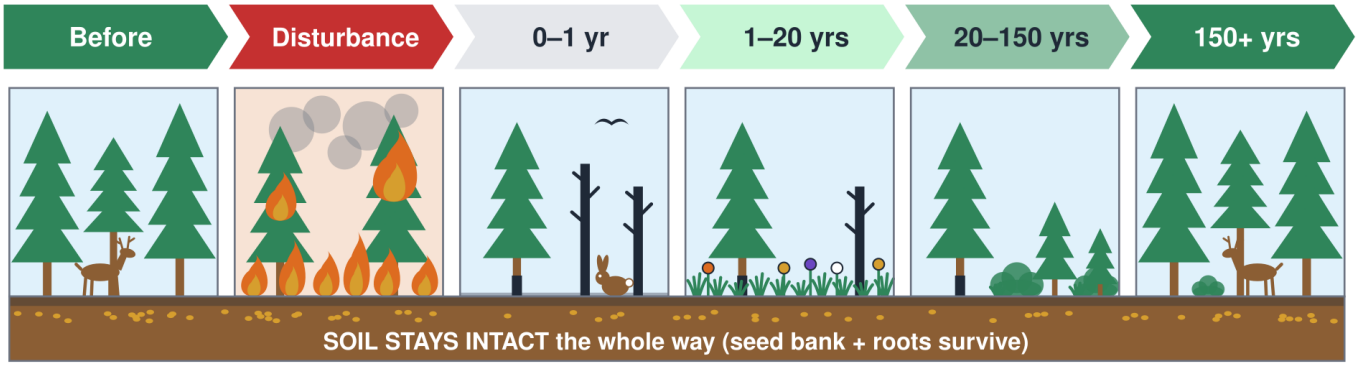
From "Ecological Succession: Life from Stone — How Yosemite was Made": Yosemite's granite formed deep underground from magma, was pushed up to the surface, and **glaciers carved huge valleys** and scraped the rock bare before melting away. That bare granite is where primary succession began.

Stage	Pioneer	Early intermediate	Late intermediate	Climax community
Approx. years	0–150	150–500	500–1,000	1,000+
Organisms	lichen, crevice-dwelling plants	grasses & wildflowers; primary consumers	shrubs & small trees; secondary consumers	large trees; tertiary consumers & apex predators
Soil	tiny rock bits — not soil yet	gravelly, shallow soil	moderate depth & quality	deep, rich soil
Light at ground	maximum	slightly less	moderate (shade from small trees)	little (shade crowds out sun-loving species)
Ecological impact	break bare rock into tiny pieces → foundation of soil	add organic matter & nitrogen; hold soil	add nitrogen, trap moisture; decomposers speed up breakdown	very diverse, stable community

Secondary & tertiary succession

Secondary succession = an established ecosystem **recovering after a disturbance that removed the community but left the SOIL** — after a fire, flood, logging, a hurricane, or an abandoned farm field. It is **faster** than primary succession because soil, buried seeds (the **seed bank**), and roots are already there.

Secondary succession: recovery after a disturbance (soil stays)



Contrast: Primary succession starts with NO soil (bare rock).
Secondary succession starts with soil already in place, so it is faster.

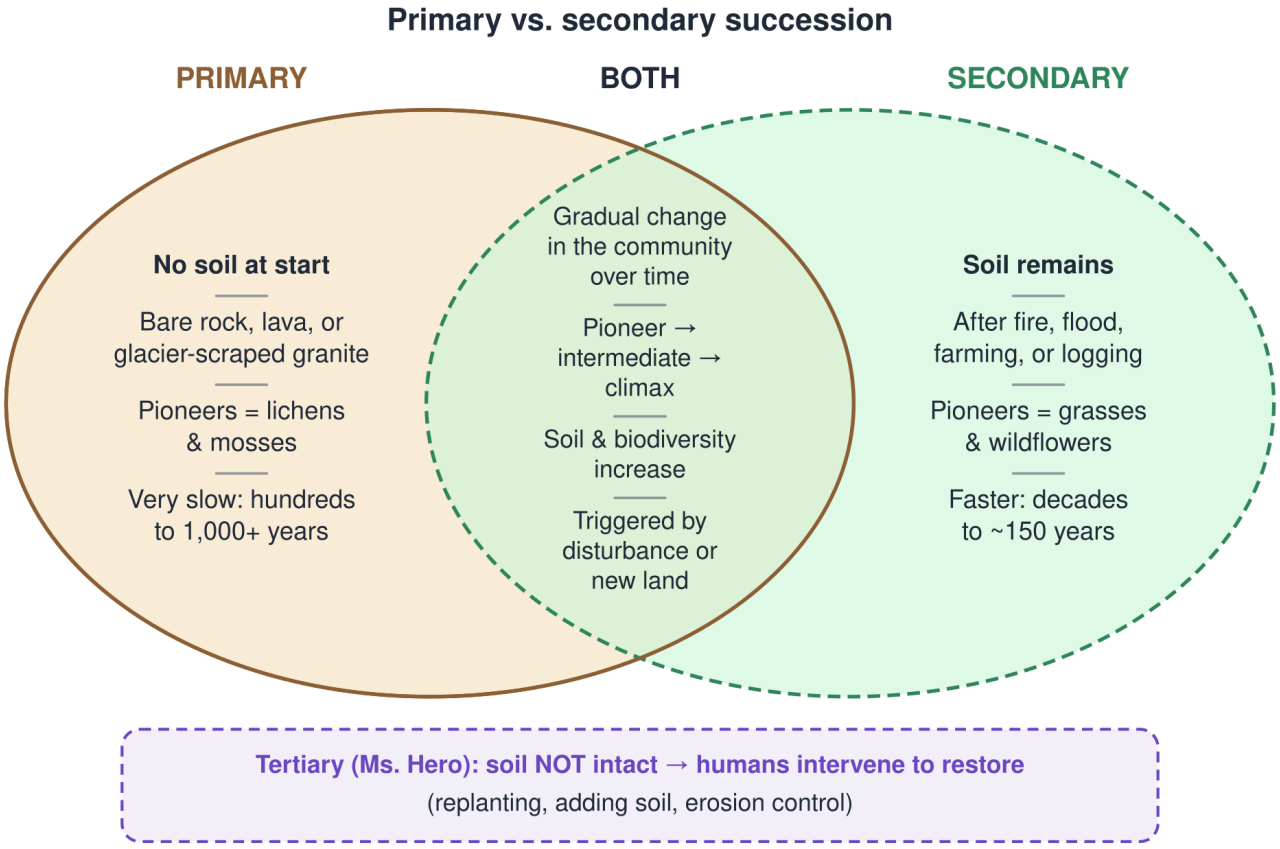
From your surface-fire disturbance card and Secondary Succession Timeline.

FROM YOUR CLASS HANDOUTS

Your Secondary Succession Timeline (after a surface fire in the Yosemite Valley climax community):

Stage	Immediate aftermath	Early recovery (pioneers)	Mid-recovery (intermediate)	Long-term recovery (climax)
Approx. years	0–1	1–20	20–150	150+
Organisms	survivors: large trees, some animals; seeds in the soil	grasses & wildflowers move back in first	shrubs & small trees	fully recovered climax community
Soil	remains healthy & moist	healthy	healthy	healthy
Light at ground	slightly more than before the fire	about the same	a little less (new trees shade)	shady again, like a mature forest
Impact	dead organic matter cleared; stored seeds sprout	new trees start to sprout		

Tertiary succession (Ms. Hero's term): the soil is **NOT intact** (severely eroded or degraded) and **humans intervene to restore** the ecosystem — replanting, adding soil or mulch, erosion control. This term is rarely in textbooks, so **use Ms. Hero's definition on her test**.



Fill in the "Both" column you left blank on your handout (Q11).

	Primary	Secondary	Tertiary (Ms. Hero)
Starting point	bare rock: lava, glacier-scraped granite	a disturbed ecosystem: burned forest, flooded area, abandoned farm	land whose soil was lost or badly degraded
Soil at the start?	**No**	**Yes**	**No / damaged**
First arrivals	lichens & mosses	grasses & wildflowers (plus seeds and roots already in the soil)	plants chosen and planted by people

	Primary	Secondary	Tertiary (Ms. Hero)
Speed	very slow — hundreds to 1,000+ years	faster — decades to ~150 years	depends on the restoration effort
Both / all	gradual change in the community over time · pioneers → intermediate → climax · soil and biodiversity increase · started by new land or a disturbance		

MODEL ANSWERS

Model answers to the questions left blank on your succession handout

- **Q13 — Does fire always mean destruction?** No. A surface fire clears dead fuel, returns nutrients to the soil, opens the canopy so sunlight reaches seedlings, lets stored seeds sprout (knobcone pine cones open in fire heat), and creates tree cavities that animals use as homes. Many Big Basin redwoods survived the 2020 fire and resprouted.
- **Q14 — How does biodiversity change during recovery?** It **increases** over time: as soil deepens and plants grow taller, more kinds of habitats (niches) form, so more species of plants and animals move back in.

Who lives there: pioneers, climax community & more

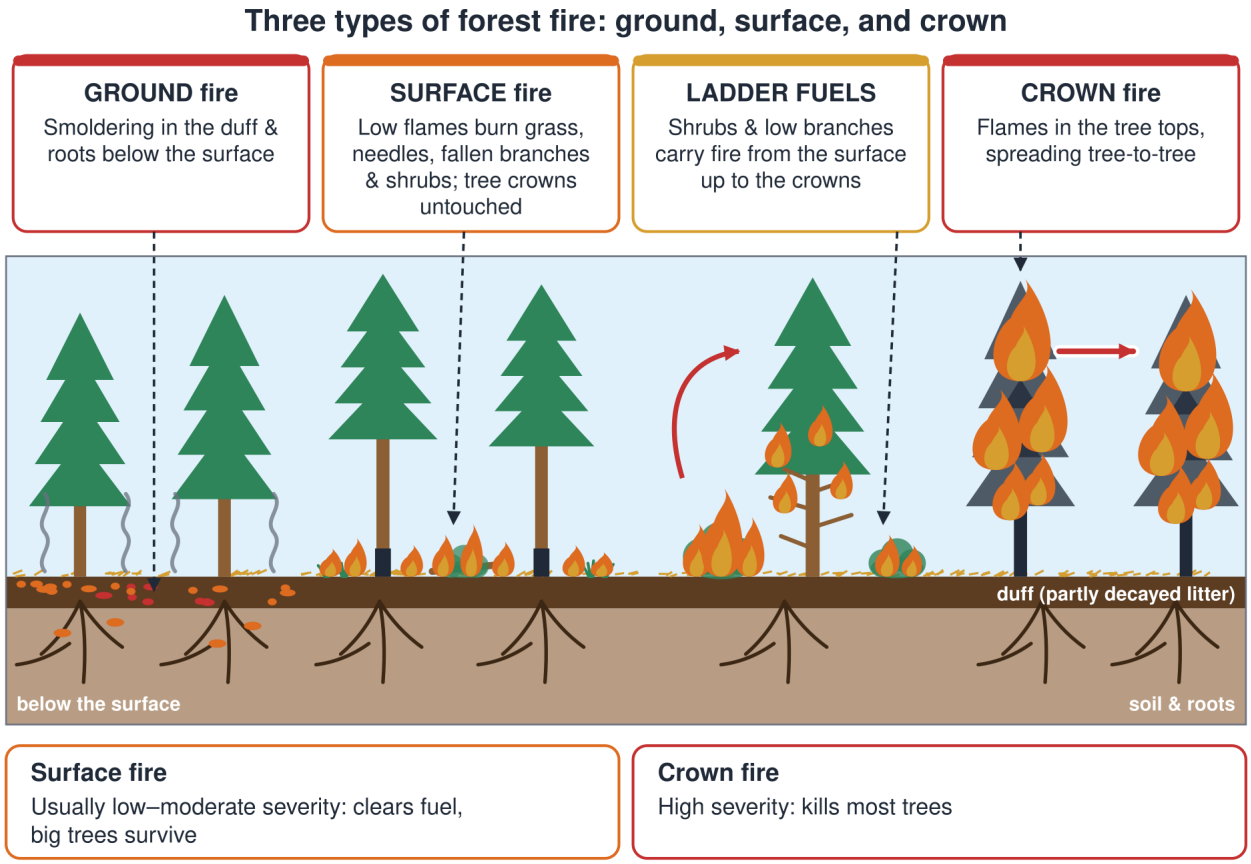
Term	Meaning	Examples
Pioneer species	First organisms to colonize; tough; build or hold soil	primary: **lichens, mosses**; secondary: **grasses, wildflowers**
Intermediate species	Move in as soil improves	shrubs, small trees, primary & secondary consumers
Climax community	The final, mature, relatively **stable** community; it lasts until the next big disturbance	old-growth redwood forest in Big Basin: large trees, tertiary consumers, apex predators

From Life from Stone Q6, organisms in the climax community fall into three groups:

Symbol	Group	Who
★	**Residents**	persist in the climax community — large trees, tertiary consumers, apex predators
✓	**Edge-dwellers**	live along edges and openings — grasses, wildflowers, shrubs, small trees, and the consumers that use them
✗	**Retired pioneers**	early colonizers that mostly disappear as shade increases — lichens, crevice-dwelling plants

Fire: surface vs. crown, severity & disturbances

A **disturbance** is an event that disrupts an ecosystem's community — **natural** (fire, flood, hurricane, volcano, landslide, drought) or **human** (logging, clearing land, development, farming, pollution). Disturbances start succession.



Ground, surface, and crown fires. Ladder fuels carry a surface fire up into the crowns.

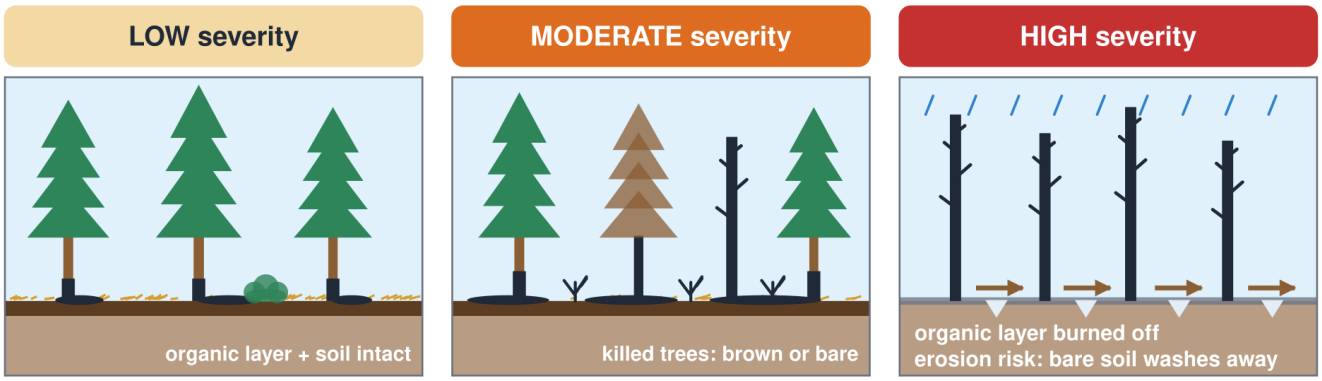
Crown fire	Both	Surface fire
Burns through the **tree tops (canopy)**, spreading tree to tree	Are wildfires and **disturbances**	Burns along the **forest floor** — grass, needles, fallen branches, shrubs
Usually very hot and fast; pushed by wind and slope	Need **fuel + heat + oxygen**	Lower flames; the **canopy is mostly untouched**
Kills most trees — usually **high severity**	Driven by weather (heat, wind, dryness) and slope	Big, thick-barked trees usually **survive** — usually **low to moderate severity**
Often climbs up **ladder fuels** (shrubs, low branches)	Release smoke and CO_2	**Clears fuel**; soil stays healthy; stored seeds can sprout
Can damage soil → slower recovery	Are followed by **succession**	Can be good for the forest (natural or **prescribed**)

A **ground fire** smolders below the surface in the duff (partly decayed litter) and roots.

Fire severity vs. fire intensity:

Term	Measures	Think of it as
Fire intensity	How much **heat/energy** the fire releases	The fire's *behavior*
Fire severity	How much the fire **changed or damaged the ecosystem** — plants killed and soil burned (low / moderate / high)	The fire's *effects*

Fire severity: low vs. moderate vs. high



- Surface fire
- Most trees survive
- Litter partly burned
- Soil intact

- A mix of effects
- Some trees killed
- Understory burned

- Crown fire
- Most of the canopy killed
- Soil organic layer burned off
- Erosion risk

SEVERITY
= how much the fire changed the ecosystem (plants & soil)

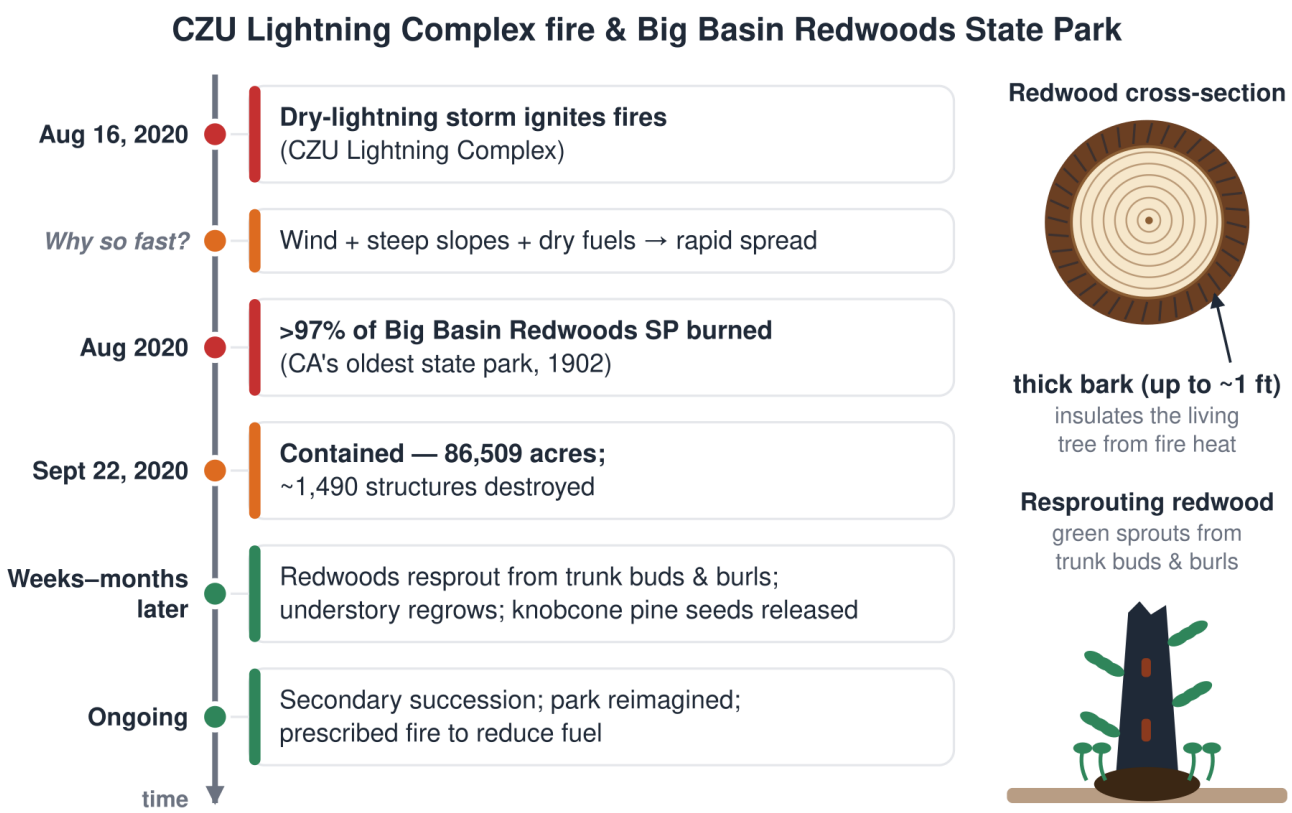
INTENSITY
= how much heat/energy the fire released

Big Basin 2020: ≈77% high, 20% moderate, 2% low severity.

Severity describes what the fire did to the ecosystem. Big Basin 2020 was mostly high severity.

Prescribed (controlled) burns are low-severity surface fires set on purpose to burn up built-up fuel, so that a future wildfire is less likely to become a devastating crown fire.

Case study: CZU Lightning Complex & Big Basin



The CZU Lightning Complex fire and Big Basin's recovery.

Topic	What to know
Big Basin ecosystem & history	**California's oldest state park** (1902); old-growth **coast redwoods** — some of the tallest, largest trees in the U.S.; lots of biodiversity; conifer forest, oak woodland, chaparral, and riparian (streamside) habitats
Climate	**Mediterranean climate**: cool, wet winters and hot, dry summers → plants grow in winter, then dry out into fuel
The fire	Started **Aug. 16, 2020** by a **dry-lightning storm**; spread fast because of **wind, steep slopes**, and dry fuel; **more than 97%** of Big Basin burned; 86,509 acres total; about 1,490 structures destroyed; contained Sept. 22, 2020
Redwood adaptations (resilience)	Very **thick bark** (up to ~1 ft) that insulates the living tree; **resprouts** from dormant buds on the trunk and from burls → many old-growth redwoods survived even after their crowns and outer bark burned
Recovery	More sunlight reaching the ground → more **undergrowth**; burned trees form **cavities** that animals use as homes; **secondary succession** (soil remained)
Climate change	Rising temperatures → **more and larger wildfires**; your class slides projected a large increase in burn area on the West Coast by the 2050s
Management	**Prescribed fire** burns up some fuel so future wildfires won't devastate the ecosystem

TEST TIP

CZU connects to almost every topic: Earth systems (lightning + wind + slope + plants + creeks), density-independent limiting factor (fire), habitat fragmentation (burned patches = the white "meadow" cells in your simulation), secondary succession, fire severity, biodiversity recovery, and graph reading (the California wildfire graphs).

FROM YOUR CLASS HANDOUTS

From "What is happening in California?" (graph analysis): the class graphs showed the average annual area burned rising by decade (biggest increase starting in the 2000s), **conifer forest and shrubland** burning the most area and hardwood forest the least, and that **acres burned do not perfectly match the number of large fires** in a year. See Topic 8 for how to analyze graphs like these.

Practice: Topic 5 drills (questions T5-xx in the Practice Questions section).

Topic 6 · Biodiversity

KNOW IT COLD
Know it cold: Richness = how many species. **Evenness** = how equal their numbers are. **Relative abundance** = one species' count ÷ total count. The most biodiverse communities have high richness **and** high evenness.

What biodiversity is and why it matters

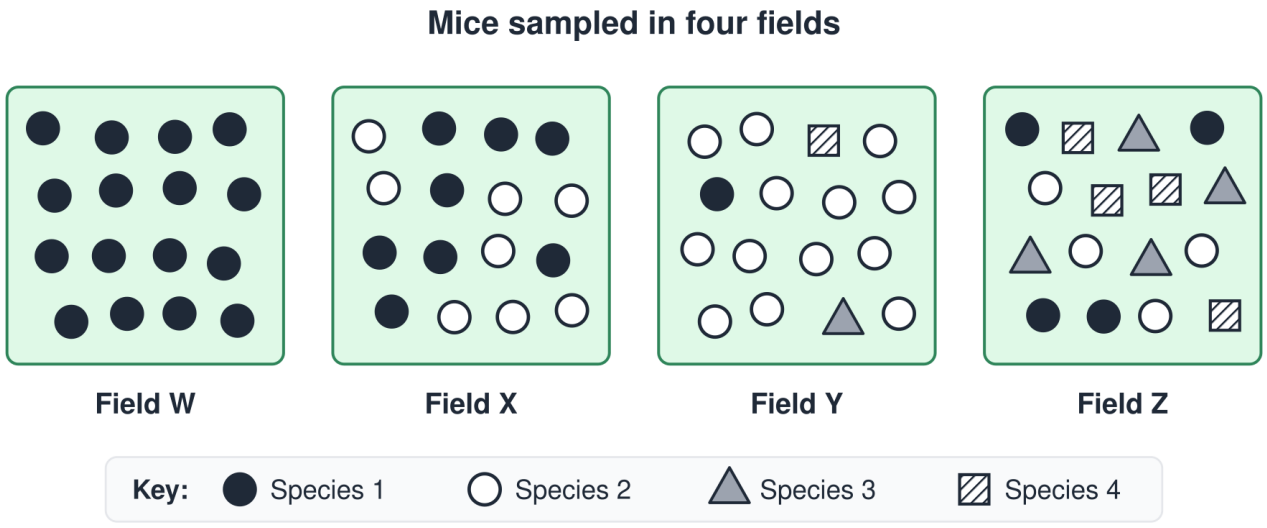
Biodiversity (biological diversity) = the **variety of life** — the variety of ecosystems, species, and genes in an area (or on Earth).

Why it matters: ecosystems with more biodiversity are usually more **stable and resilient**. If one species declines, others can fill its role (for example, a predator has other prey to switch to), so the food web doesn't collapse, and the ecosystem recovers faster after a disturbance.

How scientists measure it: they can't count every organism, so they **sample** — count individuals in sample plots (quadrats), along lines (transects), or from aerial counts (like the HHMI elephant census video) — and use the samples to estimate richness, evenness, and relative abundance.

Richness and evenness

Measure	Question it answers	High	Low
Species richness	How many **different species** are there?	rainforest — very rich	a field of only one crop
Species evenness	Are the species' population sizes **similar**, or does one dominate?	every species has about the same count	a **grass lawn**: grass far outnumbers clover, dandelions, ants, and bees



Each field has 16 mice.

Practice like Figure 1 on your Sampling Diversity handout: label each field rich, even, both, or neither.

Field	Species counts	Rich?	Even?	Label
W	16 (one species)	No	— (only one species)	**neither**
X	8, 8	No (2 species)	**Yes**	even
Y	13, 1, 1, 1	**Yes** (4 species)	No	rich
Z	4, 4, 4, 4	**Yes** (4 species)	**Yes**	**both** — most diverse

WATCH OUT

Common traps: the **total number of individuals** (16 in every field) tells you nothing about richness. Richness counts **species**, not individuals. And a community can be rich but uneven (Y) or even but not rich (X).

Relative abundance (with your class data)

Relative abundance = (number of individuals of ONE species) ÷ (total individuals of ALL species)

Multiply by 100 for a percent. All relative abundances in a community add up to 1 (100%).

Example (Community A, species B): 125 ÷ 200 = **0.625 = 62.5%**.

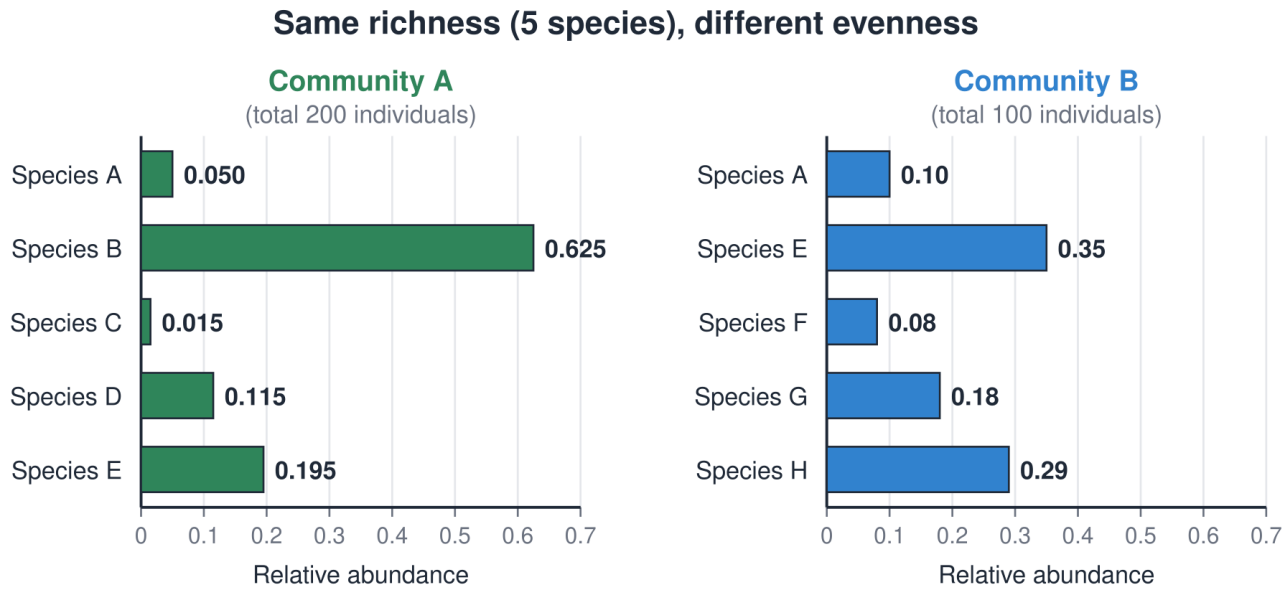
FROM YOUR CLASS HANDOUTS

Your Sampling Diversity tables:

Community A	A	B	C	D	E	Total
Individuals	10	125	3	23	39	**200**
Relative abundance	0.050	**0.625**	0.015	0.115	0.195	1.000

Community B	A	E	F	G	H	Total
Individuals	10	35	8	18	29	**100**
Relative abundance	0.10	0.35	0.08	0.18	0.29	1.00

Community C	A	B	C	D	E	F	G	H	I	J	Total
Individuals	12	1	5	31	21	4	9	8	2	2	**95**
Rel. abundance	0.126	0.011	0.053	0.326	0.221	0.042	0.095	0.084	0.021	0.021	1.000



Relative abundance = individuals of one species ÷ total individuals in the sample

Same richness (5 species each), but Community A is dominated by one species.

MODEL ANSWERS

Answers to the Sampling Diversity questions (Q1–Q2 were incomplete on your sheet):

Question	Answer	Evidence
Richness of A vs. B	**Equally rich**	Both have 5 species
Evenness of A vs. B	**B is more even**	In A, species B is 62.5% of all individuals and species C only 1.5%. In B, the range is just 8%–35%
Most rich	**Community C**	10 species (A and B have 5 — the least rich)
Most even	**Community B**	Its values are closest to an equal share (5 species → 0.20 each)
Least even	**Community A**	One species dominates (0.625)
Most diverse overall	**Community C**	Most species, and reasonably even (largest share is only 0.326)

Quick evenness check: if every species were equal, each would have a relative abundance of 1 ÷ (number of species). For 5 species, that's 0.20 each. The closer all the values are to that number, the more even the community. If one species is more than half of everything, the community is very uneven.

Q3 — richness or evenness, which matters more? A strong answer: **both** matter for a healthy ecosystem. Richness gives the food web many backup species; evenness means no single species dominates. A lawn is a weak, unstable community: if a disease hit the grass, almost everything would be gone. Use CER: *claim → evidence from the tables → reasoning about resilience*.

Practice: Topic 6 drills (questions T6-xx in the Practice Questions section).

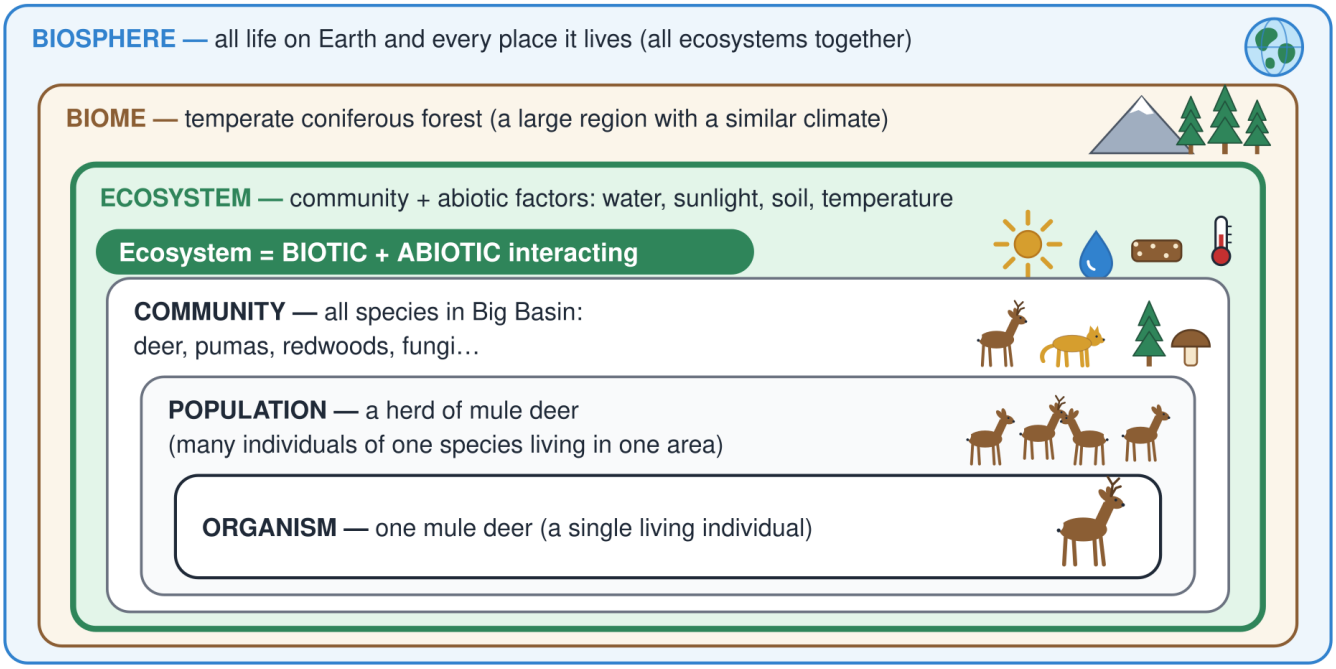
Topic 7 · What is an Ecosystem?

KNOW IT COLD
Know it cold: Ecosystem = **all the living things in an area + their nonliving environment, interacting**. A healthy ecosystem is in **equilibrium** (balance). Biotic = living (or once-living); abiotic = nonliving.

Ecosystem

An **ecosystem** is all the organisms (the **community**) in an area **interacting with each other and with the nonliving (abiotic) parts** of their environment — sunlight, water, air, rocks, and soil. "**Eco**" means **home**: an ecosystem is the system where organisms live. Ecosystems come in **different sizes** — a pond, a forest, a rotting log, or the whole Earth.

Levels of organization in ecology



Inside → out: organism → population → community → ecosystem → biome → biosphere
Each level includes all of the levels inside it.

Levels of organization. Each level includes all the levels inside it.

Level	What it includes	Big Basin example
Organism	one individual	one mule deer
Population	all members of **one species** in an area	the herd of mule deer
Community	**all the populations** (all species) in an area — living only	deer, pumas, redwoods, fungi, banana slugs...
Ecosystem	community **+ abiotic factors**	the community + creek water, sunlight, soil, temperature
Biome	a large region with a similar climate and communities	temperate coniferous forest
Biosphere	all ecosystems on Earth	the living part of the planet

WATCH OUT
Community vs. ecosystem: a **community** is only the living things. An **ecosystem** adds the **nonliving** environment. If the answer choice mentions water, soil, or sunlight, it's describing an ecosystem.

Equilibrium and resilience

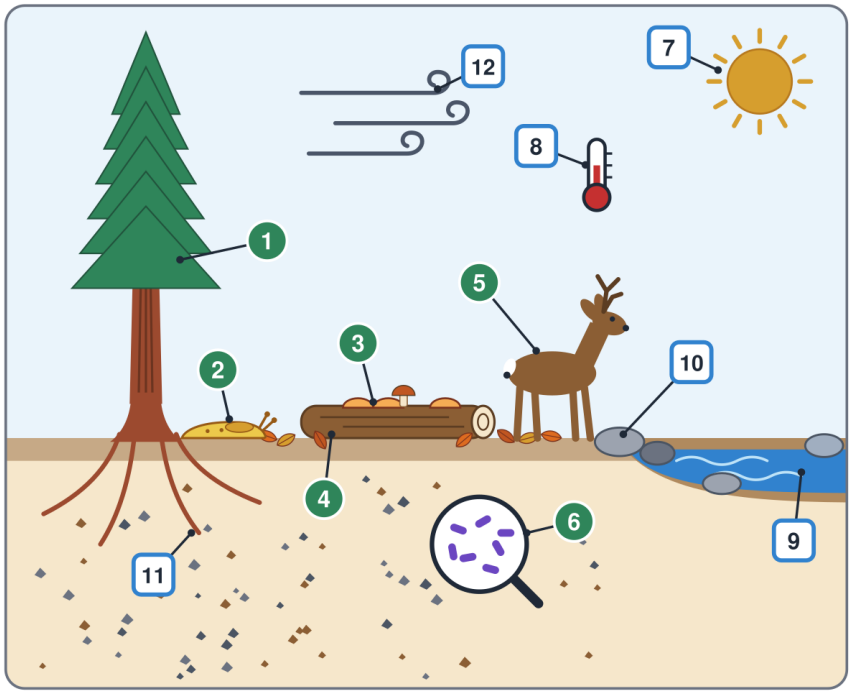
Equilibrium = a **balance** among the species in an ecosystem: populations rise and fall a little but stay within their **normal ranges** over time (a *dynamic* balance — like a population fluctuating around its carrying capacity).

Disrupts equilibrium	Restores or protects equilibrium
removing a **keystone species** (wolves gone → deer 15,000 in Atlas NP)	reintroducing the keystone species (wolves back → all populations normal)
habitat fragmentation	wildlife corridors, protecting large habitat blocks
severe disturbances (high-severity fire, flood), drought, climate change	**succession**; fire adaptations like redwood bark (**resilience mechanisms**)
pollution (waste → algae bloom), invasive species	restoration by people (**tertiary** succession, prescribed fire)

Resilience = an ecosystem's ability to **recover** after a disturbance. Higher **biodiversity** usually means higher resilience.

Biotic vs. abiotic factors

Biotic and abiotic factors: a redwood forest creek



BIOTIC — living or once living

1. Redwood tree
2. Banana slug
3. Fungi on log
4. Fallen log & dead leaves **(once living)**
5. Deer
6. Bacteria in soil

ABIOTIC — nonliving

7. Sunlight
8. Air temperature
9. Creek water
10. Rocks
11. Soil minerals
12. Wind

Trap: dead leaves and logs are BIOTIC (once living).

Living (or once-living) parts are biotic; nonliving physical and chemical parts are abiotic.

Biotic (living or once living)	Abiotic (nonliving)
plants, animals, fungi, bacteria, algae	sunlight, temperature, water, air (O_2, CO_2)
dead leaves, fallen logs (once living)	rocks, soil minerals, pH, wind

Biotic (living or once living)	Abiotic (nonliving)
biotic *interactions*: competition, predation, disease, members of the same species competing for food	weather and climate; abiotic "ecological forces or disturbances" such as **drought and fire**

WATCH OUT

Common traps

- **Dead things** (a fallen log, dead leaves, a carcass) count as **biotic** in most class materials, because they were once living.
- **Soil is a mix:** the minerals and water are abiotic; the bacteria, worms, and decaying matter (humus) are biotic.
- **Weather is abiotic** (your Limiting Factors worksheet, Q10) — and it is also a density-independent limiting factor.

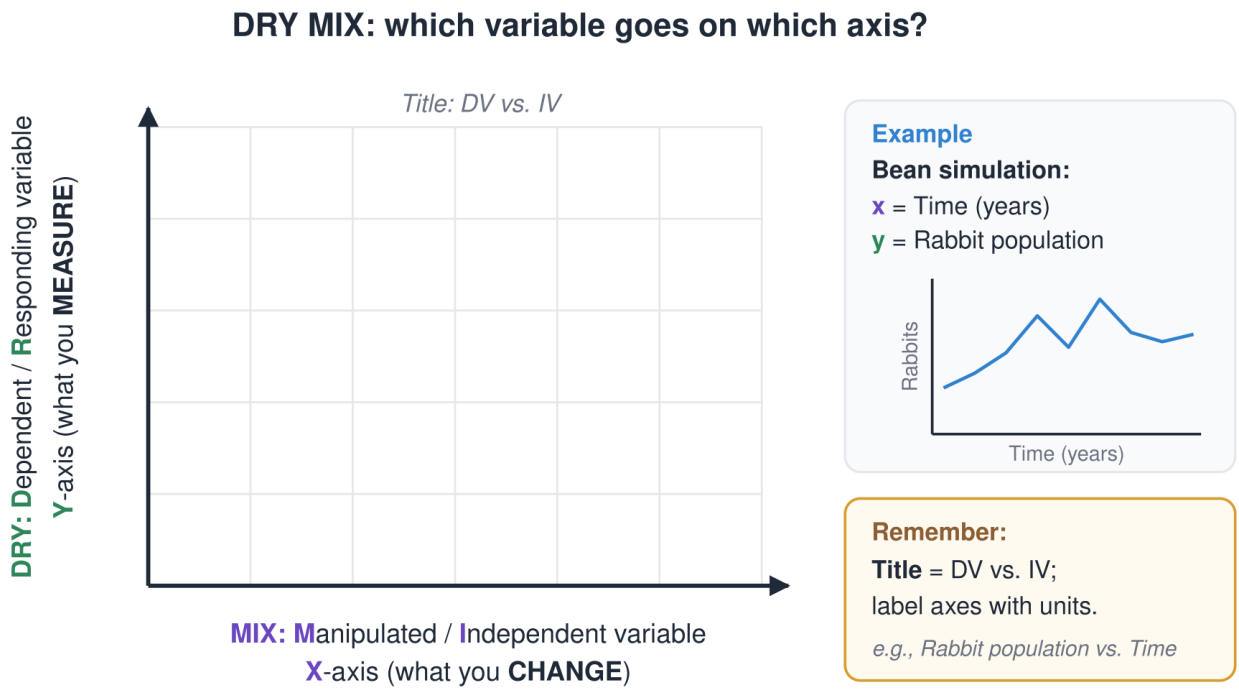
Practice: Topic 7 drills (questions T7-xx in the Practice Questions section).

Topic 8 · Graphs

KNOW IT COLD

Know it cold — DRY MIX: Dependent / **R**esponding variable goes on the **Y**-axis. **M**anipulated / **I**ndependent variable goes on the **X**-axis. **Time** is almost always the independent variable (x-axis).

DRY MIX: axes, IV & DV



Where each variable belongs on a graph.

Term	Meaning	Where it goes
Independent variable (IV) = manipulated	What the scientist **changes or chooses** on purpose (often time)	**X-axis** (horizontal)
Dependent variable (DV) = responding	What is **measured**; it *responds to* (depends on) the IV	**Y-axis** (vertical)
Constants (controlled variables)	Everything kept **the same** so the test is fair	not graphed
Control group	The normal, untreated group used for comparison	compared on the graph

Investigation	IV (x-axis)	DV (y-axis)	Some constants
Bean "rabbit" simulation	Time (years/rounds)	Rabbit population	grid size, drop height, the 1 / 2–4 / 5+ rules
Habitat fragmentation simulation	% meadow (forest burned/cleared)	# of suitable cells, or % habitat available	grid size (100 cells), the species' habitat rules
Redwood sorrel & light	Hours of light per day	Plant height (cm)	water, soil, pot size, temperature
California wildfire graphs	Year	Number of large fires (≥10,000 acres), or acres/hectares burned	—

Graph checklist: a **title** ("DV vs. IV" or "The Effect of IV on DV"), both axes **labeled with units**, an **even scale**, and data points plotted accurately.

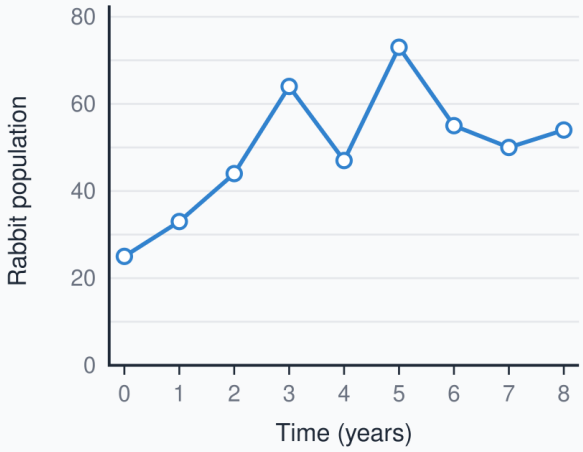
Describing a relationship: positive = as x increases, y increases (bean years 0–3: 25 → 64 rabbits). **Negative** = as x increases, y decreases (bean years 5–6: 73 → 55; more % meadow → fewer suitable murrelet cells).

Choosing the right graph

LINE graph

Use for change over time / continuous data

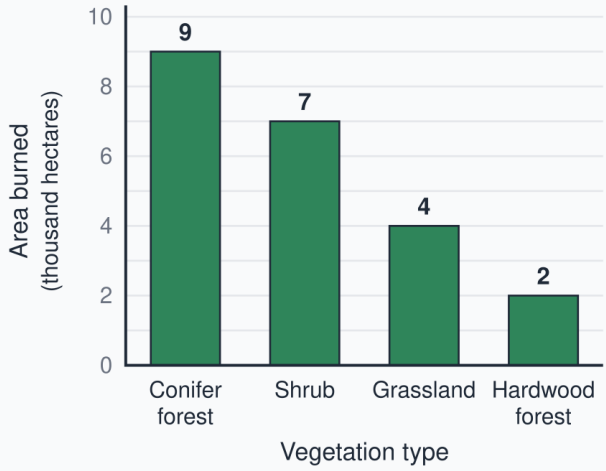
Example: bean "rabbit" population by year



BAR graph

Use for comparing categories

Example: area burned by vegetation type (illustrative)



✕ **Bean data over time should NOT be a bar graph.**

Line graphs show change over time; bar graphs compare separate categories.

Graph type	Use it for	Example from this unit
Line graph	**Change over time** or continuous data; shows trends	rabbit population per year; number of large wildfires per year
Bar graph	Comparing **separate categories**	area burned by vegetation type; individuals per species
Pie chart	Parts of a whole (adds to 100%)	relative abundance of species in one community

WATCH OUT

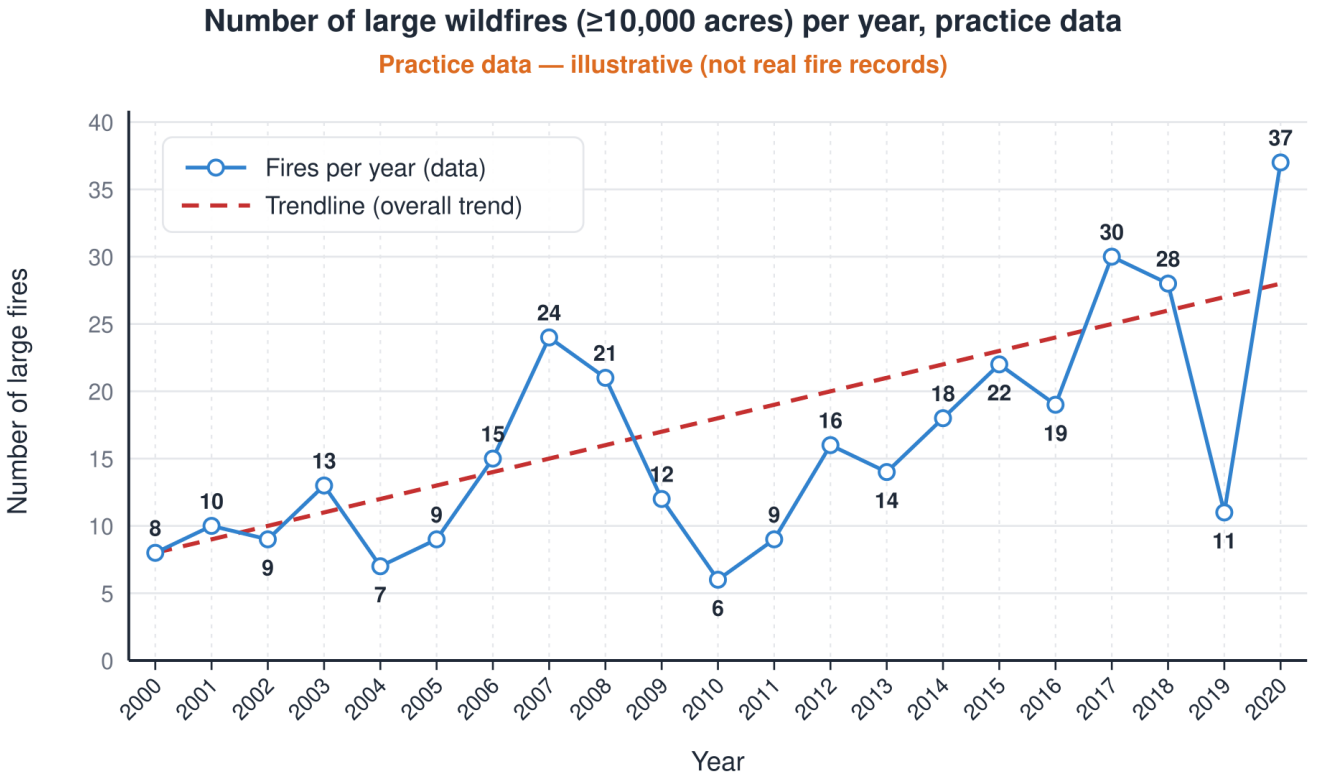
From your teacher's feedback: your bean-simulation graph was marked **"Wrong type of graph"** because it was a **bar graph**. Population **over time** should be a **line graph**.

Reading and interpreting graphs

Use the class graph-analysis template every time:

Step	What to write
Title & type	What is the graph about? Line, bar, or pie?
X-axis	label, units, range (e.g., Year · years · 2000–2020)
Y-axis	label, units, range (e.g., Number of large fires · fires · 0–40)
Trend	The general direction and shape (increasing, decreasing, cyclical)
Points of interest	Specific points — use **numbers and units from both axes**

Step	What to write
High / Low	The highest and lowest points
Pattern / Anomaly	What repeats — and what **doesn't fit** the pattern
Analysis	What you **can** conclude — and what you **cannot**



Practice data (illustrative). Try the template before reading the worked answers below.

MODEL ANSWERS

Worked reading of the practice graph:

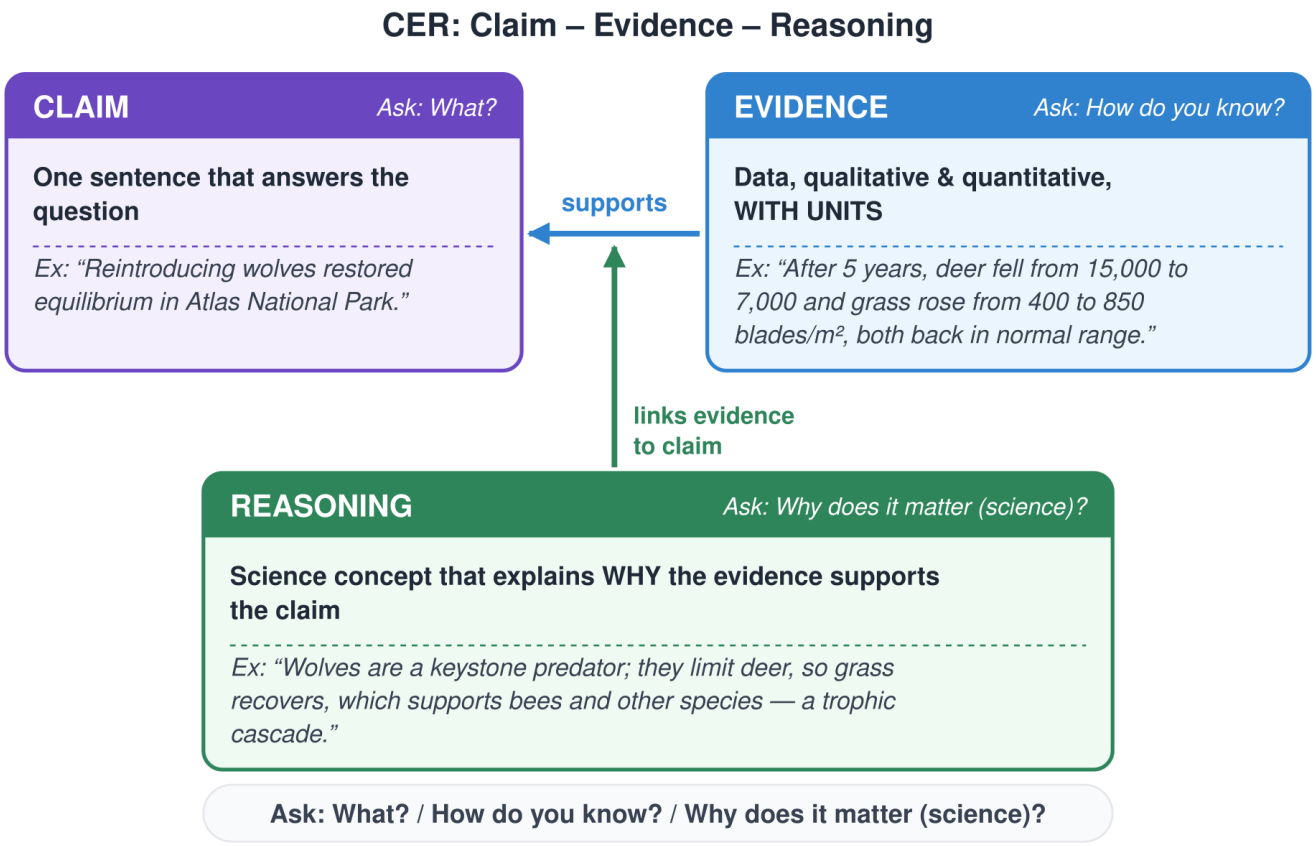
- **Trend:** the number of large fires generally **increases** from 2000 to 2020 (the dashed trendline slopes up), with ups and downs.
- **High:** 2020, with **37** large fires. **Low:** 2010, with **6**.
- **Anomaly:** 2019 drops to **11** — between two high years (28 in 2018 and 37 in 2020).
- **Can conclude:** large fires became more common over this period.
- **Cannot conclude:** *why* — the graph alone doesn't prove what **caused** the increase (correlation is not causation), and it can't tell you the exact number for a future year.

Practice: Topic 8 drills (questions T8-xx in the Practice Questions section).

Topic 9 · Scientific Argumentation (CER)

KNOW IT COLD
Know it cold: Claim = what? (one sentence that answers the question). **Evidence** = how do you know? (specific data, **with units**). **Reasoning** = why does the evidence support the claim? (a **science concept**).

Claim, evidence, reasoning



Evidence supports the claim; reasoning explains WHY, using a science idea.

Part	What it is	✓ Strong	✗ Weak
Claim	**One sentence** that directly answers the scientific question	"Wolves are the keystone species in Atlas National Park."	"Animals are important." (vague) · "I think it might be bees or wolves." (doesn't commit) · a sentence full of data (that's evidence)
Evidence	Specific data from the investigation — **quantitative (numbers with units)** and **qualitative** (observations); relevant and enough	"After wolves were reintroduced, deer fell from 15,000 to 7,000 and grass rose from 400 to 850 blades/m² in 5 years."	"The numbers changed a lot." · "Grass went from 400 to 850." (no units) · opinions or facts unrelated to the question
Reasoning	Explains **why** the evidence supports the claim using a **scientific concept**	"Wolves are a keystone predator: they limit deer, so grass recovers, which supports bees and other species — a trophic cascade."	Just repeats the evidence · "Because wolves are cool." · no science idea

Science concepts that make strong reasoning in this unit: keystone species & trophic cascade · the 10% rule & metabolism (heat) · carrying capacity & limiting factors · density-dependent vs. density-independent · habitat fragmentation (edge vs. interior) · richness & evenness → resilience · primary vs. secondary succession.

MODEL ANSWERS

Worked CER 1 — Bean simulation. Question: What is the carrying capacity of rabbits in the habitat?

- **Claim:** The carrying capacity of the habitat is about 50 rabbits.
- **Evidence:** The population overshot to 64 rabbits in year 3 and 73 rabbits in year 5, then fell each time, and it stayed between 50 and 55 rabbits from year 6 to year 8.
- **Reasoning:** When the population is above carrying capacity, limiting factors (crowding — 5 or more rabbits in a square die, a density-dependent factor) push it back down. A population that rises and falls around one level is fluctuating around its carrying capacity.

Worked CER 2 — Sampling Diversity. Question: Which community is more even, A or B?

- **Claim:** Community B is more even than Community A.
- **Evidence:** In Community B, every species makes up between 0.08 and 0.35 of the 100 individuals. In Community A, one species makes up 0.625 of the 200 individuals while another is only 0.015.
- **Reasoning:** Evenness measures how similar the population sizes of the species are. B's species are all close to an equal share (0.20 each for 5 species), while A is dominated by a single species.

Worked CER 3 — Habitat fragmentation. Question: How does losing forest affect habitat for the marbled murrelet?

- **Claim:** As more forest is burned or cleared, the habitat available to marbled murrelets drops — faster than the forest itself.
- **Evidence:** When forest fell from 90 to 20 cells (−78%), suitable cells fell from 90 to 4 (−96%), and the habitat available fell from 100% to 20%.
- **Reasoning:** Habitat fragmentation leaves small, isolated patches that are mostly edge. Species that need larger patches of intact forest can't use them, so usable habitat shrinks faster than total forest area.

TEST TIP

Multiple-choice strategy for CER questions:

- **"Best claim?"** → the one-sentence choice that answers *exactly* the question asked, with no data.
- **"Best evidence?"** → specific, relevant **numbers with units** from the data given.
- **"Best reasoning?"** → the choice that names a **science concept** and connects it to the evidence — not one that just restates the numbers.

Practice: Topic 9 drills (questions T9-xx in the Practice Questions section).

Common Traps: Frequently Confused Pairs

These are the mix-ups that multiple-choice tests are built to catch. Cover the right two columns and see whether you can explain each pair.

#	Confused pair	How to tell them apart	Quick example	Review
1	**Mutualism vs. commensalism**	Does the *second* species benefit (+/+) or is it unaffected (+/0)?	bee & flower (+/+) vs. barnacle on a whale (+/0)	Symbiosis: mutualism, commensalism, parasitism
2	**Parasitism vs. predation**	Both +/−. A parasite lives on or in its host and feeds over time; a predator kills and eats its prey.	tick on a deer vs. puma eating a deer	Symbiosis: mutualism, commensalism, parasitism
3	**Intraspecific vs. interspecific competition**	*Intra* = within one species; *inter* = between different species.	gulls fighting over nesting rocks vs. lion & tiger over a buffalo	Competition: intraspecific vs. interspecific
4	**Keystone vs. most abundant species**	A keystone species has an outsized effect; it can be rare.	Atlas NP: 97 wolves vs. 7,000 deer — the wolves are keystone	Keystone species
5	**Which way food-web arrows point**	From the food → to the eater (the direction of energy flow).	redwood sorrel → mule deer	Food chains vs. food webs: reading the arrows
6	**Autotroph vs. heterotroph**	*Auto* = self-feeder (makes its own food, producer); *hetero* = other-feeder (consumer or decomposer).	redwood sorrel vs. brush rabbit	Trophic levels: who eats whom
7	**Decomposer vs. primary consumer**	Decomposers break down dead matter from *every* level; primary consumers eat living producers.	fungi on a log vs. deer eating sorrel	Decomposers, scavengers & detritivores
8	**Where the "lost" 90% goes**	It is used for metabolism and released as **heat** — it isn't destroyed and isn't recycled.	10,000 kcal → 1,000 kcal	Energy flow, metabolism & the 10% rule
9	**Energy vs. matter**	Energy **flows** one way and leaves as heat; matter (nutrients) **cycles**.	decomposers return nutrients to soil	Decomposers, scavengers & detritivores
10	**Omnivore's trophic level**	An omnivore can be at more than one level, depending on what it eats.	black bear: level 2 eating sorrel, level 3 eating deer	Trophic levels: who eats whom
11	**Density-dependent vs. density-independent**	Ask: does crowding make the effect stronger? Yes → dependent (disease, competition). No → independent (flood, cold front, fire).	disease in a crowded herd vs. a hurricane	Density-dependent vs. density-independent
12	**"Density-independent" = "no effect"?**	NO — it can kill many; the effect just doesn't depend on crowding.	a flood killing 25% at any density	Density-dependent vs. density-independent
13	**Exponential vs. logistic growth**	Exponential (J) keeps getting steeper; logistic (S) levels off at K.	bean years 0–3 vs. the whole class model	Population growth & carrying capacity
14	**What raises vs. lowers K**	More resources or fewer threats → K goes up. Habitat loss, drought, or overgrazing → K goes down.	wildlife corridor (↑) vs. new subdivision (↓)	Population growth & carrying capacity

#	Confused pair	How to tell them apart	Quick example	Review
15	**Overshoot vs. carrying capacity**	Overshoot = the peak *above* the K line; K = the level the population settles around.	bean sim: 73 (overshoot) vs. ~50 (K)	Overshoot, decline & recovery
16	**Predator vs. prey peaks**	Predator peaks come *after* prey peaks, and predators are fewer.	lynx peaks lag hare peaks	Cyclical / periodic patterns
17	**Primary vs. secondary succession**	No soil → primary. Soil remains → secondary. A burned forest with soil is **secondary**.	Yosemite granite vs. Big Basin after the fire	Secondary & tertiary succession
18	**Secondary vs. tertiary (Ms. Hero)**	Tertiary = soil lost **and** humans step in to restore.	replanting an eroded, burned hillside	Secondary & tertiary succession
19	**Pioneer vs. climax**	Pioneer = first to arrive (lichens; grasses); climax = final, stable community.	lichen on rock vs. old-growth redwoods	Who lives there: pioneers, climax community & more
20	**Crown vs. surface fire**	Crown = burns the treetops, kills most trees. Surface = burns the forest floor, big trees survive.	fire jumping tree to tree vs. burning needles and logs	Fire: surface vs. crown, severity & disturbances
21	**Fire severity vs. intensity**	Severity = the ecosystem damage (effects); intensity = the heat released (behavior).	77% high-severity burn in Big Basin	Fire: surface vs. crown, severity & disturbances
22	**Fragmentation: amount vs. pattern**	The same total forest can be very different habitat depending on how it's broken up.	checkerboard vs. two big blocks (both 50% forest)	Habitat fragmentation
23	**Richness vs. evenness**	Richness = number of species; evenness = how equal their counts are.	Field X (even, 2 species) vs. Field Y (rich, uneven)	Richness and evenness
24	**Relative abundance formula**	(individuals of ONE species) ÷ (total individuals) — **not** number of species ÷ total.	125 ÷ 200 = 0.625	Relative abundance (with your class data)
25	**Community vs. ecosystem**	Community = living things only; ecosystem = community + abiotic factors.	all Big Basin species vs. species + creek + sunlight	Ecosystem
26	**Biotic vs. abiotic (dead things)**	Once-living things (logs, dead leaves) are **biotic**; rocks, water, sunlight, and weather are abiotic.	fallen redwood log = biotic	Biotic vs. abiotic factors
27	**IV vs. DV (DRY MIX)**	IV = what you change (x-axis); DV = what you measure (y-axis). Time → x-axis.	years (x) vs. rabbit population (y)	DRY MIX: axes, IV & DV
28	**Line vs. bar graph**	Line = change over time; bar = separate categories.	bean data → line graph	Choosing the right graph
29	**Correlation vs. cause**	A trend on a graph doesn't prove what caused it.	more fires over time ≠ proof of one cause	Reading and interpreting graphs
30	**Evidence vs. reasoning**	Evidence = the data (with units); reasoning = the science concept explaining why the data supports the claim.	"deer fell to 7,000" vs. "wolves are a keystone predator…"	Claim, evidence, reasoning

Index Card Model (3" × 5", handwritten, front & back)

Your card must be **handwritten**, with your **first & last name and period** visible. Copy what *you* need. Writing small, using abbreviations and arrows, and choosing what to include is itself great studying. Below is a model of what fits.

FRONT · Name _____ Period ___

SPHERES Atmo = air/weather/climate · Hydro = ALL water (ice, glaciers, snow, soil moisture, aquifers) · Bio = life (coral reefs, bacteria) · Geo = rock, soil minerals, volcanoes, fossils, magnetism · Sun drives most; core drives geo · Closed system: energy in/out, matter recycled

RELATIONSHIPS Mutualism +/+ (bee–flower, lichen) · Commensalism +/0 (barnacle–whale) · Parasitism +/− (tick–deer, lives ON host) · Predation +/− (kills) · Competition −/−: INTRA = same sp. (gulls/nest rock), INTER = diff. sp. (lion–tiger) · Keystone = huge effect, removal → collapse (Atlas NP wolves ≠ bees; otters)

ENERGY Producer = autotroph (1) → 1° (2, herbivore) → 2° (3) → 3° (4) · heterotroph = eats others · omnivore = 2+ levels (bear) · Arrows: food → eater · 10% up (÷10); 90% → metabolism → HEAT · 3.5M → 350k → 35k → 3.5k kcal · Decomposers (fungi, bacteria, enzymes) → nutrients → soil → producers · scavenger = vulture

POPULATIONS Limiting factor = stops growth · DD (worse when crowded): competition, predation, disease, waste, space, "5+ beans die" · DI (same at any size): weather, cold front, flood, fire, hurricane, drought · J (exponential) → overshoot → decline → recovery → ~K (S) · waves around K = cyclical/periodic; predator peaks lag prey · ↑resources → ↑K; habitat loss → ↓K · bear: 100 fish ÷ 10 = 10 bears · bean K ≈ 50

BACK

SUCCESSION Primary = NO soil (lava, Yosemite granite): lichen/moss 0–150 → grasses 150–500 → shrubs 500–1,000 → climax 1,000+ · Secondary = SOIL stays (fire, farm): grasses 1–20 → shrubs/small trees 20–150 → climax 150+, faster · Tertiary (Hero) = soil lost + humans restore · Climax = stable, final

FIRE Surface = floor, canopy OK, big trees live, low–mod severity, clears fuel · Crown = treetops, tree to tree, kills trees, high severity, ladder fuels · Severity = damage; intensity = heat · CZU: Aug 16 2020 dry lightning; wind + slope; >97% Big Basin (oldest CA park, 1902); redwood thick bark + resprouts · prescribed burns cut fuel

FRAGMENTATION big habitat → small isolated patches · more edge, less interior · same amount ≠ same habitat (checkerboard worst) · corridors/crossings help

BIODIVERSITY Richness = # species · Evenness = equal counts? · Rel. abund. = 1 sp. ÷ total (125/200 = 0.625) · A = B rich (5), B more even; C richest (10) · more diversity → more resilient

ECOSYSTEM community + abiotic · equilibrium = balance in normal ranges · biotic incl. dead logs/leaves; abiotic = sun, water, rocks, temp, weather

GRAPHS DRY = Dependent/Responding → Y · MIX = Manipulated/Independent → X · time → x · line = over time; bar = categories · correlation ≠ cause

CER Claim = 1-sentence answer · Evidence = data + UNITS · Reasoning = science concept (WHY)

Practice Questions

Answer on a separate sheet (or circle the letter). **Part A** drills are grouped by topic and run easy → hard. **Part B** is a full 25-question mock test — time yourself for 30 minutes and don't peek at hints. Stuck on a drill? The Hints section nudges you without giving the answer away. Then check the Answer Key and record misses in the Weakness Tracker.

Part A — Topic Drills (76 questions)

Topic 1 · Earth Systems Science

T1-01 · Topic 1 · core · Earth Systems Science

Which statement best describes **Earth system science**?

A. The study of Earth's rocks and minerals and how they form
B. The study of how Earth's air, water, land, and life interact
C. The study of how living things interact with one another only
D. The study of daily weather and long-term climate patterns

T1-02 · Topic 1 · core · Hydrosphere

On the class graphic organizer, **glaciers**, **soil moisture**, and **groundwater in aquifers** are all listed under the same sphere. Which sphere is it?

A. Geosphere
B. Atmosphere
C. Biosphere
D. Hydrosphere

T1-03 · Topic 1 · core · Earth Systems Science (closed system)

Earth is described as a **closed system**. What does this mean?

A. Energy flows in and out, but very little matter enters or leaves.
B. Neither energy nor matter can enter or leave Earth at all.
C. Matter from space constantly enters Earth, but energy can never leave.
D. Earth's four spheres are sealed off and cannot exchange matter with each other.

T1-04 · Topic 1 · apply · Earth Systems Science (energy drivers)

The class organizer lists two energy drivers: the **Sun** and **Earth's core**. Which event is powered mainly by heat from Earth's core rather than by the Sun?

A. Wind blowing wildfire smoke across Monterey Bay
B. Ocean water evaporating and forming clouds
C. Magma rising through the crust as plates shift
D. Redwood sorrel making sugar by photosynthesis

T1-05 · Topic 1 · apply · Atmosphere, Hydrosphere, Biosphere, Geosphere

Based on the class graphic organizer, which item is matched with the **correct** sphere?

A. Coral reef — geosphere
B. Jet stream — atmosphere
C. Glacier — geosphere
D. Fossilization — biosphere

T1-06 · Topic 1 · apply · Sphere interactions (wildfire)

Which event linked to the 2020 CZU Lightning Complex fire is an interaction between the **geosphere** and the **hydrosphere**?

A. Dry lightning from a summer storm ignited the dry forest.
B. Smoke and carbon dioxide from burning trees filled the air.
C. Winter rain washed loose, burned soil from slopes into creeks.
D. Redwoods resprouted from buds hidden under their thick, fire-resistant bark.

T1-07 · Topic 1 · challenge · Sphere interactions (volcano)

A volcano erupts, sending ash and sulfur dioxide gas into the air. The gas forms acid rain, which falls into a nearby lake. The lake water becomes more acidic, and many fish die. Which sequence of spheres matches this chain of events?

A. Geosphere → atmosphere → hydrosphere → biosphere
B. Atmosphere → geosphere → biosphere → hydrosphere
C. Geosphere → hydrosphere → atmosphere → biosphere
D. Biosphere → atmosphere → hydrosphere → geosphere

Topic 2 · Ecological Relationships

T2-01 · Topic 2 · core · Symbiosis

Symbiosis is a close, long-term relationship between two different species. Which list gives the correct effects (+ helped, − harmed, 0 not affected) for **mutualism**, **commensalism**, and **parasitism**, in that order?

A. +/+, +/−, +/0
B. +/0, +/+, +/−
C. +/+, −/−, +/−
D. +/+, +/0, +/−

T2-02 · Topic 2 · core · Inter- and Intra-specific Competition

In Ms. Hero's review example, a lion and a tiger compete for the same water buffalo. What kind of relationship exists between the **lion and the tiger**?

A. Interspecific competition
B. Intraspecific competition
C. Predation
D. Mutualism

T2-03 · Topic 2 · core · Keystone Species

Which statement best describes a **keystone species**?

A. It is always the most abundant species in its ecosystem.
B. It is always the largest top predator in the food web.
C. Removing it can upset the balance of the whole ecosystem.
D. It is the producer that supplies the most energy to consumers.

T2-04 · Topic 2 · apply · Mutualism

A **lichen** on a bare granite boulder in Yosemite is really two organisms living together. The fungus holds water and gives shelter. The alga uses sunlight to make sugar that both partners use. Which relationship is this?

A. Commensalism
B. Mutualism
C. Parasitism
D. Interspecific competition

T2-05 · Topic 2 · apply · Commensalism

Barnacles attach to the skin of gray whales migrating past the Santa Cruz coast. The barnacles get a free ride to waters full of plankton, which they filter from the sea. The whale is not noticeably helped or harmed. Which relationship is this?

A. Mutualism
B. Parasitism
C. Commensalism
D. Predation

T2-06 · Topic 2 · apply · Parasitism

A tick attaches to a mule deer in Big Basin and drinks its blood for several days. The deer loses some energy and may catch a disease, but it usually survives. Which relationship is this?

A. Parasitism
B. Predation
C. Commensalism
D. Mutualism

T2-07 · Topic 2 · apply · Inter- and Intra-specific Competition

Scenario 1: In Big Basin, coyotes and bobcats both hunt the same California brush rabbits.
Scenario 2: Western gulls crowd onto one small rocky shore and fight over nesting space.
Which choice correctly classifies the competition in each scenario?

A. 1 = intraspecific; 2 = interspecific
B. 1 = interspecific; 2 = interspecific
C. 1 = intraspecific; 2 = intraspecific
D. 1 = interspecific; 2 = intraspecific

T2-08 · Topic 2 · apply · Predator/Prey

Pumas (mountain lions) are the main predator of mule deer in Big Basin. Suppose a disease killed most of the pumas. What is the most likely **short-term** effect?

A. Deer would decline, because deer depend on pumas to survive.
B. Deer would increase and eat more of the plants they feed on.
C. Plants would increase, because fewer pumas would be eating them.
D. Nothing would change, because top predators don't affect other species.

T2-09 · Topic 2 · challenge · Keystone Species

In the Atlas National Park Gizmo, two fixes were tested for an unbalanced ecosystem. Use the data to decide which species is the **keystone species**.

Species	Normal range	Now	5 yrs after adding bees	5 yrs after adding wolves
Grass (blades/m²)	800–1,000	400	350	850
Deer	4,000–8,000	15,000	17,000	7,000
Coyotes	250–300	600	650	270
Ravens	220–270	160	145	245
Bees (hives)	4,000–5,000	2,000	3,000	4,250
Wolves	95–120	0	0	97

A. Wolves, because adding them returned every population to its normal range
B. Bees, because their hives were far below normal and adding bees raised hives to 3,000
C. Deer, because they had the largest population and affected every other species
D. Grass, because it is the producer that all of the other species depend on

T2-10 · Topic 2 · challenge · Keystone Species (trophic cascade)

In Atlas National Park, wolves do not eat grass or bees. Yet after the wolves disappeared, grass fell to 400 blades/m² (normal 800–1,000) and bee hives fell to 2,000 (normal 4,000–5,000). Which chain of events best explains this?

A. No wolves → fewer deer → less grazing → fewer hives
B. No wolves → more coyotes → coyotes ate the grass → fewer hives
C. No wolves → more deer → overgrazed plants → fewer hives
D. Bees declined first → wolves lost a food source → wolves vanished

Topic 3 · Food Chains, Webs & Energy

T3-01 · Topic 3 · core · Autotrophs vs. heterotrophs

The Gizmo handbook explains that *auto* = self, *hetero* = other, and *troph* = food. Which statement uses these terms correctly?

A. Redwood sorrel is a heterotroph, since it takes in water and minerals from the soil.
B. A mule deer is an autotroph, since it eats plants that make their own food.
C. Fungi on a fallen log are autotrophs, since they grow out of the ground like plants.
D. Redwood sorrel is an autotroph, and a mule deer is a heterotroph.

T3-02 · Topic 3 · core · Food chain arrows (direction of energy flow)

A student wants to model this: a California brush rabbit eats redwood sorrel, and a gray fox eats the rabbit. Which model and reason are correct?

A. Gray fox → brush rabbit → redwood sorrel, because arrows point to what each animal eats
B. Redwood sorrel → brush rabbit → gray fox, because arrows show energy moving from food to eater
C. Redwood sorrel → gray fox → brush rabbit, because the producer always goes first in a chain
D. Brush rabbit → redwood sorrel → gray fox, because the rabbit is at trophic level 1

T3-03 · Topic 3 · core · Metabolism and energy lost as heat

In the Gizmo example, producers hold 10,000 kcal of energy, but the primary consumers that eat them receive only about 1,000 kcal. What happened to most of the other 9,000 kcal?

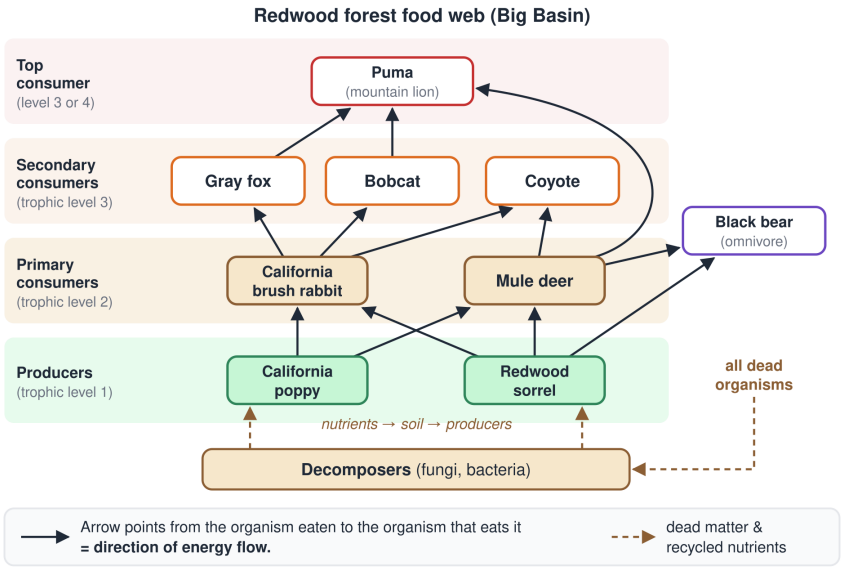
A. It was used for the producers' life processes (metabolism), and much of it left as heat.
B. It was destroyed during digestion, so that energy no longer exists anywhere at all.
C. It was stored up so the primary consumers can pass all of it on to the next level later.
D. It was sent back down to the producers so they could reuse it for more photosynthesis.

T3-04 · Topic 3 · core · Decomposers, scavengers, and detritivores

After a raccoon dies in Big Basin, vultures eat its body. Nearby, earthworms and millipedes eat dead leaves, and fungi release enzymes that break down a fallen log. Which choice names each role correctly?

A. Vultures: predators; earthworms: producers; fungi: decomposers
B. Vultures: decomposers; earthworms: primary consumers; fungi: scavengers
C. Vultures: scavengers; earthworms: detritivores; fungi: decomposers
D. Vultures: scavengers; earthworms: detritivores; fungi: producers

T3-05 · Topic 3 · apply · Trophic levels in the redwood food web (puma)



Use the redwood food web. Which statement about the puma (mountain lion) is correct?

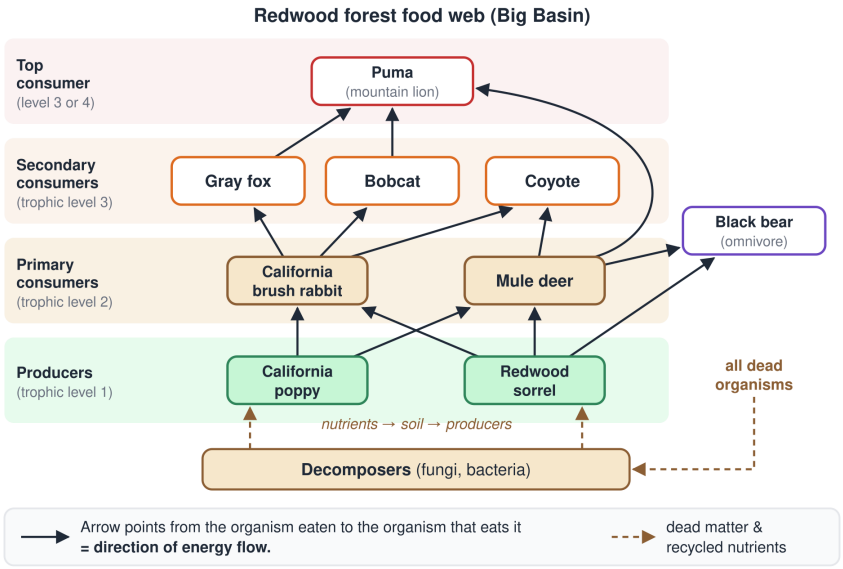
A. It is always a tertiary consumer, because it sits in the top row of the web with no predators.

B. It is a primary consumer, because it eats mule deer, and mule deer are primary consumers.

C. It is a secondary consumer when eating mule deer and a tertiary consumer when eating a gray fox.

D. It is only a secondary consumer, because every arrow that points into it comes from a primary consumer.

T3-06 · Topic 3 · apply · Omnivores at multiple trophic levels (black bear)



Use the redwood food web. Which trophic level(s) does the black bear occupy?

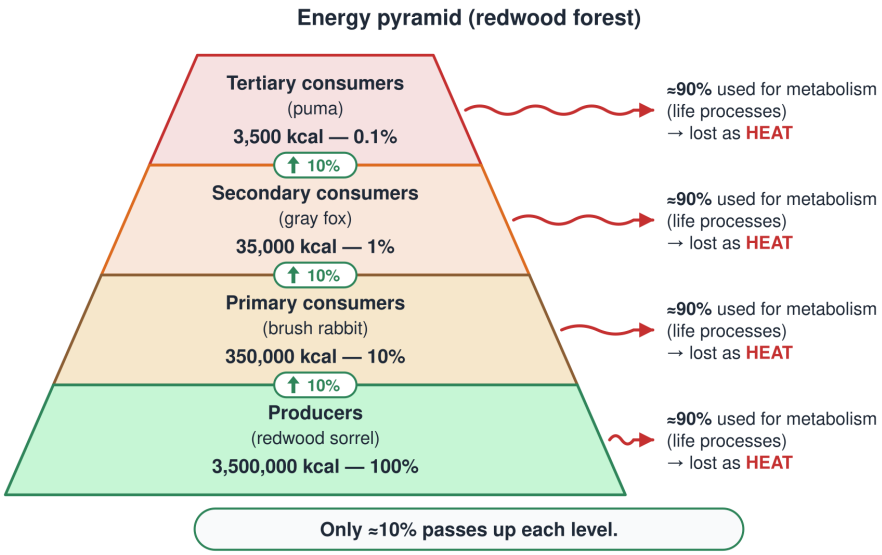
A. Levels 2 and 3 — primary consumer eating sorrel, secondary consumer eating deer
B. Levels 1 and 2 — level 1 when it eats sorrel, and level 2 when it eats mule deer
C. Level 4 only — it is the top predator, since nothing in this food web eats it
D. Level 2 only — an omnivore is always classified by the plant food that it eats

T3-07 · Topic 3 · apply · Importance of decomposers (nutrient cycling)

Imagine all the fungi and bacteria in Big Basin's soil suddenly disappeared. What would most likely happen over the next several years?

A. Nothing would change, because decomposers are usually left out of food chain drawings.
B. Only primary consumers would be affected, because decomposers feed only on producers.
C. Producers would grow faster, because there would be more dead material for them to eat.
D. Dead matter would pile up, soil nutrients would run low, and producers would decline.

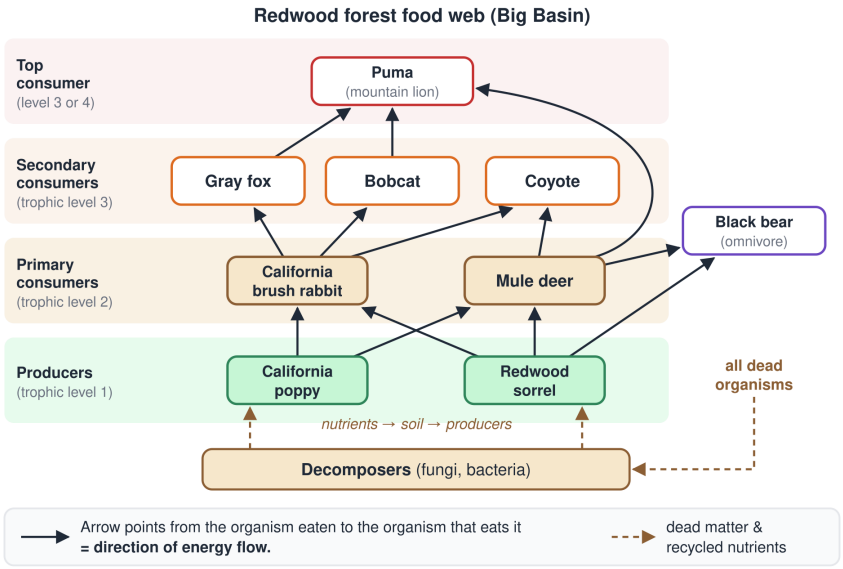
T3-08 · Topic 3 · apply · Why there are fewer top predators



Use the energy pyramid. Why can Big Basin support far fewer pumas than brush rabbits?

A. Pumas are hunted by many other predators in the forest, which keeps their numbers low.
B. Only about 0.1% of the producers' energy reaches pumas, so just a few can be supported.
C. Energy increases at each higher level, so just a few pumas already hold most of the energy.
D. Only the top level loses energy as heat; rabbits and the other lower levels lose almost none.

T3-09 · Topic 3 · challenge · Effects of removing a species from a food web



Use the redwood food web. Suppose a disease wipes out all of the California brush rabbits. Which consumers would be hurt most directly, and why?

A. Gray fox and bobcat, because the rabbit is their only food shown in this web
B. Coyote only, because it is the only animal in the web that eats brush rabbits
C. Mule deer, because they eat the same sorrel and poppies that the rabbits eat
D. Puma, because the brush rabbit is the puma's main food source in this web

T3-10 · Topic 3 · challenge · 10% rule calculation

In the chain redwood sorrel → brush rabbit → gray fox → puma, the sorrel captures 3,500,000 kcal of energy. Using the 10% rule, how much energy reaches the puma, and what percent of the sorrel's energy is that?

A. 35,000 kcal, which is 1% of the sorrel's energy
B. 350 kcal, which is 0.01% of the sorrel's energy
C. 3,500 kcal, which is 0.1% of the sorrel's energy
D. 3,500 kcal, which is 10% of the sorrel's energy

Topic 4 · Limiting Factors & Carrying Capacity

T4-01 · Topic 4 · core · Limiting factors

Gulls nest on rocky shores. Once every nesting spot on a rock is taken, extra gull pairs cannot raise chicks, so the colony stops growing. In this example, nesting space is best described as...

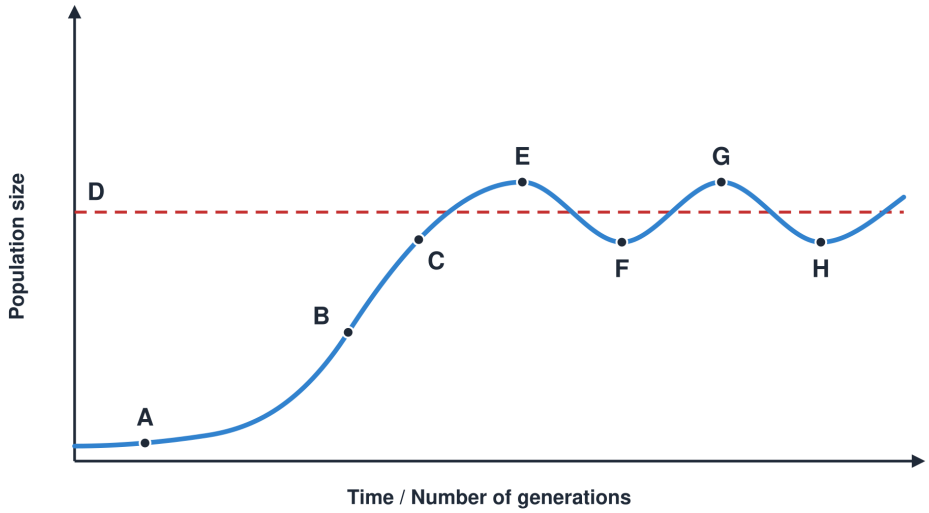
A. the carrying capacity, because it is the largest number of gulls the rock can hold
B. a limiting factor, because a shortage of it keeps the gull population from growing
C. a density-independent factor, because rocks and empty space are not alive
D. exponential growth, because more and more gull pairs keep arriving to nest

T4-02 · Topic 4 · core · Density-dependent vs. density-independent

Which situation is an example of a **density-dependent** limiting factor?

A. A hurricane washes away turtle nests along a beach.
B. A late-spring cold front kills many newly hatched birds.
C. A lightning-caused wildfire burns through a redwood forest.
D. A disease spreads faster as a deer herd becomes more crowded.

T4-03 · Topic 4 · core · Carrying capacity model (letters A–H)



Use the carrying capacity model graph. Which choice correctly describes point B, line D, and the E–H section?

A. B: exponential growth; D: carrying capacity; E–H: population rises and falls around K
B. B: carrying capacity; D: exponential growth; E–H: the population goes extinct
C. B: exponential growth; D: highest population ever reached; E–H: a steady decline
D. B: slow start (lag); D: carrying capacity; E–H: a second round of exponential growth

T4-04 · Topic 4 · apply · Carrying capacity calculation (bear & fish)

A bear must eat 10 fish per day. A river provides about 100 fish per day without harming the fish population. Shelter and water are plentiful. What is the carrying capacity for bears, and would adding more shelter raise it?

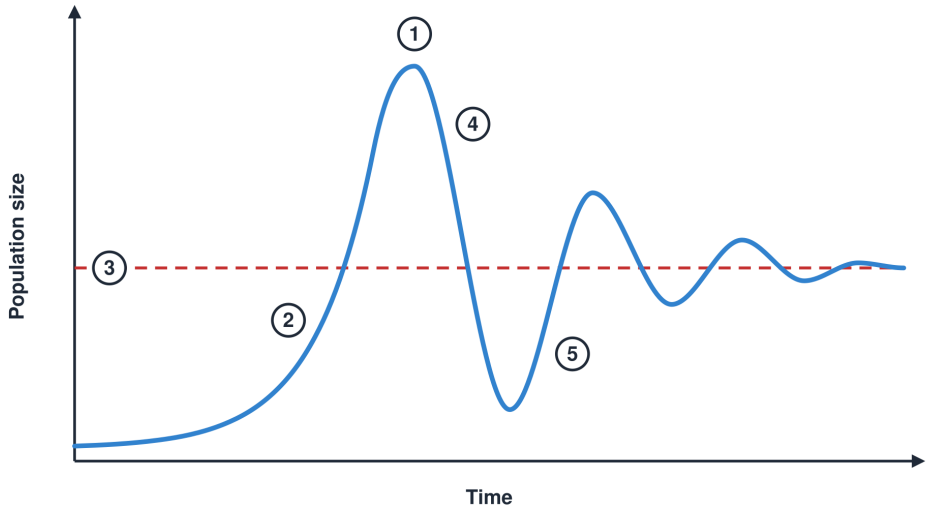
A. 1,000 bears; yes, more shelter would let even more bears live there
B. 100 bears; no, because the river makes 100 fish each day
C. 10 bears; yes, more shelter and water would raise it to about 15 bears
D. 10 bears; no, because food (fish) is the limiting factor

T4-05 · Topic 4 · apply · Density-independent does not mean 'no effect'

The worksheet calls a late-spring cold front a density-independent factor. A student concludes: 'So a cold front can't change the size of a bird population.' Which response is best?

A. The student is right; density-independent factors never change how many birds there are.
B. The student is wrong; it can kill many chicks — the term only means its effect doesn't depend on crowding.
C. The student is wrong; the cold front is actually density-dependent, because colder weather kills more birds.
D. The student is right; only living factors such as predators and disease can lower a population's size.

T4-06 · Topic 4 · apply · Exponential growth, overshoot, decline, recovery



Use the numbered graph. In the 1800s, rabbits in Australia (a) multiplied rapidly with lots of food and no predators, (b) became far more numerous than the land could support, (c) starved after eating the land bare, then (d) increased again as plants regrew. Which markers match (a)–(d), in order?

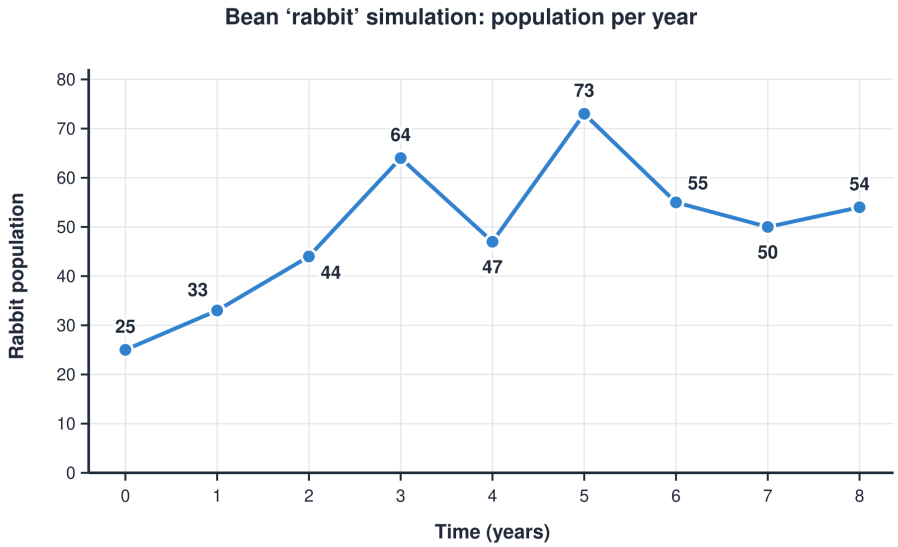
A. 2, 3, 4, 5
B. 1, 2, 4, 5
C. 2, 1, 4, 5
D. 2, 1, 5, 4

T4-07 · Topic 4 · apply · What increases vs. decreases carrying capacity

A student wrote these lists on the bean simulation worksheet:
Increases K: drought, habitat fragmentation, more competitors
Decreases K: more food, more water, habitat restoration
Which feedback is most accurate?

A. Both lists are correct, because harsh conditions make survivors stronger, so more can live there.
B. The lists don't matter, because a habitat's carrying capacity is fixed and never changes.
C. The lists are swapped: more resources raise K; drought, fragmentation, and competitors lower it.
D. Only 'more water' is misplaced, because water does not affect a habitat's carrying capacity.

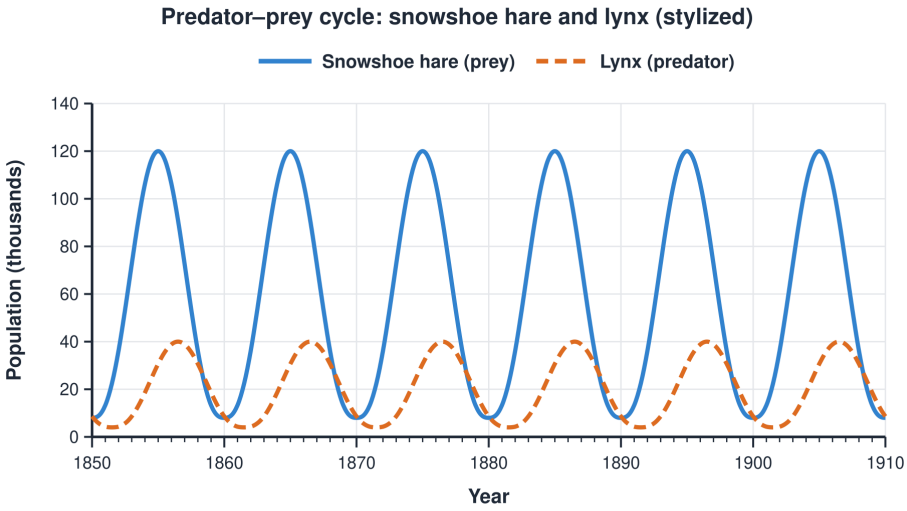
T4-08 · Topic 4 · apply · Bean simulation — estimating carrying capacity



The line graph shows the class bean 'rabbit' simulation for years 0–8. What is the best estimate of the habitat's carrying capacity?

A. About 50 rabbits, the level the population settles around in years 6–8
B. About 73 rabbits, the highest population the habitat ever reached
C. About 25 rabbits, the starting population the habitat began with in year 0
D. About 64 rabbits, the point where fast growth ended in year 3

T4-09 · Topic 4 · apply · Cyclical/periodic predator-prey pattern (lynx & hare)



Stylized after Hudson's Bay Company fur records: ~10-year cycle; predator peaks lag prey peaks.
(Illustrative pattern, not real data.)

Use the lynx–hare graph. Which statement correctly describes AND explains the pattern?

A. Lynx peak first, and then their hunting causes the hare population to grow larger.
B. Hares peak first; with more food, lynx rise a year or two later, then fall after hares drop.
C. Both peak in the same year each cycle, because lynx and hares compete for the same food.
D. Hares peak first, and then both populations level off for good at their carrying capacity.

T4-10 · Topic 4 · challenge · Density-dependent vs. density-independent (data)

Ecologists recorded the percent of a rabbit population killed by two factors at different population densities.

Rabbits per hectare	% killed by Factor P	% killed by Factor Q
10	4%	25%
40	12%	25%
80	30%	25%

Which conclusion is best supported?

A. P is density-independent (like a flood); Q is density-dependent (like disease).
B. Q has no effect on the rabbit population, because its percent never changes.
C. Both are density-dependent, because both kill rabbits at every density.
D. P is density-dependent (like disease); Q is density-independent (like a flood).

T4-11 · Topic 4 · challenge · Overshoot and crash (St. Matthew Island reindeer)

In 1944, 29 reindeer were released on St. Matthew Island, Alaska, which had plenty of lichen to eat and no predators during the whole study.

Year	Reindeer
1944	29
1957	1,350
1963	about 6,000
1966	42

Which explanation best fits the data?

A. The herd overshot carrying capacity, overgrazed the lichen (lowering K), and then crashed.
B. The herd leveled off smoothly at carrying capacity, the way an S-shaped (logistic) curve does.
C. Predators arrived after the reindeer peaked and caused the crash, like in the lynx–hare cycle.
D. The herd stayed below carrying capacity the whole time, so food was never a limiting factor.

Topic 5 · Succession, Fragmentation & Fire

T5-01 · Topic 5 · core · Primary succession & pioneer species

Glaciers scraped Yosemite Valley down to bare granite and then melted away. The first living things to grow there were lichens, which slowly broke the rock into tiny pieces that later became soil. Which pair of terms correctly names the process and the lichens' role?

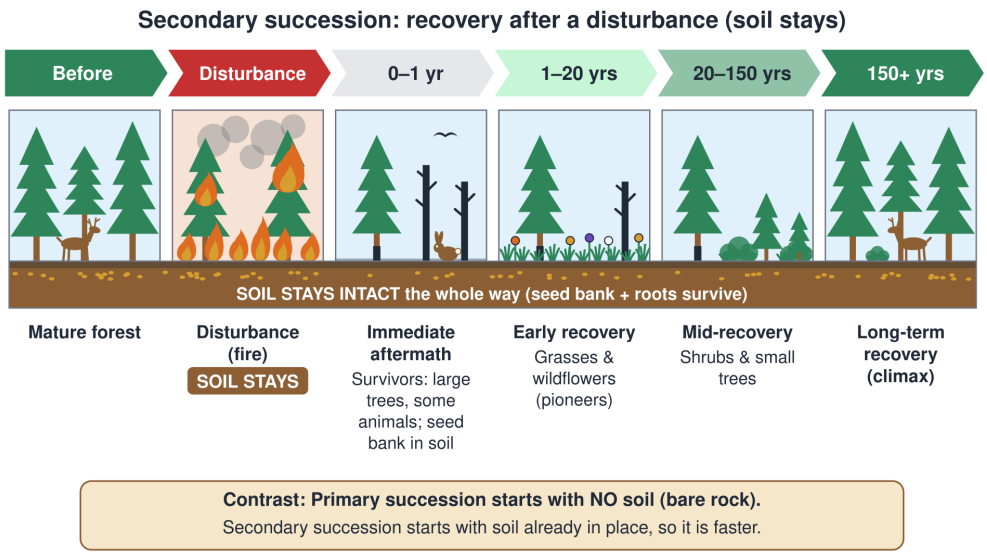
A. Secondary succession; lichens are pioneer species
B. Primary succession; lichens are a climax community
C. Primary succession; lichens are pioneer species
D. Secondary succession; lichens are a climax community

T5-02 · Topic 5 · core · Primary vs. secondary succession

Which situation would be followed by **secondary** succession rather than primary succession?

A. Lava from an eruption cools into a brand-new island of bare rock
B. A farmer stops plowing a field, and weeds sprout from the soil
C. A glacier melts back and exposes bare, scraped granite
D. A landslide strips a hillside all the way down to bare bedrock

T5-03 · Topic 5 · apply · Secondary succession after fire



The 2020 CZU fire burned more than 97% of Big Basin, but the soil and the seed bank in it stayed in place. Using the secondary-succession timeline in the figure, which organisms would be the main **new arrivals** about 10 years after the fire?

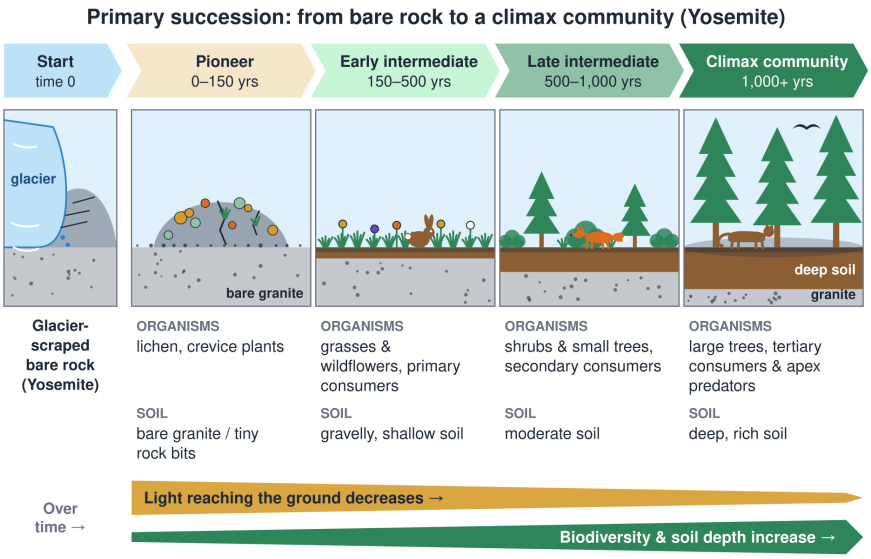
A. Lichens on bare rock, because such a huge fire restarts primary succession
B. Shrubs and small trees, the mid-recovery stage
C. Large trees and apex predators of a climax forest
D. Grasses and wildflowers, the early-recovery pioneers

T5-04 · Topic 5 · apply · Tertiary succession (Ms. Hero's term)

A very severe fire is followed by heavy rain that washes nearly all the soil off a hillside. Crews then add topsoil, plant native seedlings, and lay straw to stop erosion. Using **Ms. Hero's definition**, which type of succession is this?

A. Tertiary — the soil was lost and humans are helping restore it
B. Secondary — a fire was the disturbance that started the change
C. No succession — people, not nature, are causing the change
D. Primary — lichens must first rebuild the soil from bare rock

T5-05 · Topic 5 · apply · Organisms in communities: pioneer vs. climax



Use the primary-succession timeline in the figure. A spot in Yosemite has been developing for about 1,200 years. Compared with the pioneer stage, which description best fits this spot now?

A. Maximum sunlight reaches the ground, and tiny rock bits cover the surface
B. Gravelly, shallow soil covered mostly by grasses and wildflowers
C. Lichens and crevice plants are still the main organisms
D. Deep, rich soil, and little sunlight reaches the shaded ground

T5-06 · Topic 5 · apply · Crown & surface fires

A disturbance card says: the fire burned dry needles and fallen branches on the forest floor, the tall tree canopy was barely touched, the soil stayed healthy, and seeds stored in the soil began to sprout. Which type of fire is described?

A. Crown fire — it spread from treetop to treetop through the canopy
B. Ground fire — it smoldered in roots and duff below the soil surface
C. Surface fire — it burned fuel on the forest floor and spared the canopy
D. Crown fire — it killed most tall trees and opened up the sky

T5-07 · Topic 5 · apply · Fire severity (vs. intensity)

After the CZU fire, scientists rated each part of Big Basin as low, moderate, or high **fire severity**. Which measurement would they use to decide an area's fire severity?

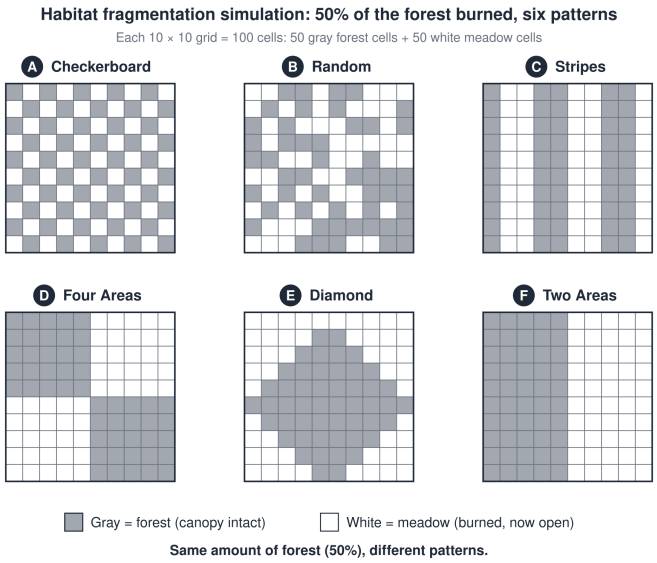
A. How much of the vegetation died and how much of the soil's organic layer burned
B. How hot the flames got and how much heat energy they released
C. How many lightning strikes started fires in that area
D. How many days the fire burned before firefighters contained it

T5-08 · Topic 5 · core · Disturbances

A disturbance is an event that suddenly changes an ecosystem and can restart succession. Which list contains **only natural** disturbances?

A. Logging, farming, building roads
B. Lightning fire, clear-cut logging, hurricane
C. Flood, housing development, drought
D. Lightning fire, flood, volcanic eruption

T5-09 · Topic 5 · apply · Habitat fragmentation: same amount, different pattern



Every grid in the figure has the same amount of forest: 50 gray cells out of 100. Which pattern is the **most fragmented**, making it the worst for a bird that needs large, connected blocks of forest?

A. A Checkerboard
B. D Four Areas
C. E Diamond
D. F Two Areas

T5-10 · Topic 5 · challenge · Habitat fragmentation (simulation data)

In the Habitat Fragmentation Simulation, more and more forest cells became meadow. A student counted the cells still suitable for the Marbled Murrelet, a forest bird.

Forest cells left (of 100)	Suitable murrelet cells
90	90
80	79
60	55
40	24
20	4

Which conclusion is **best** supported by the data?

A. Habitat fell at the same rate as forest, so only the amount of forest matters
B. Habitat rose, because more meadow made more edge for the murrelet to use
C. Habitat fell faster than forest: at 20 forest cells, only 4 (20%) were usable
D. Habitat stayed steady until the forest was almost completely gone

T5-11 · Topic 5 · core · CZU Lightning Complex & redwood fire adaptations

In August 2020, dry lightning started the CZU Lightning Complex, and more than 97% of Big Basin Redwoods State Park burned. Yet many old-growth coast redwoods survived and soon sprouted new green shoots. Which trait best explains this?

A. Resin-rich bark that coats the trunk and seals out the flames
B. Very thick bark (up to about 1 ft) plus buds that can resprout
C. Cones that stay sealed shut until the heat of a fire opens them
D. Needles kept so wet by coastal fog that they cannot catch fire

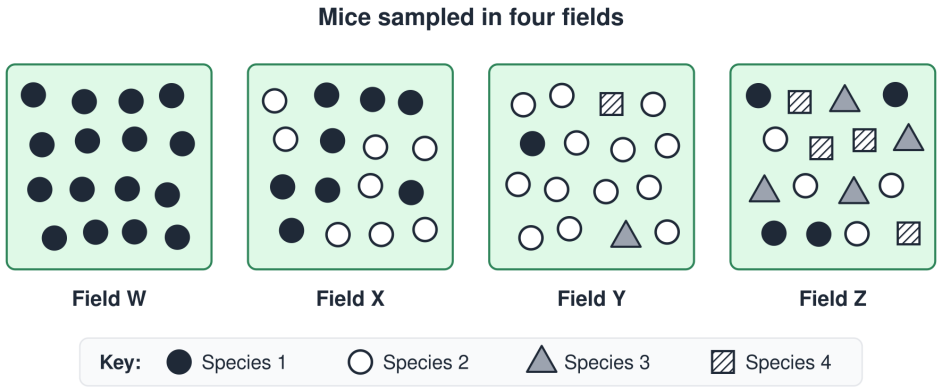
Topic 6 · Biodiversity

T6-01 · Topic 6 · core · Richness

In the Sampling Diversity lesson, rainforests were described as 'very rich.' What does high species **richness** mean?

A. The area contains a large number of different species
B. Each species in the area has about the same number of individuals
C. One species makes up most of the individuals in the area
D. The soil in the area holds a large amount of nutrients

T6-02 · Topic 6 · apply · Richness/Evenness (fields of mice)

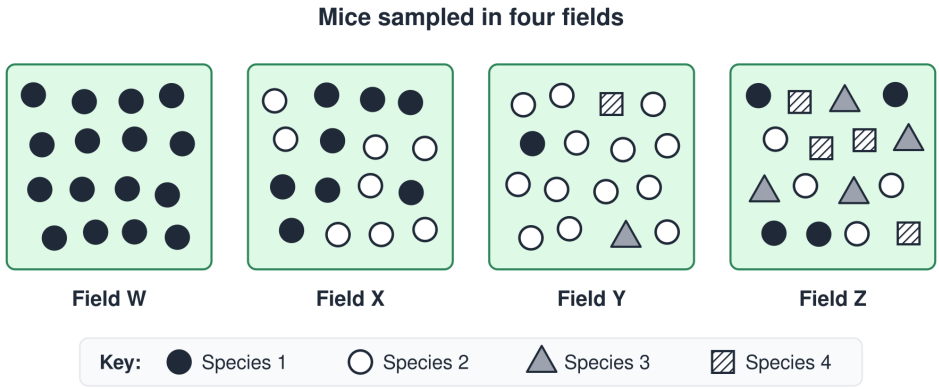


Each field has 16 mice.

Each field in the figure has 16 mice. Which field has **both** high species richness **and** high evenness?

A. Field W
B. Field X
C. Field Y
D. Field Z

T6-03 · Topic 6 · challenge · Richness/Evenness (comparing fields)



Each field has 16 mice.

Compare **Field X** and **Field Y** in the figure (16 mice each). Which statement is correct?

A. Field Y has greater richness, but Field X has greater evenness
B. Field Y has greater richness and greater evenness than Field X
C. They have equal richness, because each field has 16 mice
D. Field X has greater richness, because each species has more mice

T6-04 · Topic 6 · challenge · Relative abundance (grass lawn)

A student samples a 1 m² patch of lawn.

Species	Number of individuals
Grass	180
Clover	12
Dandelion	6
Plantain	2

What is the relative abundance of grass, and what does it show about the lawn?

A. 0.25 (25%) — each of the 4 species is one-fourth, so the lawn is even
B. 0.90 (90%) — one species dominates, so the lawn has low evenness
C. 0.90 (90%) — most plants are alike, so the lawn has high evenness
D. 180 — the lawn has so many plants that it has high richness

T6-05 · Topic 6 · challenge · Relative abundance (calculation)

Here is the sample for Community A from the Sampling Diversity lab.

Species	Number of individuals
A	10
B	125
C	3
D	23
E	39

What is the relative abundance of species D?

A. 0.200
B. 0.230
C. 0.115
D. 8.70

T6-06 · Topic 6 · apply · Richness (Communities A, B, C)

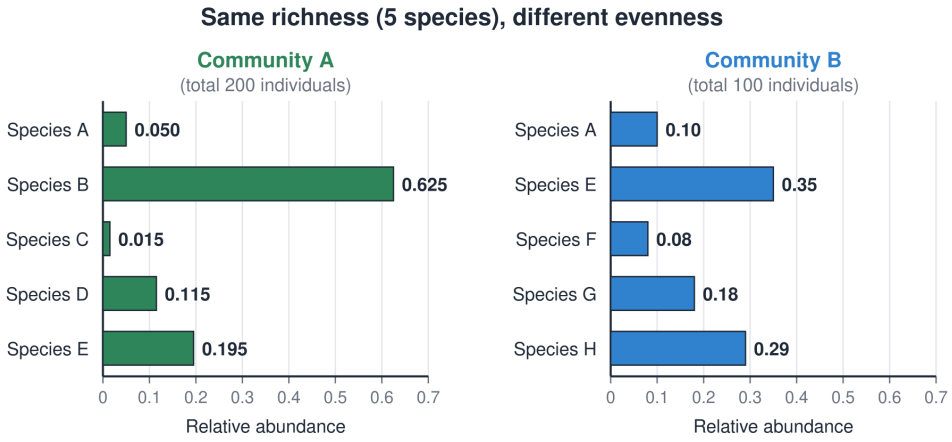
Species (letters) and numbers of individuals sampled in three communities:

Community A	Community B	Community C
A: 10	A: 10	A: 12
B: 125	E: 35	B: 1
C: 3	F: 8	C: 5
D: 23	G: 18	D: 31
E: 39	H: 29	E: 21
		F: 4
		G: 9
		H: 8
		I: 2
		J: 2
Total: 200	**Total: 100**	**Total: 95**

Which statement about species **richness** is correct?

A. A is richest, because it has the most individuals (200)
B. B is richest, because its species have the most similar numbers
C. C is least rich, because it has the fewest individuals (95)
D. C is richest (10 species); A and B are tied (5 species each)

T6-07 · Topic 6 · apply · Evenness (Community A vs. B)



Relative abundance = individuals of one species ÷ total individuals in the sample

The figure shows the relative abundance of each species in Community A and Community B. Both communities have 5 species. Which community is more **even**, and why?

A. Community A, because it has twice as many individuals (200 vs. 100)
B. Community B, because its relative abundances are closer to one another
C. Neither — both have 5 species, so their evenness is the same
D. Community A, because its top species has the highest relative abundance (0.625)

T6-08 · Topic 6 · apply · Biodiversity (why it matters)

Foxes in Forest 1 can eat 3 prey species. Foxes in Forest 2 can eat 15 prey species. A disease wipes out one prey species in each forest. Why are Forest 2's foxes more likely to be fine?

A. Biodiversity keeps diseases and other disturbances from happening
B. Forest 2 must also have more foxes, so some are sure to survive
C. Forest 2's foxes have more other prey to eat if one species is lost
D. Forest 2 is a climax community, so its populations can't change

Topic 7 · What is an Ecosystem?

T7-01 · Topic 7 · core · Ecosystem

The Gizmo handbook says 'eco' means *home*. Which statement best defines an **ecosystem**?

A. The place where a single species lives, finds food, and raises young
B. All the living things in an area plus the nonliving parts, interacting
C. All the different species living together in one area
D. A large natural area at least the size of a forest or park

T7-02 · Topic 7 · core · Equilibrium

The Gizmo handbook says healthy ecosystems have a balance, or **equilibrium**, among all species. Which description best fits an ecosystem in equilibrium?

A. Every species has exactly the same number of individuals.
B. Populations never change at all from one year to the next.
C. One species keeps growing until it outnumbers all the other species.
D. Populations rise and fall slightly but stay within normal ranges.

T7-03 · Topic 7 · apply · Ecosystem (levels of organization)

A student studies a small pond in the Santa Cruz Mountains. Which choice describes the pond at the **ecosystem** level?

A. One rough-skinned newt swimming in the shallow water near the shore
B. The pond's newts, algae, and insects plus its water, mud, and light
C. All of the rough-skinned newts that live and breed in the pond
D. All of the newts, algae, insects, and bacteria living in the pond

T7-04 · Topic 7 · apply · Biotic/Abiotic factors

Which list contains **only biotic** factors from a Big Basin redwood forest?

A. Sunlight, creek water, granite rocks, soil minerals
B. Banana slug, sunlight, mule deer, granite rocks
C. Air temperature, wind, dead leaves, fungi
D. Redwood tree, rotting log, fungi, soil bacteria

T7-05 · Topic 7 · challenge · Biotic/Abiotic factors

After the 2020 CZU fire burned much of Big Basin's tree canopy, much more sunlight reached the forest floor. Which statement correctly classifies the factors **and** predicts the most likely result?

A. Sunlight (biotic) went up, so understory plants (abiotic) grew more.
B. Sunlight (abiotic) went up, but plants didn't change, since abiotic factors can't affect life.
C. Sunlight (abiotic) went up, so understory plants (biotic) grew more.
D. Sunlight (abiotic) went up, so understory plants (biotic) grew less, since plants make food from shade.

Topic 8 · Graphs (DRY MIX)

T8-01 · Topic 8 · core · DRY MIX

Your class uses the memory trick **DRY MIX** to set up graphs. What does the **DRY** part tell you?

A. Dependent / Responding variable: what you measure; it goes on the x-axis
B. Manipulated / Independent variable: what you change; it goes on the y-axis
C. Dependent / Responding variable: what you measure; it goes on the y-axis
D. Dependent / Responding variable: what you change on purpose; it goes on the y-axis

T8-02 · Topic 8 · apply · IV, DV and constants in an experiment

A student tests how the amount of light affects redwood sorrel. She grows 5 trays of sorrel under 0, 2, 4, 6, or 8 hours of light per day. Every tray gets the same soil and 50 mL of water per day. After 3 weeks, she measures each plant's height in centimeters. Which choice correctly identifies the variables?

A. IV: hours of light per day; DV: plant height (cm); constant: amount of water (50 mL/day)
B. IV: plant height (cm); DV: hours of light per day; constant: type of soil
C. IV: hours of light per day; DV: amount of water (50 mL/day); constant: plant height (cm)
D. IV: amount of water (50 mL/day); DV: plant height (cm); constant: hours of light per day

T8-03 · Topic 8 · core · X and Y axes: where time goes

After the 2020 CZU fire, a park ranger measures the height (cm) of new sprouts growing from a burned redwood trunk once a month for 12 months. Where should **Time (months)** go on her graph, and why?

A. On the y-axis, because time is what she wrote down on her data sheet at each visit
B. On the y-axis, because time is the dependent variable that changes as the sprouts grow
C. On the x-axis, because time is the dependent variable that responds to sprout height
D. On the x-axis, because time is the independent variable; it doesn't depend on sprout height

T8-04 · **Topic 8 · challenge · IV/DV from a data table; describing the relationship**

In the Habitat Fragmentation Simulation, a student chose how much of a 100-cell forest grid to turn into meadow, then counted the cells still suitable for marbled murrelets.

% meadow	10	20	40	60	80
Suitable cells for murrelets	90	79	55	24	4

Which choice correctly sets up the graph **and** describes the relationship?

A. x-axis: Suitable cells; y-axis: % meadow; as suitable cells increase, % meadow decreases
B. x-axis: % meadow; y-axis: Suitable cells; as % meadow increases, suitable cells decrease
C. x-axis: % meadow; y-axis: Suitable cells; as % meadow increases, suitable cells increase
D. x-axis: Time (years); y-axis: Suitable cells; suitable cells decrease as time goes on

T8-05 · **Topic 8 · core · Line graph vs. bar graph**

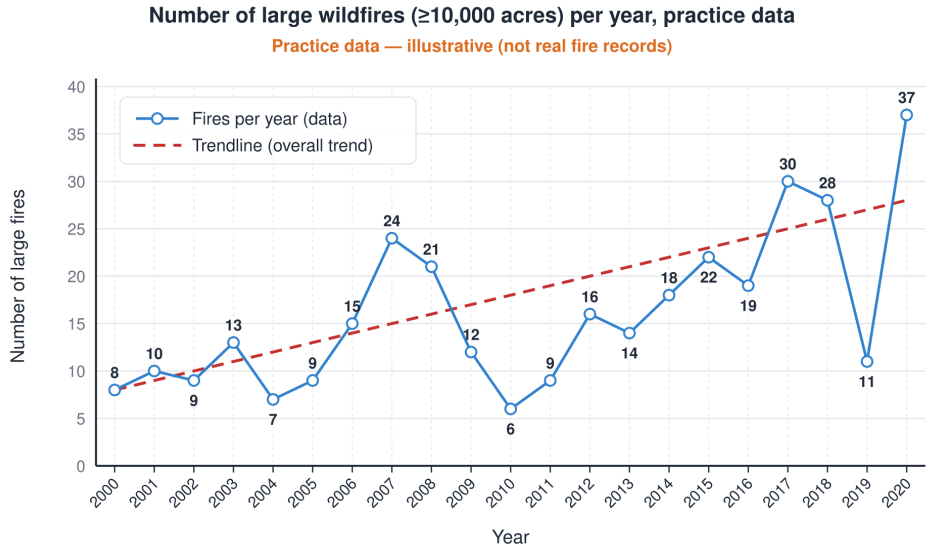
In the bean “rabbit” simulation, a student recorded the population at the end of each year:

Year	0	1	2	3	4	5	6	7	8
Rabbits	25	33	44	64	47	73	55	50	54

He made a bar graph, and the teacher wrote “Wrong type of graph.” Which explanation is correct?

A. A pie chart was needed, because each year's count is one part of the whole rabbit population.
B. A line graph was needed, because rabbit population is the independent variable.
C. A line graph was needed, because line graphs are best for comparing separate categories.
D. A line graph was needed, because the data show one population changing over time.

T8-06 · Topic 8 · apply · Reading a graph (template: axes, high, trend, anomaly)



A student fills in the class graph-analysis template for this practice graph. Which entry is correct?

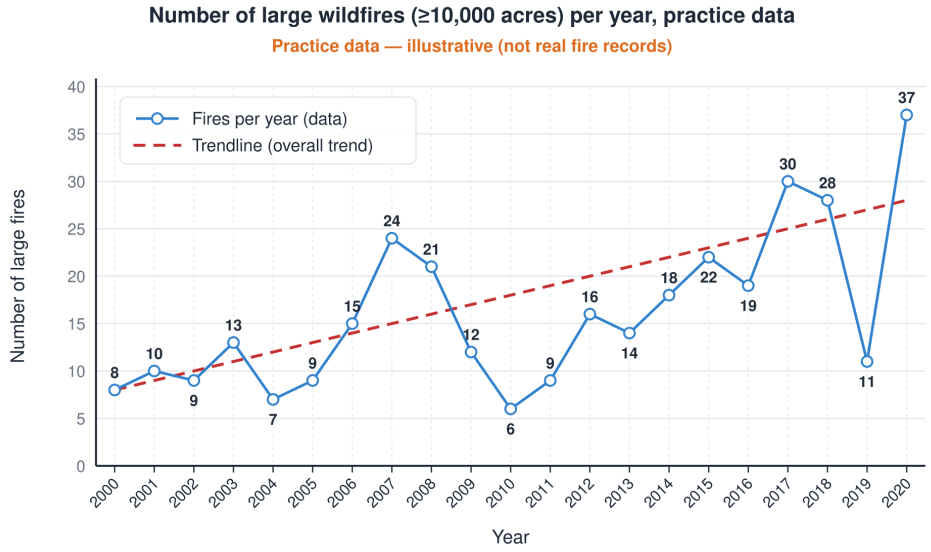
A. Y axis: label “Year,” units in years, range from 2000 to 2020

B. Anomaly: 2019, only 11 large fires, far below 2018 (28) and 2020 (37)

C. High: 2017, with 30 large fires, the most of any year on the graph

D. Trend: the number of large fires generally decreased from 2000 to 2020

T8-07 · Topic 8 · challenge · What you can and can't conclude from a graph



Using **only** this practice graph, which statement can a scientist **NOT** conclude?

A. The number of large fires per year generally increased from 2000 to 2020.
B. There were more large fires in 2020 than in any other year on the graph.
C. Rising temperatures from climate change caused the increase in large fires.
D. There were fewer large fires in 2010 than in 2007.

Topic 9 · Scientific Argumentation (CER)

T9-01 · Topic 9 · apply · Identify claim, evidence, and reasoning

Question: *How does clearing forest affect suitable habitat for marbled murrelets?* A student wrote:
(1) When forest cells dropped from 90 to 20, suitable murrelet cells dropped from 90 to 4.
(2) Clearing more forest reduces the habitat marbled murrelets can use.
(3) Murrelets need large patches of interior forest; clearing breaks the forest into small fragments with more edge, so fewer cells are suitable.
Which labels are correct?

A. (1) Evidence, (2) Claim, (3) Reasoning
B. (1) Claim, (2) Evidence, (3) Reasoning
C. (1) Evidence, (2) Reasoning, (3) Claim
D. (1) Reasoning, (2) Claim, (3) Evidence

T9-02 · Topic 9 · apply · Choosing the best claim

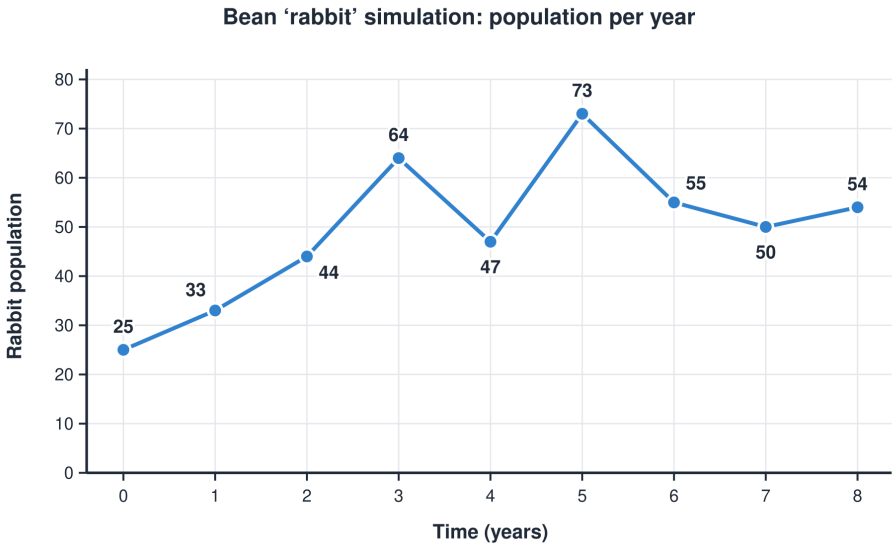
Question: *Which community is more even, Community A or Community B?*

	Community A	Community B
Number of species	5	5
Total individuals	200	100
Most common species (share of all individuals)	62.5%	35%
Rarest species (share of all individuals)	1.5%	8%

Which is the best **claim**?

A. Community A is more even, because it has more individuals (200 vs. 100).
B. Both communities are equally even, because each one has 5 species.
C. Community B seems healthier and nicer to me, so it is the better one.
D. Community B is more even than Community A.

T9-03 · Topic 9 · apply · Choosing the strongest evidence



Claim: *The bean-rabbit population leveled off near a carrying capacity of about 50 rabbits.* Using the graph, which is the **strongest** evidence for this claim?

A. The population started at 25 rabbits in year 0 and was 33 rabbits in year 1.
B. From year 6 to year 8, the population stayed between 50 and 55 rabbits.
C. The population went up and down a lot, and then it sort of stayed the same.
D. A habitat has limited food and space, so it can support only so many rabbits.

T9-04 · Topic 9 · apply · Choosing the best reasoning

Question: *Why are there far fewer pumas than brush rabbits in Big Basin?*
Claim: Fewer pumas can live there because less energy reaches the top of the food chain.
Evidence: In the class energy pyramid, producers have 3,500,000 kcal, rabbits 350,000 kcal, gray foxes 35,000 kcal, and pumas only 3,500 kcal.
Which is the best **reasoning**?

A. Only about 10% of energy passes up each level; the rest is used for metabolism and lost as heat, so fewer pumas can be supported.
B. Pumas received only 3,500 kcal while rabbits received 350,000 kcal, so the rabbits had 100 times more energy than the pumas did.
C. Pumas are apex predators, so no other animal in Big Basin hunts them, and they are able to live for many years in the forest.
D. Decomposers recycle the energy stored in dead rabbits and pass it back up to the pumas, so the pumas need to eat much less food.

T9-05 · Topic 9 · core · Weak vs. strong claims

The CZU fire spread quickly partly because of wind and steep slopes. Question: *How do wind and slope affect how fast a wildfire spreads?* A student's claim: *"Wind and slope have an effect on fires."* Why is this a **weak** claim?

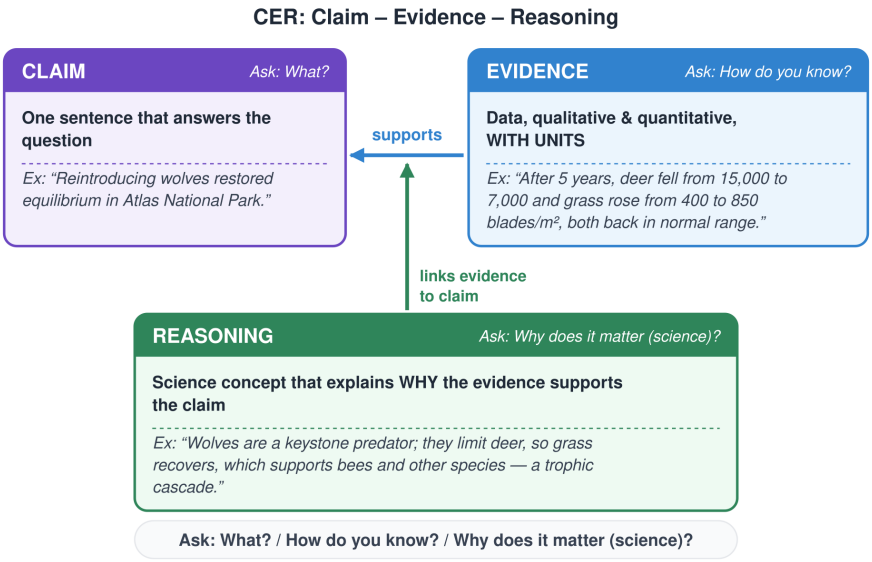
A. It has no numbers; every claim must include data with units, such as wind speed in km/h.
B. It is too short; a strong claim should be a full paragraph that explains everything.
C. It is too vague; it doesn't say how wind and slope change fire spread (faster or slower).
D. It leaves out the science of heat and fuel, and a claim must include the reasoning.

T9-06 · **Topic 9 · core · Evidence needs units**

Claim: *Reintroducing wolves helped the grass in Atlas National Park recover.* A student's evidence says: *"The grass went from 400 to 850."* Which revision turns this into the strongest evidence?

A. After the wolves came back, the grass in the park grew back a whole lot more than it was before they returned.
B. Within 5 years of wolves returning, grass rose from 400 to 850 blades/m² (normal: 800–1,000 blades/m²).
C. Wolves are a keystone species, so adding them back starts a trophic cascade that lets the grass recover.
D. I think the grass went from 400 to 850 because wolves are very important animals in the park.

T9-07 · Topic 9 · challenge · Evaluating a full CER (reasoning that restates evidence)



Use the CER model. Question: *Does a bigger habitat support more rabbits?* Groups ran the bean simulation on a large grid (100 squares) and a small grid (50 squares).
Claim: A bigger habitat supports more rabbits.
Evidence: After 8 years, the large grid had 54 rabbits and the small grid had 27 rabbits.
Reasoning: The large grid had 54 rabbits, which is more than the 27 rabbits on the small grid.
Which part is weakest, and how should it be fixed?

A. Claim: it should include the data, such as 54 rabbits vs. 27 rabbits, to be complete.
B. Evidence: it should name a science idea, such as carrying capacity, to be complete.
C. Reasoning: it just repeats the evidence; it should explain how more space raises the carrying capacity.
D. Reasoning: it should add the student's own opinion that bigger habitats are simply better for rabbits.

Part B — Mock Test (25 questions · 30 minutes)

Set a timer for 30 minutes. Answer every question — there is no penalty for guessing. You may use your index card. Score yourself with the answer key: 23–25 excellent · 18–22 solid · below 18, focus on your weakest topics in the tracker.

1. (M-01) · Topic 7 · core · Biotic vs. abiotic factors

A marine biologist is listing the factors in a California kelp forest. Which list contains **ONLY abiotic** factors?

A. Giant kelp, sea urchins, sea otters, and sunlight
B. Sunlight, water temperature, bacteria in the sand, and salinity
C. Sunlight, water temperature, dissolved oxygen, and rocky seafloor
D. Seawater, dissolved oxygen, plankton, and ocean currents

2. (M-02) · Topic 1 · apply · Sphere interactions

In Yosemite, the roots of a pine tree grow into a crack in a granite boulder. Over many years, the growing roots widen the crack until the rock splits into smaller pieces. Which Earth-system interaction does this describe?

A. The atmosphere affecting the hydrosphere
B. The biosphere affecting the geosphere
C. The geosphere affecting the atmosphere
D. The hydrosphere affecting the biosphere

3. (M-03) · Topic 2 · core · Inter- vs. intraspecific competition

Which situation is an example of **INTERSPECIFIC** competition?

A. Tree swallows fight one another over a limited number of nest boxes in a marsh.
B. A great blue heron wades into a shallow lake and catches and eats a yellow perch.
C. Redwood seedlings sprout so close together that they shade one another.
D. Bluegill and yellow perch in one lake both eat the same scarce insect larvae.

4. (M-04) · Topic 3 · core · Metabolism & flow of energy (amount passed on)

In a pond food chain (algae → mosquito larvae → bluegill → great blue heron), only about 10% of the energy a bluegill takes in is passed on to the heron that eats it. What happens to MOST of the rest of that energy?

A. The bluegill uses it for metabolism (swimming, growing, staying alive), and much leaves as heat.
B. Decomposers return it to the algae, so the same energy cycles through the pond again and again.
C. It is stored permanently in the bluegill's bones and scales and never leaves the fish's body.
D. It is destroyed when the heron digests the fish, because energy gets used up when it is eaten.

5. (M-05) · Topic 5 · apply · Primary vs. secondary succession

Two new communities are forming in Hawai'i.
Site 1: lava cooled into bare black rock; years later, lichens and mosses appear in its cracks.
Site 2: a sugarcane farm is abandoned; within a few years, grasses and weeds cover the field.
Which choice correctly classifies the succession at each site?

A. Site 1: secondary; Site 2: primary
B. Site 1: primary; Site 2: secondary
C. Both primary, since each began with almost no plants
D. Both secondary, since each followed a disturbance

6. (M-06) · Topic 8 · core · DRY MIX: IV/DV and X/Y axes

A student asks: *How does the amount of light affect the growth of redwood sorrel?* She grows sorrel under 2, 4, 6, and 8 hours of light per day, gives every pot the same water, soil, and temperature, and measures plant height (cm) after 3 weeks. Using DRY MIX, how should she set up her graph?

A. X-axis: plant height (cm); Y-axis: hours of light per day
B. X-axis: amount of water (mL); Y-axis: plant height (cm)
C. X-axis: hours of light per day; Y-axis: plant height (cm)
D. X-axis: time (weeks); Y-axis: hours of light per day

7. (M-07) · Topic 2 · apply · Symbiosis: mutualism, commensalism, parasitism

Classify each relationship.
1. A remora fish clings to a shark and eats its leftover scraps; the shark is neither helped nor harmed.
2. A tick feeds on a deer's blood for several days, then drops off; the deer survives but is weakened.
3. Algae live inside coral tissue and give the coral sugars from photosynthesis; the coral gives the algae shelter and nutrients.

A. 1 commensalism, 2 parasitism, 3 mutualism
B. 1 mutualism, 2 predation, 3 commensalism
C. 1 parasitism, 2 parasitism, 3 mutualism
D. 1 commensalism, 2 predation, 3 parasitism

8. (M-08) · Topic 4 · core · Carrying capacity: what raises vs. lowers K

A county wants to **INCREASE** the carrying capacity of a valley for black-tailed deer. Which change would most likely do this?

A. Letting the deer herd grow until it is larger than the valley's current carrying capacity
B. Building a new housing development on part of the valley's meadows
C. A three-year drought that dries up creeks and the plants that deer eat
D. Restoring a damaged creek so it has year-round water and more streamside plants

9. (M-09) · Topic 6 · apply · Richness vs. evenness

Data from the class Sampling Diversity activity:

Community A	Count	Community B	Count
Species A	10	Species A	10
Species B	125	Species E	35
Species C	3	Species F	8
Species D	23	Species G	18
Species E	39	Species H	29
Total	**200**	**Total**	**100**

Which statement is correct?

A. Community A is richer, because it has twice as many individuals.
B. They have the same richness, but Community B is more even.
C. They have the same richness, but Community A is more even.
D. Community B is richer, because its species have more similar numbers.

10. (M-10) · Topic 1 · apply · Earth system interactions (volcano)

When Mount St. Helens erupted in 1980, it blasted ash high into the sky. Winds carried the ash hundreds of miles east, where it coated crops and forest leaves and blocked sunlight from reaching them. Which sequence of Earth spheres best matches this chain of events?

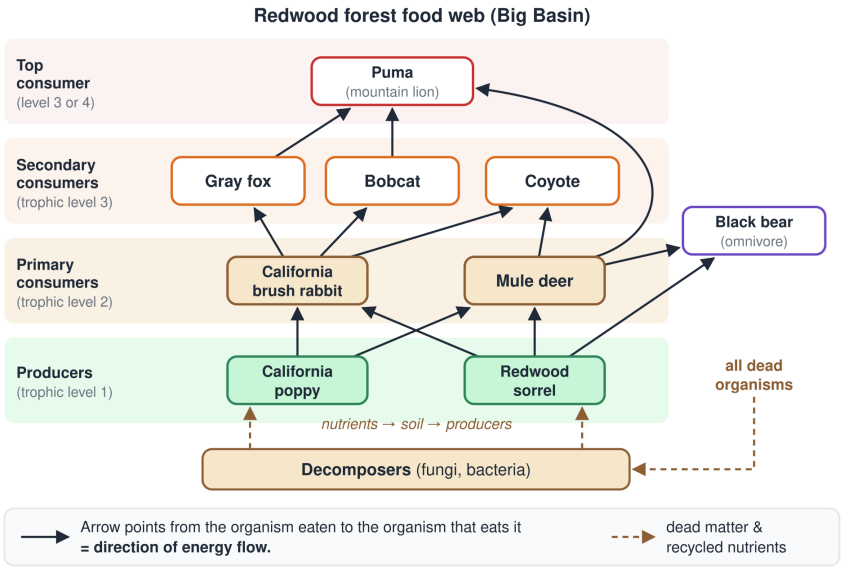
A. Biosphere → Atmosphere → Geosphere
B. Atmosphere → Geosphere → Hydrosphere
C. Geosphere → Atmosphere → Biosphere
D. Hydrosphere → Biosphere → Geosphere

11. (M-11) · Topic 2 · apply · Keystone species

In the 1700s and 1800s, fur hunters nearly wiped out sea otters along the Pacific coast. Sea otters eat sea urchins. Without otters, urchins multiplied and ate the kelp, turning many kelp forests into bare 'urchin barrens' where fish and many other species disappeared. What does this show about sea otters?

A. They are a keystone species: removing them changed the whole kelp forest community.
B. They are the producers of the kelp forest, since the kelp disappeared without them.
C. They are the most abundant species in the kelp forest, so their loss mattered most.
D. They are primary consumers, because they eat the sea urchins that feed on kelp.

12. (M-12) · Topic 3 · apply · Trophic levels & arrow direction in a food web



Use the Big Basin redwood forest food web. Which food chain appears correctly in the web AND shows the puma as a **tertiary** consumer?

A. Redwood sorrel → mule deer → puma
B. Puma → bobcat → brush rabbit → redwood sorrel
C. California poppy → mule deer → coyote → puma
D. Redwood sorrel → brush rabbit → bobcat → puma

13. (M-13) · Topic 4 · apply · Limiting factors (bean rabbit simulation)

In the class bean 'rabbit' simulation, any square with 5 or more rabbits had ALL of its rabbits removed. What real-world idea was this rule modeling?

A. A flood that kills rabbits no matter how many live in an area (density-independent)
B. Overcrowding: too many rabbits share limited food and space (density-dependent)
C. Rabbits living alone cannot find a mate, so they survive but do not reproduce
D. Carrying capacity rising as more and more rabbits are born into each square

14. (M-14) · Topic 5 · apply · Crown & surface fires; fire severity

After the 2020 CZU Lightning Complex fire burned over 97% of Big Basin Redwoods State Park, managers plan **prescribed fires**: low flames that burn only dry needles, fallen branches, and small shrubs on the forest floor. How does this help protect the forest from another high-severity fire?

A. It is a crown fire that thins out the treetops so more sunlight reaches the ground.
B. It is a ground fire that burns tree roots underground, so trees have less wood that could burn later.
C. It is a surface fire that clears floor fuel, so future flames are less likely to reach the crowns.
D. It is a high-severity fire that burns away the soil's organic layer so fewer plants can regrow.

15. (M-15) · Topic 7 · apply · Equilibrium

Fertilizer from nearby lawns washes into a pond. Algae grow so fast that they cover the surface. When the algae die, decomposers break them down and use up much of the dissolved oxygen, and many fish die. Which statement best explains what happened?

A. A change in an abiotic factor (extra nutrients) upset the pond's equilibrium, and the effects spread to living things.
B. The pond became more balanced, because more algae means more producers to feed the rest of the food web.
C. The pond's equilibrium was not affected, because nutrients are nonliving and only living things are part of an ecosystem.
D. The pond lost equilibrium only because algae now outnumber fish, and equilibrium means every population is equal.

16. (M-16) · Topic 9 · core · CER: identify strong evidence

Question: *Does adding nest boxes increase the tree swallow population at a marsh?*

Year	1	2	3	4	5	6
Nest boxes	0	0	0	40	40	40
Breeding pairs	12	11	13	25	31	34

Which sentence is the strongest **EVIDENCE** for a CER?

A. Adding nest boxes increases the tree swallow population at the marsh.
B. The swallows seemed much happier and busier after the boxes went up.
C. Nest space was a limiting factor for the swallows, so adding boxes raised the marsh's carrying capacity.
D. Breeding pairs rose from 11–13 per year with no boxes to 25–34 per year with 40 boxes.

17. (M-17) · Topic 2 · apply · Predator/prey

Ecologists tracked moose (prey) and wolves (predators) on an isolated island for many years. Each time the moose population peaked, the wolf population peaked a few years later. Which explanation best fits this pattern?

A. Wolves increase first, and having more wolves causes the moose population to grow.
B. More moose feed more wolves, so wolves increase; more wolves then eat more moose, so moose decline.
C. Moose and wolves compete for the same food supply, so whenever one population rises the other must fall.
D. The populations are unrelated; wolves and moose simply happen to share similar breeding seasons each year.

18. (M-18) · Topic 4 · challenge · Density-dependent vs. density-independent

Ecologists measured what percent of a deer mouse population was killed by two factors, in years when mice were scarce versus crowded.

Factor	% killed at LOW density (10 mice/ha)	% killed at HIGH density (80 mice/ha)
Factor X	20%	20%
Factor Y	5%	45%

Which conclusion is best supported?

A. Factor X is density-dependent; it could be a disease that spreads from one mouse to another.
B. Factor Y is density-independent; it could be a sudden winter flood that sweeps through burrows.
C. Factor Y is density-dependent; it could be a disease that spreads faster among crowded mice.
D. Neither factor is a limiting factor, because neither one kills every mouse.

19. (M-19) · Topic 5 · apply · Habitat fragmentation

Highway 17 cuts through the forests of the Santa Cruz Mountains. Mountain lions trying to cross it were often hit by cars, and lions on each side became cut off from each other. In 2023, a wildlife undercrossing was completed beneath the highway at Laurel Curve. Which statement best explains why it was built?

A. The highway split one large habitat into isolated patches; the crossing reconnects them for safe travel to mates and prey.
B. The highway only removed the thin strip of land under the pavement, so the crossing mainly gives lions more ground to hunt on.
C. The highway created more forest edge, which lions need most, so the crossing helps them reach even more edge habitat.
D. The crossing keeps lions permanently on one side of the highway, so they stay safely away from people and cars.

20. (M-20) · Topic 3 · challenge · Flow of energy: 10% rule calculation

In a kelp forest, energy flows: giant kelp → purple sea urchins → sea otters. The sea otters in one area get about 400 kcal/m² per year from eating urchins. Using the 10% rule, about how much energy did the kelp capture to support them?

A. 4 kcal/m² per year
B. 40 kcal/m² per year
C. 4,000 kcal/m² per year
D. 40,000 kcal/m² per year

21. (M-21) · Topic 9 · apply · CER: identify strong reasoning

In the Atlas National Park Gizmo, the question was: *Which species is the keystone species?* A student's claim is: **Wolves are the keystone species.** Which sentence is the best **REASONING** to support this claim?

A. After wolves were added, deer fell from 15,000 to 7,000 and grass rose from 400 to 850 blades/m².
B. Wolves prey on deer; fewer deer means less overgrazing, so plants and bees recover (a trophic cascade).
C. Wolves are the strongest and most impressive animals, so the park needs them more than any other species.
D. Bees pollinate the plants, so adding bees should have been the best way to bring the park back into balance.

22. (M-22) · Topic 6 · challenge · Relative abundance

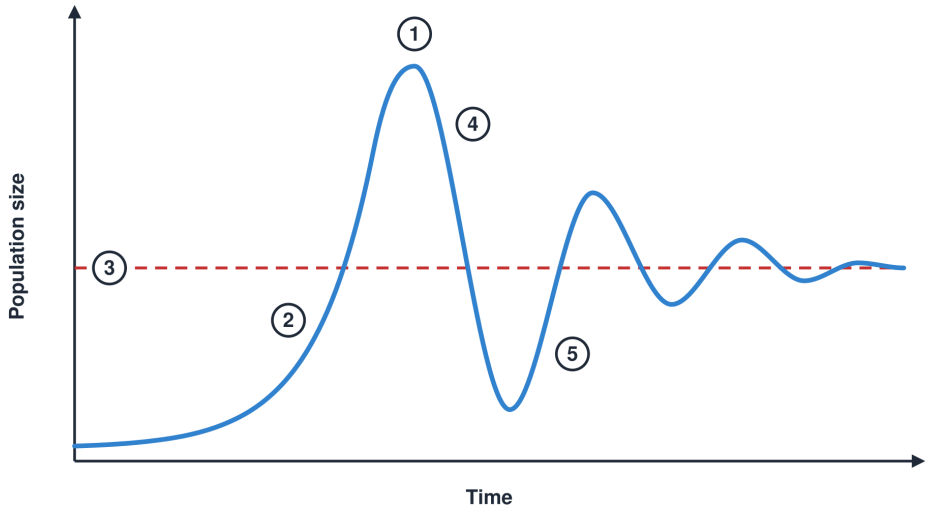
A student counts organisms in tide-pool quadrats on the Santa Cruz coast.

Species	Number counted
Aggregating anemone	54
Black turban snail	36
Hermit crab	21
Ochre sea star	9

What is the relative abundance of ochre sea stars?

A. 7.5%
B. 8.3%
C. 9%
D. 25%

23. (M-23) · Topic 4 · challenge · Overshoot, decline, recovery



In 1944, 29 reindeer were released on St. Matthew Island, Alaska, which had plenty of lichen to eat and no predators. By 1963 there were about 6,000, and the lichen was badly overgrazed. After a harsh winter, only 42 were left by 1966. Using the graph, which number matches the 1963–1966 change, and what best explains it?

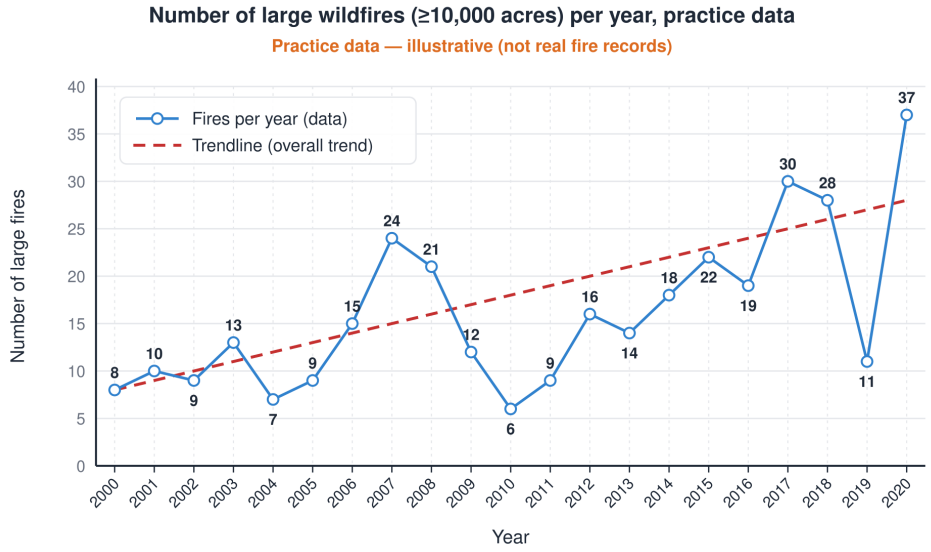
A. 5 (recovery): with fewer reindeer, the lichen grew back and the herd quickly rebounded.
B. 2 (exponential growth): with no predators, the herd kept growing without any limit.
C. 4 (decline): the herd ate lichen faster than it regrew, so food ran out and most starved.
D. 4 (decline): wolves and bears reached the island and hunted down the large, crowded herd.

24. (M-24) · Topic 5 · apply · Organisms in communities: pioneer species & climax community

In the class Yosemite primary succession timeline, lichens are the pioneer species on bare, glacier-scraped granite. By the climax community (1,000+ years), lichens are labeled 'retired pioneers.' Which statement best explains both roles?

A. Lichens need deep, rich soil, so they can only grow after large trees have built up the soil.
B. Lichens are decomposers that break down fallen trees, so they matter only at the climax stage.
C. Lichens are the climax species, because they live the longest and outlast all other organisms.
D. Lichens break granite into bits that start soil; later, shade from large trees crowds them out.

25. (M-25) · Topic 8 · challenge · Reading graphs: points of interest



Study the practice graph. Between which two **back-to-back** years did the number of large fires **increase the most**?

A. From 2006 to 2007
B. From 2016 to 2017
C. From 2019 to 2020
D. From 2011 to 2012

Hints — peek only if you're stuck

Each hint nudges your thinking without revealing the answer. On the real test there are no hints, so try first!

ID	Hint
T1-01	Think about the word 'system.' Is it about one part of Earth, or about how the parts connect?
T1-02	Ignore WHERE each item is found (in soil, underground, or frozen). Ask what the item itself is made of.
T1-03	Two things could cross Earth's boundary with space: energy and matter. Which one keeps arriving from the Sun?
T1-04	Ask where each process takes place, and where that place gets its heat or light.
T1-05	For each pair, ask what the item is mostly made of or involves: living things, rock, water, or moving air.
T1-06	Translate each sphere name into an everyday word first. Then find the choice that includes both of them.
T1-07	Take it one step at a time. Where does the eruption start? Where does the gas go? Where does the rain land? Who is harmed?
T2-01	Start with the one you're surest of: 'mutual' means shared by both. Then decide which of the other two actually harms the second species.
T2-02	Focus only on the two cats, not the buffalo. Are they the same species or different species?
T2-03	Picture the keystone at the top of a stone arch. It is only one stone, but what happens to the arch if you pull it out?
T2-04	List what each partner gets out of the deal. Does each one receive something?
T2-05	Give each species a symbol. Does the whale get a +, a −, or a 0?
T2-06	The tick benefits and the deer is harmed. But is the deer killed and eaten right away?
T2-07	For each scenario, count how many species are competing: one or two?
T2-08	Predators keep prey populations in check. What happens to the prey when that check is removed?
T2-09	Compare each of the last two columns with the normal range. Which fix actually restored balance?
T2-10	Look for the chain where every arrow is a real feeding link: predator → its prey → what the prey eats.
T3-01	Ask: which organism makes its own food from sunlight, and which has to get its food from other living things?
T3-02	Read each arrow as 'gives energy to.' Where does the energy start, and where does it go next?
T3-03	Living things use energy to live, grow, and reproduce. What form of energy do they give off while doing this?
T3-04	Ask three questions: Who eats a dead animal? Who eats bits of dead plants? Who breaks matter down with enzymes?
T3-05	Follow each arrow that points INTO the puma. What trophic level is each of those prey animals?
T3-06	Find every arrow pointing into the black bear. Is each food a producer or a consumer? The bear is one level above each food.
T3-07	Where do producers get the nutrients they take up from the soil? Who puts those nutrients there?
T3-08	Compare the kcal at the rabbit level with the kcal at the puma level. What does each level do with most of its energy?
T3-09	Trace every arrow that starts at the brush rabbit. Then check whether each of those animals has any OTHER arrow pointing into it.
T3-10	Count the arrows (steps) between the sorrel and the puma. Each step keeps only one-tenth of the energy.
T4-01	Ask: what is nesting space doing to the colony's growth?
T4-02	Ask: would this factor hit the population harder if the animals were more crowded?
T4-03	Find the steepest part of the curve and the flat dashed line. Then look at what the curve does after it reaches that line.
T4-04	Divide what the river supplies by what one bear needs. Then ask which resource is actually in short supply.
T4-05	Focus on what the word 'density' refers to in this term. Is the term about WHETHER there is an effect, or about what the effect depends on?
T4-06	First name each event: fast growth, above K, die-off, bounce back. Then find each stage on the graph.
T4-07	Carrying capacity depends on the resources a habitat provides (food, water, space). Which list ADDS resources?
T4-08	Carrying capacity isn't a single high or low point. Look for where the line stops making big jumps and stays fairly steady.
T4-09	Find one hare peak and the lynx peak closest to it. Which comes first? Then think about what lynx need in order to raise more young.

ID	Hint
T4-10	For each factor, ask: as the rabbits get more crowded, does the share that dies go up, or stay the same?
T4-11	Look at the jump from 1957 to 1963 and then the drop by 1966. Which stages from the overshoot graph does that match?
T5-01	Was there any soil when the glaciers left? And are lichens the first arrivals or the final, mature stage?
T5-02	Find the one place where the soil is still there after the change.
T5-03	Find '10 years' on the timeline. Also ask: did the soil stay or go?
T5-04	Two clues matter in Ms. Hero's definition: what happened to the soil, and who is doing the restoring?
T5-05	Which stage does 1,200 years fall in? Then follow the arrow under the timeline for light and soil.
T5-06	Where were the flames: in the treetops, on the forest floor, or underground?
T5-07	Severity is about the aftermath. What would you look at after the smoke clears?
T5-08	Check every item. One human-caused event makes the whole list wrong.
T5-09	Fragmented means broken into many small, separated pieces. Which pattern breaks the forest into the smallest pieces?
T5-10	Compare the two columns row by row. Does the gap between them stay the same or grow?
T5-11	Think about what protects the living layer just under the outside of a redwood trunk.
T6-01	Richness is a count, but a count of what?
T6-02	First count the species in each field (richness). Then check whether the species have similar counts (evenness).
T6-03	Answer two separate questions: Which field has more species? Which field's species counts are closer to equal?
T6-04	Relative abundance is about individuals, not species. Add the whole column before you divide.
T6-05	Relative abundance compares one species to the whole sample. Start by finding the total number of individuals.
T6-06	Richness and the total number of individuals are not the same thing.
T6-07	An even community has bars of similar length. Which chart's bars are closer to the same length?
T6-08	Think about the foxes' menu. What happens to a predator with few food choices when one choice disappears?
T7-01	Does an ecosystem include only living things, or also things like sunlight, water, and soil?
T7-02	Balance doesn't mean frozen. Think about the 'normal range' column in the Atlas NP data.
T7-03	Check each choice for nonliving things. Which level of organization is the first to include them?
T7-04	Biotic means living OR once living. Check every item carefully, including anything that is dead.
T7-05	First decide which factor is living and which is not. Then ask what plants use sunlight for.
T8-01	Break the word into letters: D, R and Y each stand for something. Does a “responding” variable get changed by you, or does it change in response to what you did?
T8-02	Which one did she change on purpose from tray to tray? Which one did she wait and measure at the end? Which one was the same for every tray?
T8-03	Does the passing of time depend on how tall the sprouts get, or does sprout height depend on how much time has passed?
T8-04	Which value did the student set before counting? That is the MIX variable. Then read the bottom row from left to right: is it going up or down?
T8-05	Look at what goes on the x-axis here. Are these separate groups (like types of vegetation), or the same thing measured again and again as time passes?
T8-06	Check each entry against the graph. Which axis runs along the bottom? Where is the tallest point? Is the dashed trendline going up or down?
T8-07	Does the graph include any data about temperature or weather? A graph shows WHAT happened; does it also show WHY?
T9-01	Find the sentence with the data first. Then find the one that simply answers the question. The one left over should explain WHY using a science idea.
T9-02	A strong claim directly answers the exact question that was asked. Does the choice answer “which is more even,” and does it agree with the data rather than feelings?
T9-03	Strong evidence is specific data (numbers with units) that relates directly to the claim. Which part of the graph is the claim talking about?

ID	Hint
T9-04	Good reasoning uses a science rule to explain WHY the numbers shrink at each level. It doesn't just repeat the numbers.
T9-05	Imagine reading only this claim. Could you tell what actually happens to a fire when the wind picks up?
T9-06	400 and 850 of what? Strong evidence tells the reader the units, the time frame, and how the numbers compare to normal.
T9-07	Compare each part to the model. Does the reasoning tell you anything the evidence didn't already say?
M-01	Check every single item in the list: is it living? Remember that some living things are too small to see.
M-02	Name the sphere of the thing causing the change (the roots), then the sphere of the thing being changed (the rock).
M-03	'Inter' means between different groups, like an interstate highway connecting different states.
M-04	Think about what every living thing does all day that requires energy, and what kind of energy you feel coming off a warm body.
M-05	Ask one question about each site: was there already soil when the new community started?
M-06	Which variable did she choose and change on purpose? Which one did she wait to measure?
M-07	For each pair, write + (helped), − (harmed), or 0 (no effect) for BOTH organisms before looking at the choices.
M-08	Carrying capacity depends on resources. Which choice ADDS resources instead of taking them away?
M-09	Richness = how many KINDS. Evenness = how fair the split is. Check each one separately.
M-10	Trace the ash: where did it come from, what carried it, and what did it land on?
M-11	Think of the word for a species whose removal makes far more collapse than its numbers would suggest, like pulling the center stone out of an arch.
M-12	Number the levels starting with the producer as level 1, and make sure every arrow in the chain really appears in the web.
M-13	The rule only applied to certain squares. What did those squares have in common?
M-14	Where do the flames of a prescribed fire stay, and what fuel do they remove?
M-15	Start with what was added to the water. Is it living or nonliving, and what chain of changes did it set off?
M-16	Evidence is the data. Which sentence uses numbers taken straight from the table?
M-17	Which population is the food? Think about what has to happen BEFORE predators can raise more pups.
M-18	Compare each factor's two percentages. Which one gets stronger when the mice are more crowded?
M-19	Think about what happens to a population that is split into smaller groups that cannot reach each other.
M-20	Producers always have the MOST energy. How many steps separate the otters from the kelp?
M-21	Reasoning answers 'WHY does the evidence support the claim?' using a science idea from this unit.
M-22	First add up EVERY organism counted. Then compare the sea stars to that total.
M-23	First decide whether the population was rising or falling from 1963 to 1966. Then check the cause against the facts given in the question.
M-24	Look at two rows of the timeline: what lichens DO to bare rock, and how 'light at ground' changes by the climax stage.
M-25	For each pair, subtract the earlier year's value from the later year's value. Read the numbers next to the points.

Answer Key & Explanations

Quick key

T1-01	**B**	T1-02	**D**	T1-03	**A**	T1-04	**C**	T1-05	**B**	T1-06	**C**	T1-07	**A**	T2-01	**D**
T2-02	**A**	T2-03	**C**	T2-04	**B**	T2-05	**C**	T2-06	**A**	T2-07	**D**	T2-08	**B**	T2-09	**A**
T2-10	**C**	T3-01	**D**	T3-02	**B**	T3-03	**A**	T3-04	**C**	T3-05	**C**	T3-06	**A**	T3-07	**D**
T3-08	**B**	T3-09	**A**	T3-10	**C**	T4-01	**B**	T4-02	**D**	T4-03	**A**	T4-04	**D**	T4-05	**B**
T4-06	**C**	T4-07	**C**	T4-08	**A**	T4-09	**B**	T4-10	**D**	T4-11	**A**	T5-01	**C**	T5-02	**B**
T5-03	**D**	T5-04	**A**	T5-05	**D**	T5-06	**C**	T5-07	**A**	T5-08	**D**	T5-09	**A**	T5-10	**C**
T5-11	**B**	T6-01	**A**	T6-02	**D**	T6-03	**A**	T6-04	**B**	T6-05	**C**	T6-06	**D**	T6-07	**B**
T6-08	**C**	T7-01	**B**	T7-02	**D**	T7-03	**B**	T7-04	**D**	T7-05	**C**	T8-01	**C**	T8-02	**A**
T8-03	**D**	T8-04	**B**	T8-05	**D**	T8-06	**B**	T8-07	**C**	T9-01	**A**	T9-02	**D**	T9-03	**B**
T9-04	**A**	T9-05	**C**	T9-06	**B**	T9-07	**C**	M-01	**C**	M-02	**B**	M-03	**D**	M-04	**A**
M-05	**B**	M-06	**C**	M-07	**A**	M-08	**D**	M-09	**B**	M-10	**C**	M-11	**A**	M-12	**D**
M-13	**B**	M-14	**C**	M-15	**A**	M-16	**D**	M-17	**B**	M-18	**C**	M-19	**A**	M-20	**D**
M-21	**B**	M-22	**A**	M-23	**C**	M-24	**D**	M-25	**C**						

Full explanations — Part A drills

T1-01 Answer: B · Topic 1 · Earth Systems Science

Earth system science studies the relationships between air (atmosphere), water (hydrosphere), land (geosphere), and life (biosphere). It matters because a change in one sphere ripples into the others. For example, a drought (hydrosphere) dries out plants (biosphere), which fuel a wildfire that sends smoke into the air (atmosphere).

- **A** That is geology, which covers only the geosphere. Earth system science looks at all four spheres together.
- **C** That is ecology inside the biosphere only. Earth system science also includes air, water, and land.
- **D** That is weather and climate science, which is mostly the atmosphere. Earth system science connects the atmosphere to the other three spheres.

Trap: ESS = interactions among ALL spheres, not the study of one sphere · **Review:** What is Earth system science?

T1-02 Answer: D · Topic 1 · Hydrosphere

The hydrosphere is ALL of Earth's water in every form (liquid, solid, and gas), wherever it is. Glaciers are frozen water, soil moisture is water held in soil, and aquifers hold groundwater underground. All of it is still water, so all of it belongs to the hydrosphere.

- **A** Soil and rock are geosphere, but the WATER inside them is hydrosphere. A glacier is frozen water, not rock.
- **B** The atmosphere is the layer of gases (air). Glaciers, soil moisture, and groundwater are not air.
- **C** The biosphere is all living things. Water is nonliving.

Trap: Soil moisture / glaciers / aquifers = hydrosphere, not geosphere · **Review:** The four spheres

T1-03 Answer: A · Topic 1 · Earth Systems Science (closed system)

In a closed system, energy crosses the boundary but matter almost does not. Earth receives energy as sunlight and loses energy to space as heat. Its matter (water, carbon, rock) stays on Earth and is recycled among the spheres, which is why matter has to cycle while energy keeps flowing in from the Sun.

- **B** Sunlight clearly enters Earth and heat radiates back out to space, so energy is not trapped. A system with no exchange at all is called 'isolated,' not 'closed.'
- **C** This is backward. Only tiny amounts of matter (like meteorites) cross the boundary, and energy DOES leave Earth as heat.
- **D** 'Closed' describes Earth's boundary with space, not the spheres. Inside Earth the spheres constantly exchange matter. For example, ocean water evaporates into the air.

Trap: Closed ≠ isolated: energy crosses the boundary, matter (mostly) doesn't · **Review:** What is Earth system science?

T1-04 Answer: C · Topic 1 · Earth Systems Science (energy drivers)

Heat from Earth's core drives many geosphere processes: plate tectonics, volcanoes, earthquakes, and the rising magma that formed Yosemite's granite. The Sun drives most other Earth-system interactions. It heats air unevenly to make wind, evaporates water in the water cycle, and powers photosynthesis.

A Wind happens because the Sun heats Earth's surface and air unevenly: warm air rises and cooler air moves in.
B Evaporation in the water cycle is powered by the Sun's heat.
D Photosynthesis captures light energy from the Sun.

Trap: Sun drives most interactions; Earth's core drives geosphere processes · **Review:** What is Earth system science?

T1-05 Answer: B · Topic 1 · Atmosphere, Hydrosphere, Biosphere, Geosphere

The jet stream is a fast, high-altitude band of wind, so it belongs to the atmosphere, along with clouds, weather, and the atmosphere's role as a solar-radiation buffer. On the organizer, coral reefs (built by living corals) are biosphere, glaciers (frozen water) are hydrosphere, and fossilization (remains turning into rock) is a geosphere process.

A Reefs look rocky, but they are built by living coral animals, so the organizer lists coral reefs under the biosphere.
C A glacier is frozen water. The hydrosphere includes water in solid form.
D A fossil comes from a once-living thing, but fossilization is the process of remains turning into rock in sediment. The organizer lists it under the geosphere, next to sedimentation.

Trap: Classifying by appearance (rocky reef, ice on land, fossil from life) instead of the organizer's sphere · **Review:** The four spheres

T1-06 Answer: C · Topic 1 · Sphere interactions (wildfire)

The fire burned away plants and the soil's organic layer. Rain (hydrosphere) then eroded the bare soil (geosphere) and carried sediment and ash into streams. This is why burned slopes face higher risk of erosion and landslides: one change (the fire) chains into other spheres.

A Lightning and storms are atmosphere, and the forest fuel is biosphere. No land and water link is described.
B This is the biosphere (burning trees) adding gases to the atmosphere.
D Resprouting redwoods are the biosphere recovering. No interaction between water and land is described.

Trap: Identifying which spheres are involved in one step of a multi-sphere event · **Review:** How the spheres interact

T1-07 Answer: A · Topic 1 · Sphere interactions (volcano)

The volcano is part of the geosphere. Its gas enters the atmosphere, acid rain falls into the lake (hydrosphere), and the fish that die are part of the biosphere. This is why Earth system science matters: a change in one sphere can ripple through all four.

B The chain starts with the volcano (geosphere), not the air. The fish (biosphere) are affected last, after the lake water changes.
C The gas goes into the air (atmosphere) BEFORE it falls as rain into the lake (hydrosphere).
D This order is backward. The chain begins with the volcano (geosphere) and ends with the fish (biosphere).

Trap: Tracing a multi-step chain of sphere interactions in order · **Review:** How the spheres interact

T2-01 Answer: D · Topic 2 · Symbiosis

Mutualism means both species benefit (+/+), like bees and flowers. Commensalism means one benefits and the other is unaffected (+/0), like barnacles on a whale. Parasitism means the parasite benefits and the host is harmed (+/−), like a tick on a deer.

A Commensalism and parasitism are swapped. Commensalism is +/0 (no harm), and parasitism is +/− (the host is harmed).
B Mutualism is +/+, not +/0. This list starts with commensalism.
C −/− (both harmed) is competition, not commensalism.

Trap: Commensalism (+/0) vs. mutualism (+/+) vs. parasitism (+/−) · **Review:** Symbiosis: mutualism, commensalism, parasitism

T2-02 Answer: A · Topic 2 · Inter- and Intra-specific Competition

Interspecific competition happens between DIFFERENT species that need the same limited resource, here the same prey. Both are harmed (−/−) because each gets less food. Intraspecific competition happens within ONE species, like seagulls fighting over nesting space on a rock.

B 'Intra-' means within, so intraspecific competition is within the same species. Lions and tigers are different species, even though both are big cats.
C Each cat preys on the buffalo, but the question asks about the lion and the tiger. With each other, they compete for a shared resource.
D Neither cat benefits from the other. Both lose food, so the relationship is −/−, not +/+.

Trap: Inter- (between species) vs. intra- (within one species); don't get distracted by the prey · **Review:** Competition: intraspecific vs. interspecific

T2-03 Answer: C · Topic 2 · Keystone Species

A keystone species has a much bigger effect on its ecosystem than its numbers suggest. If it is removed, the balance (equilibrium) can collapse, as happened with the wolves in the Atlas National Park Gizmo. Keystones can be predators, pollinators like bees, or ecosystem engineers like beavers, and one ecosystem can have more than one.

A Keystones are often NOT abundant. Atlas NP normally has only about 95–120 wolves but 4,000–8,000 deer. Importance does not equal numbers.

B Many keystones are predators, but not all. Pollinators such as bees and dam-builders such as beavers can also be keystone species.

D A producer can sometimes be a keystone, but the definition is about how much the ecosystem changes when the species is removed, not about supplying the most energy.

Trap: Keystone ≠ most abundant, and ≠ always the top predator · **Review:** Keystone species

T2-04 Answer: B · Topic 2 · Mutualism

Both partners benefit (+/+). The alga gets water and shelter, and the fungus gets food from photosynthesis. This mutualism lets lichens survive on bare rock with no soil, which is why lichens are pioneer species in primary succession.

A In commensalism only one partner benefits and the other is unaffected. Here both partners get something.

C Neither partner is harmed. Parasitism requires one species (the host) to be hurt.

D Competition harms both species (−/−). Here the two partners help each other.

Trap: Lichen = mutualism · **Review:** Symbiosis: mutualism, commensalism, parasitism

T2-05 Answer: C · Topic 2 · Commensalism

Commensalism is +/0. The barnacle benefits (transport and food) and the whale is basically unaffected. The key clue is that the whale is neither helped nor harmed.

A Mutualism requires BOTH species to benefit. The whale gets nothing from the barnacles.

B Parasites harm their host, usually by feeding on it. Barnacles filter plankton from the water, not from the whale's body.

D In predation, one animal kills and eats the other. Barnacles don't eat whales.

Trap: Commensalism (+/0) vs. mutualism (+/+) · **Review:** Symbiosis: mutualism, commensalism, parasitism

T2-06 Answer: A · Topic 2 · Parasitism

This is parasitism (+/−). The tick (the parasite) benefits by feeding on the deer (the host), which is harmed but usually not killed, because a parasite needs its host alive to keep feeding. Predation is also +/−, but a predator kills and eats its prey.

B Predation is also +/−, but a predator (like a puma) kills and eats its prey. The tick feeds on a living host for days.

C In commensalism the second species is unaffected. This deer is harmed by blood loss and possible disease.

D Mutualism requires both species to benefit. The deer gains nothing from the tick.

Trap: Parasite vs. predator: parasites usually don't kill their host quickly · **Review:** Symbiosis: mutualism, commensalism, parasitism

T2-07 Answer: D · Topic 2 · Inter- and Intra-specific Competition

Coyotes and bobcats are different species competing for the same prey, so Scenario 1 is interspecific (between species). Gulls of one species competing for nest sites is intraspecific (within a species). Both kinds of competition are −/− and help limit population size. A gull pair that finds no nesting room raises no chicks.

A This is reversed. 'Inter-' means between different species (coyote vs. bobcat), and 'intra-' means within one species (gull vs. gull).

B Scenario 2 involves only one species (Western gulls), so it is intraspecific.

C Scenario 1 involves two species (coyote and bobcat), so it is interspecific.

Trap: Inter- vs. intra-specific competition · **Review:** Competition: intraspecific vs. interspecific

T2-08 Answer: B · Topic 2 · Predator/Prey

With fewer predators, more deer survive and reproduce, so the deer population grows. More deer then eat more plants such as redwood sorrel. This shows why predator–prey relationships matter: predators keep prey from overgrazing their food supply.

A Prey don't depend on predators for food; it's the other way around. Losing predators usually lets prey numbers rise.

C Pumas are carnivores and don't eat plants. Plants would more likely DECREASE as the growing deer herd eats them.

D Predators can strongly affect other species. The Atlas NP wolves showed how losing a top predator ripples through a whole food web.

Trap: Removing a predator → prey INCREASE (not decrease) · **Review:** Predator and prey

T2-09 Answer: A · Topic 2 · Keystone Species

After wolves were added, all six populations returned to their normal ranges (deer 15,000 → 7,000; grass 400 → 850 blades/m²), so equilibrium was restored. Adding bees made most things worse: deer rose to 17,000 and grass fell to 350 blades/m². Wolves were the missing keystone. By controlling deer and coyotes, they indirectly protected the grass, bees, and ravens.

B Bees are important pollinators, and many people guess bees. But adding bees did NOT fix the park: hives stayed below normal (3,000), and deer and grass got worse.

C Deer were the most abundant species, but a keystone isn't the most abundant species. The deer boom was a SYMPTOM of losing the wolves.

D Grass is a vital producer, but it was low because too many deer were eating it. Adding wolves, not grass, fixed the system.

Trap: Atlas NP keystone = wolves, NOT bees (and not the most abundant species) · **Review:** Keystone species

T2-10 Answer: C · Topic 2 · Keystone Species (trophic cascade)

This is a trophic cascade. Without wolves, the deer population exploded (15,000 vs. a normal 4,000–8,000) and overgrazed the plants — grass fell to 400 blades/m² (normal 800–1,000), and the flowering plants that bees depend on for nectar and pollen were eaten too, so bee hives declined. Losing one keystone predator rippled down through the food web.

A Removing a predator makes its prey INCREASE, not decrease. Less grazing would also leave MORE grass, but grass actually fell to 400 blades/m².

B Coyotes did increase, but coyotes are meat-eating predators, not grass grazers. The grazers in this park are the deer.

D This reverses cause and effect, and the stem says wolves don't eat bees. Wolves were the first species to drop out of their normal range (around 1920).

Trap: Trophic cascade: a keystone's effects travel indirectly through the food web · **Review:** Keystone species

T3-01 Answer: D · Topic 3 · Autotrophs vs. heterotrophs

Redwood sorrel uses photosynthesis to turn light energy into chemical energy (sugar), so it is a 'self-feeder' — an autotroph and a producer at trophic level 1. A mule deer must get its food from other organisms, so it is a heterotroph ('other-feeder'). Every consumer and every decomposer is a heterotroph.

A Taking in water and minerals is not the same as eating food. Sorrel makes its own sugar from sunlight, so it is an autotroph.

B Eating plants does not make the deer a food-maker. The deer gets its food from another organism, so it is a heterotroph.

C Fungi are not plants and cannot photosynthesize. They get food by breaking down dead organisms, so they are heterotrophs (decomposers).

Trap: Fungi look like plants but are heterotrophs; eating plants does not make an animal an autotroph · **Review:** Trophic levels: who eats whom

T3-02 Answer: B · Topic 3 · Food chain arrows (direction of energy flow)

Arrows in food chains and webs point FROM the organism that is eaten TO the organism that eats it — the direction energy flows (Gizmo handbook; redwood sorrel → mule deer in class). Energy starts in the producer (redwood sorrel), passes to the primary consumer (brush rabbit), and then to the secondary consumer (gray fox).

A This is the most common arrow mistake. Arrows do not point to the food; they point to the eater, showing where the energy goes.

C The producer does go first, but the gray fox does not eat sorrel. The rabbit eats the sorrel, and then the fox eats the rabbit.

D The rabbit is a consumer (level 2), not a producer. Trophic level 1 is always the producer — here, the redwood sorrel.

Trap: Arrows point to the eater (energy flow), not to the food · **Review:** Food chains vs. food webs: reading the arrows

T3-03 Answer: A · Topic 3 · Metabolism and energy lost as heat

Only about 10% of the energy moves up a level: 10,000 kcal × 0.10 = 1,000 kcal. The other ~90% is used by the organisms at that level for metabolism (living, growing, reproducing). During metabolism some chemical energy is converted to heat, which leaves the organism, goes into the environment, and leaves the food chain. Energy is not destroyed, and it does not cycle back — it flows one way.

B Energy cannot be destroyed. It changes form — mostly into heat that escapes into the environment.

C Energy does not pile up to be passed on later. At every level, most energy is used and lost as heat, so each level passes on only about 10%.

D Energy flows one way (sun → producers → consumers) and ends up as heat. Nutrients (matter) cycle back to producers, but usable energy does not.

Trap: Energy is not 'destroyed' and does not cycle back — about 90% is used in metabolism and leaves as heat · **Review:** Energy flow, metabolism & the 10% rule

T3-04 Answer: C · Topic 3 · Decomposers, scavengers, and detritivores

Scavengers are animals that feed on dead animals (vultures eating a dead raccoon). Detritivores, like earthworms and millipedes, eat dead and decaying material such as leaf litter. Decomposers (fungi and bacteria) use enzymes to break dead matter down. Together they return nutrients to the soil, where producers take them up again.

A Vultures did not kill the raccoon, so they are not predators here, and earthworms cannot make food from sunlight, so they are not producers.

B Vultures eat whole carcasses instead of breaking matter down with enzymes, earthworms eat dead (not living) plants, and fungi are decomposers, not scavengers.

D Fungi are not plants and cannot photosynthesize. They are decomposers that break down dead matter with enzymes.

Trap: Fungi are decomposers, not producers; scavengers eat dead animals but do not kill them · **Review:** Decomposers, scavengers & detritivores

T3-05 Answer: C · Topic 3 · Trophic levels in the redwood food web (puma)

An organism's trophic level depends on what it is eating — it is one level above its food. Mule deer are primary consumers (level 2), so a puma eating deer is a secondary consumer (level 3). Gray foxes and bobcats are secondary consumers (level 3), so a puma eating them is a tertiary consumer (level 4). Top predators often feed at more than one level.

A Its spot in the top row only shows its highest level. When the puma eats a mule deer (a primary consumer), it is acting as a secondary consumer.

B A consumer is one level ABOVE what it eats. Eating a primary consumer makes the puma a secondary consumer, not a primary one.

D Not every arrow into the puma comes from a primary consumer — arrows also come from the gray fox and the bobcat, which are secondary consumers.

Trap: A top predator can sit at more than one trophic level; you are one level above what you eat · **Review:** Trophic levels: who eats whom

T3-06 Answer: A · Topic 3 · Omnivores at multiple trophic levels (black bear)

In the web, arrows run from redwood sorrel → black bear and from mule deer → black bear. Eating a producer (level 1) makes the bear a primary consumer (level 2); eating a primary consumer (deer, level 2) makes it a secondary consumer (level 3). An animal that eats both plants and animals is an omnivore, and omnivores can occupy more than one trophic level.

B Level 1 is only for producers. Eating a producer puts the bear one level higher, at level 2 (primary consumer); eating deer puts it at level 3.

C Having no predators does not set the trophic level. The bear eats sorrel and deer, so it feeds at levels 2 and 3, not level 4.

D The bear also eats mule deer. Omnivores are counted at every level they eat from, so the bear is at level 3 too.

Trap: Omnivores occupy more than one trophic level; primary consumer = level 2, not level 1 · **Review:** Trophic levels: who eats whom

T3-07 Answer: D · Topic 3 · Importance of decomposers (nutrient cycling)

Decomposers use enzymes to break down dead organisms from every trophic level and return their nutrients to the soil, where producers take them up again. Without decomposers, dead leaves, logs, and bodies would build up, and their nutrients would stay locked inside. Producers would decline — and because every consumer depends on producers, the whole food web would suffer.

A Decomposers are often left off food web drawings, but they are essential: they recycle nutrients from all dead organisms back into the soil.

B Decomposers are not primary consumers. They break down dead organisms from ALL trophic levels, so losing them would hurt the whole web.

C Producers do not eat dead material — they make food from sunlight. They need decomposers to turn dead matter into nutrients they can absorb.

Trap: Decomposers are not primary consumers; they recycle nutrients from all levels · **Review:** Decomposers, scavengers & detritivores

T3-08 Answer: B · Topic 3 · Why there are fewer top predators

At each step about 90% of the energy is used for metabolism and lost as heat, so only about 10% moves up. Rabbits (level 2) receive 350,000 kcal, but pumas (level 4) receive only 3,500 kcal — just 0.1% of the producers' 3,500,000 kcal. With so little energy at the top, only a small population of top predators can survive.

A Pumas are top predators with practically no predators of their own. Their low numbers come from limited energy, not from being hunted.

C Energy decreases at each level; it never increases. That is why the pyramid gets narrower toward the top.

D Every level, including the rabbits, uses about 90% of its energy for metabolism and loses it as heat.

Trap: Energy decreases up the pyramid; every level loses about 90% as heat · **Review:** Energy flow, metabolism & the 10% rule

T3-09 Answer: A · Topic 3 · Effects of removing a species from a food web

Three arrows leave the brush rabbit — to the gray fox, the bobcat, and the coyote. The coyote also eats mule deer, but the gray fox and bobcat have no other food in this web, so they would be hit hardest. The effects would ripple: fewer foxes and bobcats means less food for the puma, while the producers (sorrel, poppy) might increase with fewer rabbits eating them.

B Three predators eat brush rabbits (gray fox, bobcat, and coyote). The coyote also eats mule deer, so it has a backup food.

C Mule deer don't eat rabbits. Losing the rabbits removes a competitor for sorrel and poppies, so the deer would more likely benefit.

D No arrow goes from the rabbit to the puma. The puma would be affected only indirectly, and it can still eat mule deer.

Trap: Reading a web: follow the arrows out of the removed species and check for other food sources · **Review:** Food chains vs. food webs: reading the arrows

T3-10 Answer: C · Topic 3 · 10% rule calculation

There are 3 steps, and each keeps 10% (÷ 10): 3,500,000 kcal → 350,000 kcal (rabbit) → 35,000 kcal (gray fox) → 3,500 kcal (puma). As a percent: 3,500 ÷ 3,500,000 = 0.001 = 0.1% (or 10% × 10% × 10% = 0.1%). This matches the class pyramid: 100% → 10% → 1% → 0.1%.

A 35,000 kcal (1%) is the gray fox level, after only 2 steps. The puma is one more step up, so divide by 10 once more.

B This divides by 10 four times (once per organism). There are only 3 arrows (steps) from the sorrel to the puma.

D The energy amount is right, but 10% applies to ONE step. After 3 steps, 10% × 10% × 10% = 0.1% of the sorrel's energy remains.

Trap: Count steps (arrows), not organisms; 10% applies per step, not overall · **Review:** Energy flow, metabolism & the 10% rule

T4-01 Answer: B · Topic 4 · Limiting factors

A limiting factor is an environmental factor that keeps a population from growing (worksheet). When nesting space runs out, pairs without room have no offspring, so space limits the gull population. The largest colony that the space can support is the carrying capacity — that is a population size, not the factor itself.

A Carrying capacity is a population SIZE (the maximum number of gulls). Nesting space is the factor that sets that limit.

C Being nonliving makes space abiotic, not density-independent. Crowding for nesting space gets worse as the colony grows, so it is density-dependent.

D Exponential growth means a population increasing faster and faster. Here the colony has stopped growing.

Trap: Limiting factor (the cause) vs. carrying capacity (the max population); abiotic does not mean density-independent · **Review:** Limiting factors

T4-02 Answer: D · Topic 4 · Density-dependent vs. density-independent

A density-dependent factor has a stronger effect as a population gets more crowded. Disease spreads more easily when animals are close together, so a larger, denser herd loses a greater share of its members. Hurricanes, cold fronts, and wildfires are density-independent — the worksheet says such factors are not affected by population size.

A A hurricane washes away nests no matter how many turtles nest there, so it is density-independent (a worksheet example).

B This is the worksheet's own example of a density-independent factor: a cold front's effect is not affected by population size.

C Fire is a physical event (lightning, weather, dry fuel). It burns whether the population is large or small, so it is density-independent.

Trap: Weather and natural disasters = density-independent; disease and competition = density-dependent · **Review:** Density-dependent vs. density-independent

T4-03 Answer: A · Topic 4 · Carrying capacity model (letters A–H)

B is the steep rise where the population grows faster and faster — exponential growth. D, the dashed horizontal line, is the carrying capacity (K), the largest population the habitat can support over time. E–H show the population going slightly above and below K again and again; the class model calls this repeating up-and-down pattern around K the cyclical/periodic relationship.

B B and D are switched: the steep curve is growth, and the flat dashed line is K. The curve keeps going after D, so the population does not go extinct.

C Peaks E and G rise above D, so D is not the highest point — K is the level the population fluctuates around. E–H go up and down, not steadily down.

D The slow start (lag) is A, at the bottom left. E–H are small ups and downs near K, not another burst of exponential growth.

Trap: K is the dashed line the population fluctuates around, not the highest peak · **Review:** Population growth & carrying capacity

T4-04 Answer: D · Topic 4 · Carrying capacity calculation (bear & fish)

100 fish per day ÷ 10 fish per bear per day = 10 bears. Five bears would eat only 50 fish; fifteen bears would need 150 fish, so not all of them could survive. Because food is the limiting factor, the worksheet notes that no matter how much shelter and water there was, the population would not get larger than 10 bears.

A Multiplying (10 × 100) is the wrong operation. Divide the fish supply by what one bear needs: 100 ÷ 10 = 10 bears.

B 100 is the number of fish, not bears. Each bear needs 10 fish, so 100 fish per day feeds only 10 bears.

C 10 bears is right, but more shelter or water won't help. The bears are limited by fish, so K rises only if the food supply rises.

Trap: Adding a resource that is not limiting does not raise K · **Review:** Limiting factors

T4-05 Answer: B · Topic 4 · Density-independent does not mean 'no effect'

A late-spring cold front can kill the offspring of many species, so it can sharply reduce a population. 'Density-independent' means the size of the effect is not affected by population size or crowding — a similar share dies whether the population is small or large. Weather, floods, and fires are density-independent, yet they can still cause big population drops.

A Density-independent factors DO change population size — a cold front, flood, or hurricane can kill many individuals. They just act regardless of crowding.

C Temperature is not density. A factor is density-dependent only if its effect gets stronger as the population gets more crowded.

D Nonliving factors like weather also lower populations. Predators and disease are density-dependent, but they are not the only factors that reduce a population.

Trap: Density-independent ≠ 'doesn't affect the population' · **Review:** Density-dependent vs. density-independent

T4-06 Answer: C · Topic 4 · Exponential growth, overshoot, decline, recovery

(a) Rapid growth with plenty of resources = exponential growth (2, the steep rise). (b) Rising far above what the land can support = overshoot (1, the big peak above the dashed K line). (c) Running out of food and dying = decline (4, the fall after the peak). (d) Rising again once food recovers = recovery (5). Afterward the population levels off near carrying capacity (3).

A Marker 3 is on the dashed carrying-capacity line. Stage (b) happens ABOVE that line, at the overshoot peak (1).

B Order matters: the population grows exponentially (2) BEFORE it overshoots (1), not after.

D Decline (4) comes before recovery (5). The rabbits starved first, then increased again as the plants regrew.

Trap: Overshoot is the peak ABOVE K, not K itself; decline comes before recovery · **Review:** Overshoot, decline & recovery

T4-07 Answer: C · Topic 4 · What increases vs. decreases carrying capacity

Carrying capacity depends on the resources a habitat provides. More food, more water, and habitat restoration give the habitat more resources, so they RAISE K. Drought, habitat fragmentation, and more competitors reduce the resources available to each individual, so they LOWER K. K changes whenever the habitat changes.

A Harsh conditions reduce the food, water, and space available, so the habitat can support FEWER individuals, not more.

B K can change. Drought, fire, or habitat loss can lower it, and restoration or more food can raise it.

D Water is a key resource that helps set carrying capacity — and both whole lists are reversed, not just one item.

Trap: Swapping what raises vs. lowers K; K is not fixed · **Review:** Population growth & carrying capacity

T4-08 Answer: A · Topic 4 · Bean simulation — estimating carrying capacity

The population grew from 25 to 64 by year 3, then swung below and above the same level (47 in year 4, 73 in year 5) before settling at 55, 50, and 54 in years 6–8. The level a population fluctuates around is the carrying capacity, so K ≈ 50 rabbits (the class estimate). A line graph fits these data because it shows continuous change over time.

B 73 was a peak (an overshoot) in year 5, and the population quickly fell back. K is the level the population settles around, not its highest point.

C 25 is just the starting number. The population quickly grew above it, so the habitat can support more than 25 rabbits.

D 64 was a temporary high in year 3; the population dropped to 47 the next year. K is where the population levels off over time.

Trap: K is the level the population fluctuates around, not the highest point · **Review:** Overshoot, decline & recovery

T4-09 Answer: B · Topic 4 · Cyclical/periodic predator–prey pattern (lynx & hare)

The graph shows a repeating rise and fall about every 10 years — a cyclical pattern. Hare peaks come first. Lots of hares means lots of food for lynx, so lynx numbers climb a year or two later. Then many lynx eat many hares, the hares drop, and with less food the lynx drop too — and the cycle repeats. Predator peaks lag behind prey peaks.

A Each lynx (dashed) peak comes AFTER a hare peak. And predators eating more prey makes the prey decrease, not grow.

C The peaks don't line up — lynx peak 1–2 years after hares. Also, lynx eat hares; they do not compete with them for the same food.

D Both lines keep rising and falling from 1850 to 1910; they never level off for good. That repeating rise and fall is the cycle.

Trap: Predator peaks lag behind prey peaks · **Review:** Cyclical / periodic patterns

T4-10 Answer: D · Topic 4 · Density-dependent vs. density-independent (data)

Factor P kills a bigger share as crowding increases (4% → 12% → 30%), so its effect depends on density — density-dependent, like disease or competition for food. Factor Q kills 25% no matter the density, so it is density-independent, like a flood or a cold snap. Notice that Q still kills a quarter of the rabbits: density-independent does not mean harmless.

A This is backwards. P's death rate rises with crowding (density-dependent), while Q's stays the same (density-independent).

B Q kills 25% of the rabbits at every density — a big effect. Its percent stays the same because it does not depend on crowding.

C Killing rabbits at every density isn't the test. Only P gets stronger as density rises; Q stays at 25%.

Trap: Density-independent ≠ no effect; look at the trend with density, not just whether deaths happen · **Review:** Density-dependent vs. density-independent

T4-11 Answer: A · Topic 4 · Overshoot and crash (St. Matthew Island reindeer)

With plenty of food and no predators, the herd grew exponentially (29 → 1,350 → about 6,000). It overshot the island's carrying capacity and overgrazed the slow-growing lichen, which lowered K. During a severe winter in 1963–64 (a density-independent factor), the reindeer starved, and only 42 were left by 1966 — about a 99% crash: (6,000 − 42) ÷ 6,000 ≈ 0.99.

B A smooth S-curve levels off at K and stays near it. This herd crashed from about 6,000 to 42 — overshoot and decline, not a smooth level-off.

C The island had no predators during the study. The crash came from running out of food (plus a harsh winter), not from predation.

D Starvation and a crash to 42 show the herd went far ABOVE what its food supply could support; food was the limiting factor.

Trap: Overshoot leads to a crash, not a smooth S-curve; K can drop after overgrazing · **Review:** Overshoot, decline & recovery

T5-01 Answer: C · Topic 5 · Primary succession & pioneer species

Bare, glacier-scraped granite has NO soil, so building an ecosystem there is primary succession. Lichens are pioneer species: tough first arrivals that don't need soil and break bare rock into tiny pieces, the foundation of soil. In the class timeline the pioneer stage lasts about 0–150 years; the climax community (large trees, apex predators, deep soil) comes after 1,000+ years.

A Lichens are pioneers, but secondary succession starts where soil is still present (for example, after a fire). Glacier-scraped granite has no soil at all.

B Primary is right, but a climax community is the final, mature stage (large trees, deep rich soil), not the first organisms on bare rock.

D Both parts are wrong: there was no soil (so it is not secondary), and lichens are the first arrivals, not the final stage.

Trap: Pioneer species ≠ climax community; no soil = primary · **Review:** Primary succession

T5-02 Answer: B · Topic 5 · Primary vs. secondary succession

An abandoned farm field still has its soil (and seeds in it), so plants move back quickly. That is secondary succession. New lava rock, glacier-scraped granite, and bare bedrock have no soil, so life must start over with pioneers like lichens (primary succession). Secondary succession is faster because the soil is already there.

A New lava rock has no soil, so this would be primary succession.
C Glacier-scraped granite has no soil. This is the Yosemite example of primary succession.
D Bare bedrock has no soil left, so succession would start over as primary succession.

Trap: Soil present = secondary; no soil = primary · **Review:** Secondary & tertiary succession

T5-03 Answer: D · Topic 5 · Secondary succession after fire

Because the soil stayed, Big Basin is going through secondary succession. On the class timeline, years 1–20 are early recovery, when grasses and wildflowers (the pioneers of secondary succession) move back in. Shrubs and small trees come in at about 20–150 years, and long-term (climax) recovery takes 150+ years.

A Even a huge fire did not remove the soil here. Soil and seed bank remained, so this is secondary succession. Lichens on bare rock belong to primary succession.
B Shrubs and small trees are the mid-recovery stage, about 20–150 years after the fire. That is too late for year 10.
C Climax (long-term) recovery takes 150+ years on the timeline, not 10.

Trap: Fire where soil remains = secondary, even if almost everything burned · **Review:** Secondary & tertiary succession

T5-04 Answer: A · Topic 5 · Tertiary succession (Ms. Hero's term)

Ms. Hero defines tertiary succession as recovery where the soil is NOT intact and humans intervene to restore the ecosystem (adding soil, replanting, controlling erosion). This is her term, not a standard textbook term, so use her definition on the test. If the soil had stayed, it would be secondary; if the soil were gone and nature rebuilt it alone, it would be primary.

B Fire does not automatically mean secondary succession. Secondary requires the soil to stay intact, and here it washed away.
C Human-led restoration is exactly what Ms. Hero calls tertiary succession. The community still changes over time.
D The crews added topsoil, so the hillside is not starting over from bare rock with lichens, and humans are stepping in to help.

Trap: Fire ≠ automatically secondary; soil lost + human restoration = tertiary (teacher's term) · **Review:** Secondary & tertiary succession

T5-05 Answer: D · Topic 5 · Organisms in communities: pioneer vs. climax

At 1,000+ years the area is a climax community: large trees, tertiary consumers and apex predators, and deep, rich soil. The tall trees shade the ground, so ground-level light is at its lowest and sun-loving species die out. Over succession, soil depth and biodiversity increase while light at the ground decreases. The lichens have become 'retired pioneers' that were largely replaced.

A That describes the pioneer stage (0–150 years): bare granite breaking into tiny rock bits, with maximum light at the ground.
B Gravelly, shallow soil with grasses and wildflowers is the early intermediate stage (150–500 years).
C Lichens and crevice plants are pioneers. By the climax stage they are 'retired pioneers,' largely replaced by large trees.

Trap: Pioneer ≠ climax; ground light DECREASES as succession goes on · **Review:** Who lives there: pioneers, climax community & more

T5-06 Answer: C · Topic 5 · Crown & surface fires

A surface fire burns low fuels on the forest floor (needles, fallen branches, grass, shrubs) and leaves tall tree crowns mostly untouched. It is usually low to moderate severity: it clears dead fuel, keeps the soil healthy, and lets the seed bank sprout, so fire does not always mean destruction. A crown fire burns through the treetops, spreads tree to tree, and kills most trees (high severity).

A A crown fire burns in the treetops, but the card says the canopy was barely touched.
B A ground fire smolders in duff and roots below the surface and damages the soil. Here the soil stayed healthy and seeds sprouted.
D A crown fire does kill most tall trees, but the card says the tall canopy was barely touched.

Trap: Surface fire (forest floor) vs. crown fire (treetops) · **Review:** Fire: surface vs. crown, severity & disturbances

T5-07 Answer: A · Topic 5 · Fire severity (vs. intensity)

Fire severity is how much the fire changed the ecosystem — how much of the vegetation (trees, canopy, understory) was killed and how much of the soil's organic layer burned. Fire intensity is different: it is how much heat or energy the fire released. In Big Basin about 77% of the area burned at high severity (most of the canopy killed or scorched) — yet many old-growth redwoods still resprouted afterward.

B Heat and energy released describe fire INTENSITY, not severity. A very hot fire usually, but not always, causes high severity.
C The number of lightning strikes explains how the fires started, not how much they changed the ecosystem.
D How long a fire burned is not how severity is rated. Severity is judged by the damage to plants and soil.

Trap: Severity (ecosystem damage) ≠ intensity (heat released) · **Review:** Fire: surface vs. crown, severity & disturbances

T5-08 Answer: D · Topic 5 · Disturbances

Lightning fires (like the 2020 CZU Lightning Complex), floods, and volcanic eruptions happen without people. Logging, farming, roads, and housing development are human disturbances. Both kinds matter: disturbances can destroy or fragment habitat and reset succession, but some, like surface fires, also renew an ecosystem.

A All three are human disturbances, not natural ones.
B Clear-cut logging is a human disturbance, so this list is not all natural.
C Housing development is a human disturbance, so this list is not all natural.

Trap: Disturbances can be natural OR human-caused · **Review:** Fire: surface vs. crown, severity & disturbances

T5-09 Answer: A · Topic 5 · Habitat fragmentation: same amount, different pattern

In the checkerboard, every forest cell is surrounded by meadow, so no two forest cells share a side. The forest is all edge and has no interior, which makes it the most fragmented pattern. Four Areas, Diamond, and Two Areas keep large connected blocks that interior-forest species need. Key idea: the same AMOUNT of forest can give very different habitat depending on its PATTERN.

B Four Areas keeps the forest in two solid 5×5 blocks of 25 cells each, so some interior forest remains.
C The Diamond is one large, connected central forest patch with plenty of interior cells.
D Two Areas keeps all 50 forest cells in one big connected block. That is the best case for a forest-interior bird, not the worst.

Trap: Same total forest ≠ same habitat; pattern matters · **Review:** Habitat fragmentation

T5-10 Answer: C · Topic 5 · Habitat fragmentation (simulation data)

At first, suitable habitat tracked the forest closely (80 forest cells → 79 suitable). As the forest shrank, suitable cells crashed: 40 → 24 (60%) and 20 → 4 (20%). The remaining forest was broken into small, scattered patches with lots of edge and little interior, which a forest-interior bird can't use. This is the habitat-fragmentation effect: habitat is lost AND broken apart.

A Early on the numbers are close (80 → 79), but later they split far apart: 20 forest cells gave only 4 suitable cells. Pattern matters, not just amount.
B Suitable cells went DOWN at every step (90 → 4). The murrelet is a forest bird; the meadow species in the simulation was Say's Phoebe.
D Suitable habitat fell at every step, and it was down to 4 cells while 20 forest cells still remained.

Trap: Fragmentation shrinks usable habitat faster than area loss alone · **Review:** Habitat fragmentation

T5-11 Answer: B · Topic 5 · CZU Lightning Complex & redwood fire adaptations

Coast redwoods have bark up to about a foot thick that insulates the living tissue from heat, and they can resprout from dormant buds and burls even after their crowns burn. That is why many redwoods survived even though more than 97% of the park burned. Fire also creates tree cavities that animals use as homes, and prescribed fire can reduce fuel so future wildfires are less destructive.

A Resin is flammable, so it would feed a fire, not stop it. Redwood bark actually has very little resin.
C Sealed cones that open with fire heat belong to knobcone pines. They help a pine reseed; they don't keep a redwood trunk alive.
D Many redwood crowns did burn in 2020, so fog-wet needles didn't stop the fire. Thick bark and resprouting explain survival.

Trap: Redwood survival = thick bark + resprouting (not sealed cones) · **Review:** Case study: CZU Lightning Complex & Big Basin

T6-01 Answer: A · Topic 6 · Richness

Species richness is the number of different species in an area. Rainforests have huge numbers of different species, so they are very rich. Evenness is a different measure: how similar the numbers of each species are. Richness and evenness are both parts of biodiversity.

B Similar numbers of each species describes high EVENNESS, not richness.
C One species dominating describes LOW evenness (like a grass lawn), not richness.
D 'Rich' here means many species, not nutrient-rich soil.

Trap: Richness (number of species) ≠ evenness · **Review:** Richness and evenness

T6-02 Answer: D · Topic 6 · Richness/Evenness (fields of mice)

Field Z has 4 species with 4 mice each: the most species of any field AND perfectly equal numbers. Field Y also has 4 species but is very uneven (13, 1, 1, 1). Field X is perfectly even but has only 2 species. Field W has just 1 species, so it is neither rich nor even.

A All 16 mice in Field W are one species, so it is neither rich nor even.
B Field X is perfectly even (8 and 8), but with only 2 species its richness is low.
C Field Y has 4 species (rich), but one species has 13 of the 16 mice, so it is very uneven.

Trap: Richness ≠ evenness; you need both for high diversity · **Review:** Richness and evenness

T6-03 Answer: A · Topic 6 · Richness/Evenness (comparing fields)

Field Y has 4 species and Field X has 2, so Y is richer. But Field X's two species are equal (8 and 8), while Field Y is dominated by one species (13, 1, 1, 1), so X is more even. A community can be rich but uneven, or even but not rich. They are separate measures.

B Y is richer, but one species has 13 of its 16 mice, so Y is very uneven. X (8 and 8) is more even.
C Richness counts different SPECIES, not total individuals. X has 2 species; Y has 4.
D More mice per species is about abundance, not richness. Y has more species (4 vs. 2).

Trap: Richness = number of species, not number of individuals · **Review:** Richness and evenness

T6-04 Answer: B · Topic 6 · Relative abundance (grass lawn)

Total = 180 + 12 + 6 + 2 = 200 plants. Relative abundance of grass = 180 ÷ 200 = 0.90 (90%). When one species makes up 90% of a community, the community is very uneven. That is why Ms. Hero calls a grass lawn 'very uneven.'

A 0.25 comes from 1 species ÷ 4 species. Relative abundance uses numbers of individuals (180 ÷ 200), not the number of species.
C The math (0.90) is right, but one species making up 90% means LOW evenness, not high.
D 180 is just the count of grass, not a proportion of the total. And many individuals ≠ high richness; there are only 4 species.

Trap: Relative abundance = individuals of one species ÷ total individuals (not species ÷ species) · **Review:** Relative abundance (with your class data)

T6-05 Answer: C · Topic 6 · Relative abundance (calculation)

Total = 10 + 125 + 3 + 23 + 39 = 200 individuals. Relative abundance of species D = 23 ÷ 200 = 0.115 (11.5%). This matches the class table, where species D ranks 3rd in Community A.

A 0.200 is 1 species ÷ 5 species. Relative abundance uses numbers of individuals, not the number of species.
B 0.230 is 23 ÷ 100. Community A's total is 200 individuals, not 100.
D 8.70 is 200 ÷ 23, which is upside down. Put the one species on top and the total on the bottom.

Trap: Divide individuals of ONE species by the TOTAL (right total, right order) · **Review:** Relative abundance (with your class data)

T6-06 Answer: D · Topic 6 · Richness (Communities A, B, C)

Richness is the number of different species. Community C has 10 species (A–J), while A and B each have 5, so A and B are equally rich and C is the richest. The totals (200, 100, 95) don't matter for richness. Note: B is more even than A, but evenness is a different measure.

A The total number of individuals is abundance, not richness. Community A has only 5 species.
B Similar numbers describe EVENNESS. B is more even than A, but it still has only 5 species.
C C has the fewest individuals but the MOST species (10). Richness counts species, not individuals.

Trap: Richness counts species, not individuals; richness ≠ evenness · **Review:** Richness and evenness

T6-07 Answer: B · Topic 6 · Evenness (Community A vs. B)

In Community B the shares range from 0.08 to 0.35, fairly close together. In Community A one species makes up 0.625 (62.5%) while another is only 0.015, so A is dominated by one species and is uneven. Same richness does not mean same evenness, and B is the more diverse community of the two.

A The number of individuals doesn't measure evenness. Evenness compares how similar the species' shares are.
C Equal richness (5 species each) does not mean equal evenness. That is the point of this comparison.
D One species having 0.625 means it DOMINATES the community, which makes A LESS even, not more.

Trap: Same richness ≠ same evenness; one dominant species = low evenness · **Review:** Richness and evenness

T6-08 Answer: C · Topic 6 · Biodiversity (why it matters)

Higher biodiversity usually means more resilience and stability. With 15 prey species, Forest 2's foxes have many other food sources (backup, or redundancy, in the food web), so losing one prey species barely changes their food supply. In Forest 1, losing 1 of only 3 prey species removes a third of the menu, and the disruption ripples through the food web.

A Biodiversity doesn't stop disturbances from happening. The disease hit both forests; biodiversity helps an ecosystem absorb and recover from it.

B More prey SPECIES doesn't mean more foxes. The advantage is more food choices, not more predators.

D Nothing says Forest 2 is a climax community, and even climax communities change after disturbances.

Trap: Biodiversity = resilience through backup food sources, not immunity to disturbance · **Review:** What biodiversity is and why it matters

T7-01 Answer: B · Topic 7 · Ecosystem

An ecosystem is a community of living things (biotic factors) PLUS the nonliving environment (abiotic factors such as sunlight, water, air, rocks, and soil), all interacting. Ecosystems come in different sizes: a pond, a forest like Big Basin, or even the whole Earth.

A That is a habitat, one species' home. An ecosystem includes many species and the nonliving environment they share.

C That is a community, which includes only the living things. An ecosystem also includes the nonliving parts.

D Ecosystems can be any size. A pond, or even a rotting log, can be an ecosystem.

Trap: Ecosystem (biotic + abiotic) vs. community (biotic only) vs. habitat · **Review:** Ecosystem

T7-02 Answer: D · Topic 7 · Equilibrium

Equilibrium is a dynamic balance. Populations go up and down slightly (births, deaths, seasons) but stay within a normal range over time, as in Atlas NP after the wolves returned (deer at 7,000, inside the normal 4,000–8,000). A healthy, resilient ecosystem can return to this balance after small disturbances.

A Balance doesn't mean equal numbers. A healthy Atlas NP has 4,000–8,000 deer but only about 95–120 wolves, because less energy reaches the top of the food chain.

B Real populations always change a little. Equilibrium allows small ups and downs around a stable level.

C One species growing out of control, like deer reaching 15,000 in Atlas NP, is a sign that the ecosystem is OUT of balance.

Trap: Equilibrium ≠ equal numbers, and ≠ no change at all · **Review:** Equilibrium and resilience

T7-03 Answer: B · Topic 7 · Ecosystem (levels of organization)

An ecosystem includes the living community AND the abiotic environment (water, mud, sunlight), interacting. Even a small pond counts, because ecosystems come in different sizes, from a pond to a forest like Big Basin to the whole Earth.

A One individual is an organism, the smallest level.

C All the members of ONE species in an area form a population.

D All the living species together form a community. It leaves out the nonliving parts, so it is not yet an ecosystem.

Trap: Community vs. ecosystem: the ecosystem must include the abiotic parts · **Review:** Ecosystem

T7-04 Answer: D · Topic 7 · Biotic/Abiotic factors

All four items are biotic. Redwoods, fungi, and bacteria are alive. A fallen log is also BIOTIC because it came from a once-living tree, and it is food for decomposers like fungi and banana slugs. Abiotic factors were never alive: sunlight, water, air, temperature, rocks, and soil minerals.

A Every item here is ABIOTIC. None of them is alive or was ever alive.

B Banana slugs and deer are biotic, but sunlight and rocks are abiotic, so this list is mixed.

C Dead leaves and fungi are biotic, but air temperature and wind are abiotic, so this list is mixed.

Trap: Dead logs and dead leaves are BIOTIC (once living) · **Review:** Biotic vs. abiotic factors

T7-05 Answer: C · Topic 7 · Biotic/Abiotic factors

Sunlight is abiotic (never alive), and understory plants are biotic. Plants use sunlight for photosynthesis, so once the canopy burned away, shrubs, wildflowers, and seedlings grew thickly beneath the burned trees, just as the class CZU notes describe. A change in an abiotic factor can change the whole living community.

A The labels are reversed. Sunlight is abiotic (nonliving), and plants are biotic (living).

B Abiotic factors strongly affect living things. Sunlight, water, and temperature control how well plants grow.

D Plants need light, not shade, to make food by photosynthesis. More light at the forest floor lets understory plants grow more.

Trap: Abiotic changes affect biotic factors (and correct classification of each) · **Review:** Biotic vs. abiotic factors

T8-01 Answer: C · Topic 8 · DRY MIX

DRY = Dependent / Responding variable goes on the Y-axis. It is what you measure; it "responds" to the change you make. MIX = Manipulated / Independent variable goes on the X-axis; that is what you change on purpose. Example: in the bean simulation, rabbit population (DV) is on the y-axis and time in years (IV) is on the x-axis.

A The DV is the variable you measure, but the Y in DRY means it goes on the y-axis, not the x-axis.

B This describes MIX (Manipulated / Independent), and MIX goes on the x-axis, not the y-axis.

D The dependent variable is not changed on purpose; it responds to the change. The variable you change on purpose is the manipulated (independent) variable.

Trap: Mixing up "responding" and "manipulated"; putting the DV on the x-axis · **Review:** DRY MIX: axes, IV & DV

T8-02 Answer: A · Topic 8 · IV, DV and constants in an experiment

The manipulated/independent variable is what she changed on purpose: hours of light (x-axis). The responding/dependent variable is what she measured: plant height in cm (y-axis). Water (50 mL/day) and soil were the same for every tray, so they are constants (controlled variables). Keeping constants the same makes it a fair test, so differences in height can be linked to light.

B Swapped: height is the result she measured (DV), and hours of light is what she changed (IV). Soil is a constant, but the IV and DV are backwards.

C Water was kept the same (50 mL/day) for every tray, so it is a constant, not the DV. Plant height is what she measured, so it is the DV, not a constant.

D Water was the same for every tray, so it can't be the IV. Light was the variable she changed, so it is not a constant.

Trap: Swapping IV and DV; calling a constant a variable · **Review:** DRY MIX: axes, IV & DV

T8-03 Answer: D · Topic 8 · X and Y axes: where time goes

Time is almost always the independent variable, so it goes on the x-axis. Sprout height depends on how much time has passed (not the other way around), so height in cm is the DV on the y-axis. The bean simulation followed the same rule: Time (years) on the x-axis, rabbit population on the y-axis.

A She does write down the date, but recording something doesn't make it the DV. The measured outcome is sprout height; time is the IV on the x-axis.

B Time doesn't change because the sprouts grow; sprout growth depends on time. Time is independent, so it goes on the x-axis.

C Right axis, wrong reason: time is the INDEPENDENT variable. Sprout height is the variable that responds.

Trap: Putting time on the y-axis or calling time the dependent variable · **Review:** DRY MIX: axes, IV & DV

T8-04 Answer: B · Topic 8 · IV/DV from a data table; describing the relationship

The student manipulated % meadow (IV, x-axis) and measured the number of suitable cells (DV, y-axis). As meadow rose from 10% to 80%, suitable cells fell from 90 to 4, a negative (inverse) relationship. Suitable habitat also dropped faster than forest area: at 60% meadow, 40 forest cells were left but only 24 were suitable, because fragmentation leaves small patches murrelets can't use.

A The axes are swapped. Suitable cells were counted (responding, y-axis); % meadow was set by the student (manipulated, x-axis).

C The axes are right, but check the data: suitable cells drop from 90 to 4 as meadow increases. That is a negative relationship, not a positive one.

D This experiment didn't measure time. The variable the student changed was % meadow, so % meadow belongs on the x-axis.

Trap: IV on the y-axis; assuming time is always the IV · **Review:** DRY MIX: axes, IV & DV

T8-05 Answer: D · Topic 8 · Line graph vs. bar graph

Change over time (continuous data) should be shown with a LINE graph. Connecting the points shows the trend from year to year (rising to 73 rabbits, then leveling off near 50–55). Bar graphs are for comparing separate categories, such as area burned in conifer forest vs. shrubland vs. grassland. This was the exact feedback on the class bean-simulation graph.

A Pie charts show parts of one whole at one moment. These are counts of the same population at different times, not parts of a total.

B Rabbit population is what was measured, so it is the dependent variable (y-axis). Time is the independent variable.

C Comparing separate categories is what BAR graphs are for. Line graphs are for showing change over time.

Trap: Using a bar graph for change over time · **Review:** Choosing the right graph

T8-06 Answer: B · Topic 8 · Reading a graph (template: axes, high, trend, anomaly)

An anomaly is a point that doesn't fit the pattern. In 2019 the count dropped to 11 large fires, even though the years on either side were high (28 in 2018, 37 in 2020). Good template entries use numbers and units from both axes. The other entries mix up the axes, the high point (2020, 37 fires), or the trend (upward).

A Year runs along the bottom, so it is the x-axis label (years, 2000–2020). The y-axis is Number of large fires, range 0–40.

C 2017 (30 fires) is high, but 2020 is higher with 37 large fires, so the high is 2020.

D The points and the dashed trendline slope upward (about 8 to 28 fires), so the general trend is an increase, not a decrease.

Trap: Confusing the axes; picking a local peak as the overall high · **Review:** Reading and interpreting graphs

T8-07 Answer: C · Topic 8 · What you can and can't conclude from a graph

The graph only shows the number of large fires per year. It has no temperature data, so it can't show that climate change was the cause. A trend shows a pattern (a correlation), not a cause. Class notes link climate change to more fires, but to make that claim you would need evidence about temperature, drought, or fuels too. The other three statements can be read directly from the graph.

A This can be concluded: the points and the dashed trendline rise overall (trendline from about 8 to about 28 fires per year).

B This can be concluded: 2020 is the highest point on the graph, at 37 large fires.

D This can be concluded: 2010 had 6 large fires and 2007 had 24 large fires.

Trap: Correlation is not causation: a graph with no cause data can't prove a cause · **Review:** Reading and interpreting graphs

T9-01 Answer: A · Topic 9 · Identify claim, evidence, and reasoning

Sentence 2 is the claim, a one-sentence answer to the question. Sentence 1 is the evidence: quantitative data with units (cells) from the simulation. Sentence 3 is the reasoning: it uses the concept of habitat fragmentation (more edge, less interior) to explain why the data support the claim. The order of the sentences doesn't matter; what each sentence does is what counts.

B Sentence 1 reports data, so it is evidence, not the claim. Sentence 2 is the short answer to the question (the claim), and it has no data in it.

C Sentence 3 doesn't just answer the question; it explains WHY using a science concept (fragmentation, edge, interior). That makes it reasoning. Sentence 2 is the claim.

D Sentence 1 reports numbers, which makes it evidence, not reasoning. Sentence 3 explains the science, which makes it reasoning, not evidence.

Trap: Assuming the first sentence is always the claim; confusing evidence with reasoning · **Review:** Claim, evidence, reasoning

T9-02 Answer: D · Topic 9 · Choosing the best claim

A claim is a one-sentence answer to the scientific question. Community B is more even because no single species dominates (its most common species is 35%), while in Community A one species makes up 62.5% of all individuals. The claim states the answer; the percentages are the evidence that comes next.

A Evenness is about how equal the species' counts are, not how many individuals there are in total. A is dominated by one species (62.5%), so it is LESS even.

B Having the same number of species means equal RICHNESS, not equal evenness. Their individuals are spread among the species very differently.

C This is an opinion (“nicer to me”), and it doesn't answer the question about evenness.

Trap: Confusing richness with evenness; an opinion used as a claim · **Review:** Claim, evidence, reasoning

T9-03 Answer: B · Topic 9 · Choosing the strongest evidence

This evidence is quantitative, has units (rabbits), and comes from the part of the graph that actually shows leveling off: 55, 50, and 54 rabbits in years 6–8. Strong evidence is specific, relevant data from the investigation that supports the claim.

A These are real numbers with units, but they describe the start, when the population was still growing, not leveling off. Evidence must be relevant to the claim.

C This is vague: no numbers, no units, no years. Strong evidence gives specific data from the graph.

D This is a science idea that explains WHY a population levels off. That makes it reasoning, not evidence (data).

Trap: Evidence that is irrelevant, vague, or actually reasoning · **Review:** Claim, evidence, reasoning

T9-04 Answer: A · Topic 9 · Choosing the best reasoning

Reasoning links the evidence to the claim with a science concept, here the 10% rule. Each trophic level uses about 90% of its energy for metabolism (living, growing, reproducing), and that energy leaves as heat, so only about 10% passes up. With so little energy at the top, the ecosystem can support only a few pumas.

B This only restates the evidence (with a little math). It doesn't explain WHY there is less energy at the top.

C It's true that pumas have almost no predators, but that would lead you to expect MORE pumas, not fewer. It doesn't explain the energy data.

D Energy doesn't cycle; it flows one way and leaves as heat. Decomposers recycle nutrients (matter), not energy back up to pumas.

Trap: Reasoning that only restates the evidence; thinking energy cycles · **Review:** Claim, evidence, reasoning

T9-05 Answer: C · Topic 9 · Weak vs. strong claims

A claim must clearly answer the question. “Have an effect” doesn't say WHAT the effect is. A strong claim would be: “Stronger wind and steeper slopes make a wildfire spread faster.” Numbers belong in the evidence and the science explanation belongs in the reasoning; the claim just needs to be a clear, one-sentence answer.

A Data with units belongs in the EVIDENCE. A claim can be a clear sentence with no numbers.

B The teacher's rule is that a claim is ONE sentence. Length isn't the problem; vagueness is.

D Explaining the science (heat, fuel, oxygen) is the job of the REASONING, not the claim.

Trap: Thinking a claim must contain data or reasoning; vague claims · **Review:** Claim, evidence, reasoning

T9-06 Answer: B · Topic 9 · Evidence needs units

“400 to 850” has no units (400 what?). The revision adds units (blades/m²), a time frame (5 years), and a comparison to the normal range (800–1,000 blades/m²). Evidence should be specific data, with units, that clearly supports the claim.

A This removes the numbers entirely. “A whole lot” is vague; strong evidence keeps the data and adds units.

C This is reasoning (science concepts: keystone species, trophic cascade), not evidence. It also doesn't fix the missing units.

D “I think” turns it into an opinion, and the numbers still have no units.

Trap: Evidence without units; mixing reasoning into evidence · **Review:** Claim, evidence, reasoning

T9-07 Answer: C · Topic 9 · Evaluating a full CER (reasoning that restates evidence)

The reasoning just repeats the numbers from the evidence. Strong reasoning uses a science concept to explain WHY: a bigger grid has more squares (more space and resources), so fewer squares get overcrowded (5+ rabbits). That gives the habitat a higher carrying capacity, so it supports more rabbits. The claim (one clear sentence) and the evidence (numbers with units) are already fine.

A Data belongs in the evidence. The claim is already a clear, one-sentence answer to the question.

B Science ideas belong in the reasoning. The evidence already has specific numbers with units (rabbits).

D Opinions don't belong in a CER. Reasoning needs a science concept (carrying capacity, limiting factors), not a personal view.

Trap: Reasoning that only restates the evidence · **Review:** Claim, evidence, reasoning

Full explanations — Part B mock test

M-01 Answer: C · Topic 7 · Biotic vs. abiotic factors

Abiotic factors are the nonliving parts of an ecosystem, such as sunlight, temperature, water, dissolved gases like oxygen, and rock. Biotic factors are living things, including microscopic ones like bacteria and plankton. An ecosystem is the biotic community interacting with these abiotic factors.

A Kelp, urchins, and otters are all living (biotic). Only sunlight in this list is abiotic.

B Bacteria are living organisms, so they are biotic. One biotic item means the list is not ONLY abiotic.

D Plankton are tiny living organisms (biotic), so this list is not all abiotic.

Trap: Microscopic organisms (bacteria, plankton) are biotic · **Review:** Biotic vs. abiotic factors

M-02 Answer: B · Topic 1 · Sphere interactions

The tree roots are living (biosphere), and the granite is rock (geosphere). Living things breaking rock into smaller pieces is the biosphere affecting the geosphere — the same kind of interaction as lichens breaking down bare rock to start soil formation in primary succession.

A Neither air nor water is the main actor here. The roots (living) are doing the work on the rock.

C The rock is being changed, not causing a change in the air. The cause is the living roots.

D Water may be in the crack, but the scenario describes living roots changing rock — not water changing living things.

Trap: Name the CAUSE sphere first, then the EFFECT sphere · **Review:** How the spheres interact

M-03 Answer: D · Topic 2 · Inter- vs. intraspecific competition

Interspecific competition happens between DIFFERENT species that need the same limited resource. Here bluegill and perch both need the same insect larvae, just like the class example of a lion and tiger competing for water buffalo. Intraspecific competition is within ONE species (gulls, swallows, or redwood seedlings). Competition is −/− because both competitors end up with less.

A All the tree swallows are the same species, so this is INTRAspecific competition (like gulls competing for nesting space on a rock).
B The heron kills and eats the perch. That is predation (+/−), not competition for a shared resource.
C The seedlings are all redwoods (one species), so this is INTRAspecific competition for light.

Trap: Inter (different species) vs. intra (same species); predation is not competition · **Review:** Competition: intraspecific vs. interspecific

M-04 Answer: A · Topic 3 · Metabolism & flow of energy (amount passed on)

Each organism uses about 90% of the energy it takes in for its own metabolism (living, moving, growing, reproducing). During metabolism some chemical energy is converted to heat, which leaves the organism and the food chain. Only about 10% ends up stored in body tissue that the next trophic level can eat.

B Decomposers recycle NUTRIENTS (matter), not energy. Energy flows one way through a food chain and leaves as heat; the algae get new energy from sunlight.
C Most of the energy is used up by the bluegill's life processes, not stored. Only the ~10% that IS stored can pass to the heron.
D Energy is never destroyed; it changes form, mostly into heat. And the 90% is 'lost' at the bluegill's own level, before the heron ever eats it.

Trap: Energy flows one way and leaves as heat; it does not cycle · **Review:** Energy flow, metabolism & the 10% rule

M-05 Answer: B · Topic 5 · Primary vs. secondary succession

Primary succession starts where there is NO soil, like new lava or glacier-scraped granite; pioneers such as lichens must break down rock to start soil, so it is very slow. Secondary succession happens where a disturbance (fire, farming, logging) removed the community but left the soil, so grasses and weeds come back quickly.

A Reversed. New lava has no soil (primary); the old farm field still has its soil (secondary).
C Having few plants is not the test; soil is. The farm field kept its soil and seed bank, so it is secondary succession.
D Both had a disturbance, but the lava flow left NO soil at all, which makes Site 1 primary succession.

Trap: Any disturbance ≠ secondary; the deciding factor is whether soil remains · **Review:** Primary succession

M-06 Answer: C · Topic 8 · DRY MIX: IV/DV and X/Y axes

DRY MIX means the Dependent/Responding variable goes on the Y-axis and the Manipulated/Independent variable goes on the X-axis. She changed the hours of light (IV → X-axis) and measured plant height in cm (DV → Y-axis). Water, soil, and temperature are constants, kept the same so the test is fair.

A Swapped. Height is what she MEASURES (dependent, Y-axis); light is what she CHANGES (manipulated, X-axis).
B Water was kept the same for every pot, so it is a constant (controlled variable), not the independent variable.
D Hours of light is the manipulated variable, so it belongs on the X-axis, and the measured result (height) is missing entirely.

Trap: Swapping IV/DV axes; mistaking a constant for the IV · **Review:** DRY MIX: axes, IV & DV

M-07 Answer: A · Topic 2 · Symbiosis: mutualism, commensalism, parasitism

The remora and shark are commensalism (+/0): one benefits, the other is unaffected. The tick and deer are parasitism (+/−) because the parasite feeds on a living host that usually survives. Coral and algae are mutualism (+/+) because both benefit. Predation is also +/−, but a predator kills and eats its prey.

B The shark gets no benefit (not mutualism), the tick does not kill the deer (not predation), and both coral and algae benefit (not commensalism).
C The remora does not harm the shark, so #1 is commensalism (+/0), not parasitism (+/−).
D The tick feeds on a living host without killing it, which is parasitism, not predation. Coral and algae BOTH benefit, which is mutualism, not parasitism.

Trap: Parasitism vs. predation; commensalism (+/0) vs. mutualism (+/+) · **Review:** Symbiosis: mutualism, commensalism, parasitism

M-08 Answer: D · Topic 4 · Carrying capacity: what raises vs. lowers K

Carrying capacity (K) is the largest population an area can support with its resources (food, water, shelter, space). Adding resources, like restoring water and plants, raises K. Habitat destruction, fragmentation, drought, and fire lower K. These two lists got swapped on the bean simulation sheet, so double-check them!

A More deer does not raise K. A herd above K has overshot; it overuses food and then declines, and overgrazing can even LOWER K.
B Development destroys and fragments habitat, reducing food and space, which LOWERS carrying capacity.
C Drought reduces water and food, which LOWERS carrying capacity.

Trap: Swapping what raises vs. lowers K; more animals ≠ higher K · **Review:** Population growth & carrying capacity

M-09 Answer: B · Topic 6 · Richness vs. evenness

Richness is the number of different species, and both communities have 5, so they are equally rich. Evenness is how similar the species' numbers are. In A, one species makes up 125 of 200 (62.5%) of all individuals; in B the biggest species is only 35 of 100 (35%), so B is more even. (Community C, with 10 species, was the richest.)

A Richness counts the number of different SPECIES, not the number of individuals. Both communities have 5 species.
C Community A is dominated by Species B (62.5% of all individuals), so A is LESS even, not more.
D Similar numbers describe EVENNESS, not richness. Both communities have 5 species, so their richness is equal.

Trap: Richness = number of species, not number of individuals · **Review:** Richness and evenness

M-10 Answer: C · Topic 1 · Earth system interactions (volcano)

A volcano is part of the geosphere (driven by heat from Earth's interior). The ash entered the atmosphere and was carried by wind, then affected the biosphere by blocking the sunlight plants need for photosynthesis. Earth system science is the study of exactly these relationships between land, air, water, and life.

A Backward. The chain starts with the volcano (geosphere) and ends with harm to plants (biosphere).
B The eruption begins in the geosphere, not the atmosphere, and the ash ends up affecting living crops and trees, not water.
D No step here is mainly about water. The ash comes from the geosphere, travels through the air, and affects plants.

Trap: Identifying which sphere starts an interaction and which is affected · **Review:** How the spheres interact

M-11 Answer: A · Topic 2 · Keystone species

A keystone species has an effect on its ecosystem much larger than its numbers; removing it can destroy the balance. Losing otters → urchins boom → kelp is overgrazed → fish and other species lose habitat. This ripple down the food web is a trophic cascade, just like wolves in the Atlas National Park Gizmo.

B Kelp is the producer (it photosynthesizes). Otters affect kelp only indirectly, by eating the urchins that graze on it.
C Keystone species are often NOT the most abundant. What makes them keystone is that their effect is far bigger than their numbers.
D Urchins eat kelp, so URCHINS are primary consumers. Otters eat urchins, which makes otters secondary consumers.

Trap: Keystone ≠ most abundant species · **Review:** Keystone species

M-12 Answer: D · Topic 3 · Trophic levels & arrow direction in a food web

Trophic levels go producer (1) → primary consumer (2) → secondary consumer (3) → tertiary consumer (4). In sorrel → rabbit → bobcat → puma, the puma is at level 4, and every arrow points from food to eater. The puma's level depends on what it eats: it is secondary when it eats deer, but tertiary when it eats a bobcat or gray fox.

A This chain is in the web, but here the puma eats a primary consumer (deer), so the puma is a SECONDARY consumer, not tertiary.
B The arrows are backward. Arrows point from the organism eaten to the organism that eats it (the direction energy flows).
C The web has no coyote → puma arrow. Pumas eat mule deer, gray foxes, and bobcats, not coyotes.

Trap: Arrow direction; one consumer at different trophic levels · **Review:** Food chains vs. food webs: reading the arrows

M-13 Answer: B · Topic 4 · Limiting factors (bean rabbit simulation)

The 5-or-more rule models a density-dependent limiting factor. The more crowded an area, the more competition for food, water, and space, until resources run out and individuals die. Limits like this kept the class population from growing forever, so it leveled off and fluctuated around a carrying capacity of about 50 rabbits.

A The rule ONLY affected crowded squares. A density-independent event like a flood would hit rabbits whether a square was crowded or not.
C That was a different rule: a square with exactly 1 rabbit survived but did not reproduce (no mate).
D Crowding does not raise carrying capacity. The rule shows what happens when a local population goes OVER what its resources can support.

Trap: Density-dependent (crowding) vs. density-independent · **Review:** Limiting factors

M-14 Answer: C · Topic 5 · Crown & surface fires; fire severity

A surface fire burns grass, needles, branches, and shrubs on the forest floor, while big trees usually survive (low-to-moderate severity). Removing this fuel, especially 'ladder fuels' like shrubs and low branches, makes it harder for a future fire to climb into the treetops. A crown fire spreads tree-to-tree through the canopy and is high severity, killing most trees.

A Prescribed fires are kept LOW on the forest floor. A crown fire burns the treetops and is the most destructive kind of fire.

B A ground fire smolders in roots and duff below the surface and would harm trees, which is not the goal.

D Managers want LOW severity. Burning off the soil's organic layer is a sign of HIGH severity and leads to erosion.

Trap: Crown vs. surface vs. ground fire; low vs. high severity · **Review:** Fire: surface vs. crown, severity & disturbances

M-15 Answer: A · Topic 7 · Equilibrium

An ecosystem is the living (biotic) community interacting with its nonliving (abiotic) environment. In equilibrium, populations stay fairly stable within normal ranges. Extra nutrients (abiotic) caused an algae boom, then oxygen (abiotic) dropped and fish died, so the disturbance rippled through the system. The Limiting Factors worksheet warns about this same chain: pollution → dangerous algae growth → risk to fish.

B A sudden boom in one species is a sign of imbalance, not health; here it led to low oxygen and a fish die-off.

C An ecosystem includes BOTH biotic and abiotic factors. Changing nutrients or oxygen affects every living thing in the pond.

D Equilibrium does not mean equal numbers; producers normally far outnumber consumers. It means populations stay relatively stable.

Trap: Equilibrium ≠ equal population sizes; ecosystems include abiotic factors · **Review:** Equilibrium and resilience

M-16 Answer: D · Topic 9 · CER: identify strong evidence

In a CER, the claim is a one-sentence answer to the question, the evidence is specific data (quantitative, with units), and the reasoning explains WHY using a science concept. The correct sentence quotes the measured data from both conditions so a reader can compare them.

A This is the CLAIM, a one-sentence answer to the question, not the evidence.

B This is an opinion with no data. Strong evidence uses measured data with numbers and units.

C This is REASONING: the science concept (limiting factor, carrying capacity) that explains why the evidence supports the claim.

Trap: Mixing up claim, evidence, and reasoning · **Review:** Claim, evidence, reasoning

M-17 Answer: B · Topic 2 · Predator/prey

In a predator–prey cycle, prey numbers rise first. With more food, predators survive and reproduce more, so predator numbers rise a little later. More predators then eat more prey, prey decline, and predators decline afterward because food is scarce. That is why predator peaks lag behind prey peaks, as in the classic lynx–snowshoe hare cycle.

A The data show the moose peak comes FIRST, and more predators make prey numbers fall, not rise.

C Moose eat plants and wolves eat moose, so this is a predator–prey relationship, not competition for the same food.

D The lag is exactly what we expect when predators depend on prey for food, so the two populations are linked.

Trap: Predator peaks come AFTER prey peaks · **Review:** Predator and prey

M-18 Answer: C · Topic 4 · Density-dependent vs. density-independent

A density-dependent factor has a bigger effect as a population gets more crowded. Disease, competition, predation, and parasites spread or intensify when individuals are close together, and Factor Y jumps from 5% to 45%. Factor X kills 20% no matter the density, like weather, floods, or fire (density-independent). Density-independent factors still affect populations, just not based on crowding.

A Factor X kills the SAME percent no matter how crowded the mice are. That is the pattern of a density-independent factor, like a flood or cold snap.

B Factor Y's effect grows as crowding grows (5% → 45%). That is the signature of a density-DEPENDENT factor.

D A limiting factor does not have to kill every individual. Anything that slows or stops population growth is a limiting factor.

Trap: Density-independent ≠ 'doesn't affect the population'; read dependence from data · **Review:** Density-dependent vs. density-independent

M-19 Answer: A · Topic 5 · Habitat fragmentation

Habitat fragmentation breaks a large, continuous habitat into smaller, isolated patches (by roads, development, or fire). Isolated groups have less room to hunt and fewer possible mates, which lowers genetic diversity. Wildlife crossings and corridors reconnect patches so animals can move safely, like the Laurel Curve undercrossing on Highway 17.

B Fragmentation harms wildlife far more than the paved area suggests: it isolates populations, blocks movement, and causes roadkill.

C More edge is a side effect of fragmentation. Wide-ranging animals like lions need large, connected areas, not more edge.

D The goal is the opposite: let animals move BETWEEN both sides safely, reconnecting the split populations.

Trap: Fragmentation is about isolation and pattern, not just lost area · **Review:** Habitat fragmentation

M-20 Answer: D · Topic 3 · Flow of energy: 10% rule calculation

Only about 10% of energy passes up each level, so going DOWN toward the producers you multiply by 10 at each step: otters 400 → urchins 4,000 → kelp 40,000 kcal/m² per year. This is why an ecosystem needs a huge amount of producer energy to support just a few top consumers.

A You divided by 10 twice. Going DOWN the chain toward the producers, the energy gets bigger, so multiply.

B Dividing gives a smaller number, but producers always have the MOST energy. Also, there are two steps, not one.

C That is only one step (the urchins' energy). The kelp is one more level down, so multiply by 10 again.

Trap: Going up a level ÷10, going down ×10; count the steps · **Review:** Energy flow, metabolism & the 10% rule

M-21 Answer: B · Topic 9 · CER: identify strong reasoning

Reasoning connects the evidence to the claim using a scientific concept. Here the concept is a keystone predator causing a trophic cascade: wolves limit deer (and coyotes), so grass and flowering plants recover, which supports bees and ravens and restores equilibrium. The numbers about deer and grass are evidence, not reasoning.

A This is EVIDENCE: specific data with numbers and units. Reasoning explains the science behind why that data supports the claim.

C This is an opinion, not a science concept. Being strong does not make a species keystone.

D This argues for a different claim (bees), and the Gizmo data showed that adding bees did NOT restore balance.

Trap: Evidence (data) vs. reasoning (science concept) · **Review:** Claim, evidence, reasoning

M-22 Answer: A · Topic 6 · Relative abundance

Relative abundance = individuals of one species ÷ total individuals of all species. Total = 54 + 36 + 21 + 9 = 120, so 9 ÷ 120 = 0.075 = 7.5%. Notice that ochre sea stars are a famous keystone species even though they are the least abundant here; keystone does not mean most common.

B That is 9 ÷ 108: you left the sea stars out of the total. The total must include ALL individuals (120).

C 9 is the COUNT of sea stars, not a percent. Divide by the total of 120 first.

D That is 1 ÷ 4 species. Relative abundance uses the number of individuals, not the number of species.

Trap: Dividing by number of species, or leaving the species out of the total · **Review:** Relative abundance (with your class data)

M-23 Answer: C · Topic 4 · Overshoot, decline, recovery

With lots of lichen and no predators, the reindeer grew exponentially (2) and overshot (1) the island's carrying capacity. Overgrazing destroyed their food supply, and a harsh winter (a density-independent factor) hit a starving, crowded herd, so the population declined (4) by about 99%. The herd never recovered and died out by the 1980s, showing that overshoot can damage resources and lower K.

A Recovery (5) is a rise after the lowest point. From 1963 to 1966 the herd was FALLING (about 6,000 → 42), and it never recovered.

B Exponential growth (2) describes 1944–1963 (29 → about 6,000). The 1963–1966 change was a steep drop.

D The island had no predators. The cause was overgrazed food plus a harsh winter, not predation.

Trap: Overshoot leads to a crash (decline), not a smooth leveling off · **Review:** Overshoot, decline & recovery

M-24 Answer: D · Topic 5 · Organisms in communities: pioneer species & climax community

Pioneer species are the first organisms on bare surfaces. Lichens attach to rock and break it into tiny pieces, the foundation of soil. As soil deepens, grasses, shrubs, and finally large trees move in. At the climax stage little light reaches the ground, so sun-loving pioneers like lichens are mostly replaced, which is why they are called 'retired pioneers.'

A Reversed. Pioneers are tough and do NOT need soil; they help MAKE soil from bare rock.

B Lichens grow on bare rock at the very start (the pioneer stage), breaking it down; they are not mainly decomposers of fallen trees.

C The climax community is large trees and apex predators. Lichens are 'retired' because they are largely replaced.

Trap: Pioneers don't need soil; climax = large trees, not pioneers · **Review:** Who lives there: pioneers, climax community & more

M-25 Answer: C · Topic 8 · Reading graphs: points of interest

From 2019 (11 fires) to 2020 (37 fires) the number rose by 26 large fires — the biggest one-year jump on the graph. The other jumps are smaller: 2016→2017 rose by 11 (19 → 30), 2006→2007 by 9 (15 → 24), and 2011→2012 by 7 (9 → 16). 'Points of interest' answers should always use the numbers and units from both axes.

A 15 → 24 is an increase of only 9 fires.

B 19 → 30 is a big increase (11 fires), but not the biggest.

D 9 → 16 is an increase of only 7 fires.

Trap: Compare the size of each change, not just which year is high · **Review:** Reading and interpreting graphs

Weakness Tracker

Step 1. After checking your answers, circle every question ID you missed below. **Step 2.** Count your misses per topic. **Step 3.** If you missed **2 or more** drills in a topic, or **any** mock question in it, re-read the sections listed for the questions you missed (page 2 of this tracker), re-read the class handouts, and redo those questions.

Topic	Drill question IDs (circle misses)	Missed	Mock IDs	Missed	Class handouts to re-read
1. Earth Systems Science	T1-01 T1-02 T1-03 T1-04 T1-05 T1-06 T1-07	___ / 7	M-02 M-10	___ / 2	Earth Systems Science Graphic Organizer
2. Ecological Relationships	T2-01 T2-02 T2-03 T2-04 T2-05 T2-06 T2-07 T2-08 T2-09 T2-10	___ / 10	M-03 M-07 M-11 M-17	___ / 4	Ecological Relationships notes; Gizmos Ecosystems STEM Case (Atlas National Park)
3. Food Chains, Webs & Energy	T3-01 T3-02 T3-03 T3-04 T3-05 T3-06 T3-07 T3-08 T3-09 T3-10	___ / 10	M-04 M-12 M-20	___ / 3	Creating Chains and Webs (Redwood Forest); Gizmos Ecosystems Handbook
4. Limiting Factors & Carrying Capacity	T4-01 T4-02 T4-03 T4-04 T4-05 T4-06 T4-07 T4-08 T4-09 T4-10 T4-11	___ / 11	M-08 M-13 M-18 M-23	___ / 4	Limiting Factors & Carrying Capacity worksheet; Carrying Capacity Simulation & Explanatory Model
5. Succession, Fragmentation & Fire	T5-01 T5-02 T5-03 T5-04 T5-05 T5-06 T5-07 T5-08 T5-09 T5-10 T5-11	___ / 11	M-05 M-14 M-19 M-24	___ / 4	Habitat Fragmentation Simulation; Life from Stone (Yosemite); CZU / Big Basin notes; Wildfires: What is happening in California?
6. Biodiversity	T6-01 T6-02 T6-03 T6-04 T6-05 T6-06 T6-07 T6-08	___ / 8	M-09 M-22	___ / 2	Sampling Diversity
7. What is an Ecosystem?	T7-01 T7-02 T7-03 T7-04 T7-05	___ / 5	M-01 M-15	___ / 2	Gizmos Ecosystems Handbook ("What is an Ecosystem")
8. Graphs (DRY MIX)	T8-01 T8-02 T8-03 T8-04 T8-05 T8-06 T8-07	___ / 7	M-06 M-25	___ / 2	DRY MIX Graphing Vocabulary Notes; Carrying Capacity Simulation graph; wildfire graph analysis
9. Scientific Argumentation (CER)	T9-01 T9-02 T9-03 T9-04 T9-05 T9-06 T9-07	___ / 7	M-16 M-21	___ / 2	Sampling Diversity; Habitat Fragmentation recommendations; Gizmo case study

Status guide

Mastered — 80%+ right on the first try · **Getting there** — 60–79% · **Needs review** — under 60%. Work on "Needs review" topics first; they are the fastest points to gain.

Every question → the exact section to re-read

✗?	ID	What it tests	Re-read this section	Trap
☐	**T1-01**	Earth Systems Science	What is Earth system science?	ESS = interactions among ALL spheres, not the study of one sphere
☐	**T1-02**	Hydrosphere	The four spheres	Soil moisture / glaciers / aquifers = hydrosphere, not geosphere
☐	**T1-03**	Earth Systems Science (closed system)	What is Earth system science?	Closed ≠ isolated: energy crosses the boundary, matter (mostly) doesn't
☐	**T1-04**	Earth Systems Science (energy drivers)	What is Earth system science?	Sun drives most interactions; Earth's core drives geosphere processes
☐	**T1-05**	Atmosphere, Hydrosphere, Biosphere, Geosphere	The four spheres	Classifying by appearance (rocky reef, ice on land, fossil from life) instead of the organizer's sphere
☐	**T1-06**	Sphere interactions (wildfire)	How the spheres interact	Identifying which spheres are involved in one step of a multi-sphere event
☐	**T1-07**	Sphere interactions (volcano)	How the spheres interact	Tracing a multi-step chain of sphere interactions in order
☐	**T2-01**	Symbiosis	Symbiosis: mutualism, commensalism, parasitism	Commensalism (+/0) vs. mutualism (+/+) vs. parasitism (+/−)
☐	**T2-02**	Inter- and Intra-specific Competition	Competition: intraspecific vs. interspecific	Inter- (between species) vs. intra- (within one species); don't get distracted by the prey
☐	**T2-03**	Keystone Species	Keystone species	Keystone ≠ most abundant, and ≠ always the top predator
☐	**T2-04**	Mutualism	Symbiosis: mutualism, commensalism, parasitism	Lichen = mutualism
☐	**T2-05**	Commensalism	Symbiosis: mutualism, commensalism, parasitism	Commensalism (+/0) vs. mutualism (+/+)
☐	**T2-06**	Parasitism	Symbiosis: mutualism, commensalism, parasitism	Parasite vs. predator: parasites usually don't kill their host quickly
☐	**T2-07**	Inter- and Intra-specific Competition	Competition: intraspecific vs. interspecific	Inter- vs. intra-specific competition
☐	**T2-08**	Predator/Prey	Predator and prey	Removing a predator → prey INCREASE (not decrease)
☐	**T2-09**	Keystone Species	Keystone species	Atlas NP keystone = wolves, NOT bees (and not the most abundant species)
☐	**T2-10**	Keystone Species (trophic cascade)	Keystone species	Trophic cascade: a keystone's effects travel indirectly through the food web
☐	**T3-01**	Autotrophs vs. heterotrophs	Trophic levels: who eats whom	Fungi look like plants but are heterotrophs; eating plants does not make an animal an autotroph
☐	**T3-02**	Food chain arrows (direction of energy flow)	Food chains vs. food webs: reading the arrows	Arrows point to the eater (energy flow), not to the food
☐	**T3-03**	Metabolism and energy lost as heat	Energy flow, metabolism & the 10% rule	Energy is not 'destroyed' and does not cycle back — about 90% is used in metabolism and leaves as heat
☐	**T3-04**	Decomposers, scavengers, and detritivores	Decomposers, scavengers & detritivores	Fungi are decomposers, not producers; scavengers eat dead animals but do not kill them
☐	**T3-05**	Trophic levels in the redwood food web (puma)	Trophic levels: who eats whom	A top predator can sit at more than one trophic level; you are one level above what you eat
☐	**T3-06**	Omnivores at multiple trophic levels (black bear)	Trophic levels: who eats whom	Omnivores occupy more than one trophic level; primary consumer = level 2, not level 1
☐	**T3-07**	Importance of decomposers (nutrient cycling)	Decomposers, scavengers & detritivores	Decomposers are not primary consumers; they recycle nutrients from all levels
☐	**T3-08**	Why there are fewer top predators	Energy flow, metabolism & the 10% rule	Energy decreases up the pyramid; every level loses about 90% as heat

✗?	ID	What it tests	Re-read this section	Trap
☐	**T3-09**	Effects of removing a species from a food web	Food chains vs. food webs: reading the arrows	Reading a web: follow the arrows out of the removed species and check for other food sources
☐	**T3-10**	10% rule calculation	Energy flow, metabolism & the 10% rule	Count steps (arrows), not organisms; 10% applies per step, not overall
☐	**T4-01**	Limiting factors	Limiting factors	Limiting factor (the cause) vs. carrying capacity (the max population); abiotic does not mean density-independent
☐	**T4-02**	Density-dependent vs. density-independent	Density-dependent vs. density-independent	Weather and natural disasters = density-independent; disease and competition = density-dependent
☐	**T4-03**	Carrying capacity model (letters A–H)	Population growth & carrying capacity	K is the dashed line the population fluctuates around, not the highest peak
☐	**T4-04**	Carrying capacity calculation (bear & fish)	Limiting factors	Adding a resource that is not limiting does not raise K
☐	**T4-05**	Density-independent does not mean 'no effect'	Density-dependent vs. density-independent	Density-independent ≠ 'doesn't affect the population'
☐	**T4-06**	Exponential growth, overshoot, decline, recovery	Overshoot, decline & recovery	Overshoot is the peak ABOVE K, not K itself; decline comes before recovery
☐	**T4-07**	What increases vs. decreases carrying capacity	Population growth & carrying capacity	Swapping what raises vs. lowers K; K is not fixed
☐	**T4-08**	Bean simulation — estimating carrying capacity	Overshoot, decline & recovery	K is the level the population fluctuates around, not the highest point
☐	**T4-09**	Cyclical/periodic predator–prey pattern (lynx & hare)	Cyclical / periodic patterns	Predator peaks lag behind prey peaks
☐	**T4-10**	Density-dependent vs. density-independent (data)	Density-dependent vs. density-independent	Density-independent ≠ no effect; look at the trend with density, not just whether deaths happen
☐	**T4-11**	Overshoot and crash (St. Matthew Island reindeer)	Overshoot, decline & recovery	Overshoot leads to a crash, not a smooth S-curve; K can drop after overgrazing
☐	**T5-01**	Primary succession & pioneer species	Primary succession	Pioneer species ≠ climax community; no soil = primary
☐	**T5-02**	Primary vs. secondary succession	Secondary & tertiary succession	Soil present = secondary; no soil = primary
☐	**T5-03**	Secondary succession after fire	Secondary & tertiary succession	Fire where soil remains = secondary, even if almost everything burned
☐	**T5-04**	Tertiary succession (Ms. Hero's term)	Secondary & tertiary succession	Fire ≠ automatically secondary; soil lost + human restoration = tertiary (teacher's term)
☐	**T5-05**	Organisms in communities: pioneer vs. climax	Who lives there: pioneers, climax community & more	Pioneer ≠ climax; ground light DECREASES as succession goes on
☐	**T5-06**	Crown & surface fires	Fire: surface vs. crown, severity & disturbances	Surface fire (forest floor) vs. crown fire (treetops)
☐	**T5-07**	Fire severity (vs. intensity)	Fire: surface vs. crown, severity & disturbances	Severity (ecosystem damage) ≠ intensity (heat released)
☐	**T5-08**	Disturbances	Fire: surface vs. crown, severity & disturbances	Disturbances can be natural OR human-caused
☐	**T5-09**	Habitat fragmentation: same amount, different pattern	Habitat fragmentation	Same total forest ≠ same habitat; pattern matters
☐	**T5-10**	Habitat fragmentation (simulation data)	Habitat fragmentation	Fragmentation shrinks usable habitat faster than area loss alone
☐	**T5-11**	CZU Lightning Complex & redwood fire adaptations	Case study: CZU Lightning Complex & Big Basin	Redwood survival = thick bark + resprouting (not sealed cones)
☐	**T6-01**	Richness	Richness and evenness	Richness (number of species) ≠ evenness
☐	**T6-02**	Richness/Evenness (fields of mice)	Richness and evenness	Richness ≠ evenness; you need both for high diversity

✗?	ID	What it tests	Re-read this section	Trap
☐	**T6-03**	Richness/Evenness (comparing fields)	Richness and evenness	Richness = number of species, not number of individuals
☐	**T6-04**	Relative abundance (grass lawn)	Relative abundance (with your class data)	Relative abundance = individuals of one species ÷ total individuals (not species ÷ species)
☐	**T6-05**	Relative abundance (calculation)	Relative abundance (with your class data)	Divide individuals of ONE species by the TOTAL (right total, right order)
☐	**T6-06**	Richness (Communities A, B, C)	Richness and evenness	Richness counts species, not individuals; richness ≠ evenness
☐	**T6-07**	Evenness (Community A vs. B)	Richness and evenness	Same richness ≠ same evenness; one dominant species = low evenness
☐	**T6-08**	Biodiversity (why it matters)	What biodiversity is and why it matters	Biodiversity = resilience through backup food sources, not immunity to disturbance
☐	**T7-01**	Ecosystem	Ecosystem	Ecosystem (biotic + abiotic) vs. community (biotic only) vs. habitat
☐	**T7-02**	Equilibrium	Equilibrium and resilience	Equilibrium ≠ equal numbers, and ≠ no change at all
☐	**T7-03**	Ecosystem (levels of organization)	Ecosystem	Community vs. ecosystem: the ecosystem must include the abiotic parts
☐	**T7-04**	Biotic/Abiotic factors	Biotic vs. abiotic factors	Dead logs and dead leaves are BIOTIC (once living)
☐	**T7-05**	Biotic/Abiotic factors	Biotic vs. abiotic factors	Abiotic changes affect biotic factors (and correct classification of each)
☐	**T8-01**	DRY MIX	DRY MIX: axes, IV & DV	Mixing up “responding” and “manipulated”; putting the DV on the x-axis
☐	**T8-02**	IV, DV and constants in an experiment	DRY MIX: axes, IV & DV	Swapping IV and DV; calling a constant a variable
☐	**T8-03**	X and Y axes: where time goes	DRY MIX: axes, IV & DV	Putting time on the y-axis or calling time the dependent variable
☐	**T8-04**	IV/DV from a data table; describing the relationship	DRY MIX: axes, IV & DV	IV on the y-axis; assuming time is always the IV
☐	**T8-05**	Line graph vs. bar graph	Choosing the right graph	Using a bar graph for change over time
☐	**T8-06**	Reading a graph (template: axes, high, trend, anomaly)	Reading and interpreting graphs	Confusing the axes; picking a local peak as the overall high
☐	**T8-07**	What you can and can't conclude from a graph	Reading and interpreting graphs	Correlation is not causation: a graph with no cause data can't prove a cause
☐	**T9-01**	Identify claim, evidence, and reasoning	Claim, evidence, reasoning	Assuming the first sentence is always the claim; confusing evidence with reasoning
☐	**T9-02**	Choosing the best claim	Claim, evidence, reasoning	Confusing richness with evenness; an opinion used as a claim
☐	**T9-03**	Choosing the strongest evidence	Claim, evidence, reasoning	Evidence that is irrelevant, vague, or actually reasoning
☐	**T9-04**	Choosing the best reasoning	Claim, evidence, reasoning	Reasoning that only restates the evidence; thinking energy cycles
☐	**T9-05**	Weak vs. strong claims	Claim, evidence, reasoning	Thinking a claim must contain data or reasoning; vague claims
☐	**T9-06**	Evidence needs units	Claim, evidence, reasoning	Evidence without units; mixing reasoning into evidence
☐	**T9-07**	Evaluating a full CER (reasoning that restates evidence)	Claim, evidence, reasoning	Reasoning that only restates the evidence
☐	**M-01**	Biotic vs. abiotic factors	Biotic vs. abiotic factors	Microscopic organisms (bacteria, plankton) are biotic
☐	**M-02**	Sphere interactions	How the spheres interact	Name the CAUSE sphere first, then the EFFECT sphere
☐	**M-03**	Inter- vs. intraspecific competition	Competition: intraspecific vs. interspecific	Inter (different species) vs. intra (same species); predation is not competition

✗?	ID	What it tests	Re-read this section	Trap
☐	**M-04**	Metabolism & flow of energy (amount passed on)	Energy flow, metabolism & the 10% rule	Energy flows one way and leaves as heat; it does not cycle
☐	**M-05**	Primary vs. secondary succession	Primary succession	Any disturbance ≠ secondary; the deciding factor is whether soil remains
☐	**M-06**	DRY MIX: IV/DV and X/Y axes	DRY MIX: axes, IV & DV	Swapping IV/DV axes; mistaking a constant for the IV
☐	**M-07**	Symbiosis: mutualism, commensalism, parasitism	Symbiosis: mutualism, commensalism, parasitism	Parasitism vs. predation; commensalism (+/0) vs. mutualism (+/+)
☐	**M-08**	Carrying capacity: what raises vs. lowers K	Population growth & carrying capacity	Swapping what raises vs. lowers K; more animals ≠ higher K
☐	**M-09**	Richness vs. evenness	Richness and evenness	Richness = number of species, not number of individuals
☐	**M-10**	Earth system interactions (volcano)	How the spheres interact	Identifying which sphere starts an interaction and which is affected
☐	**M-11**	Keystone species	Keystone species	Keystone ≠ most abundant species
☐	**M-12**	Trophic levels & arrow direction in a food web	Food chains vs. food webs: reading the arrows	Arrow direction; one consumer at different trophic levels
☐	**M-13**	Limiting factors (bean rabbit simulation)	Limiting factors	Density-dependent (crowding) vs. density-independent
☐	**M-14**	Crown & surface fires; fire severity	Fire: surface vs. crown, severity & disturbances	Crown vs. surface vs. ground fire; low vs. high severity
☐	**M-15**	Equilibrium	Equilibrium and resilience	Equilibrium ≠ equal population sizes; ecosystems include abiotic factors
☐	**M-16**	CER: identify strong evidence	Claim, evidence, reasoning	Mixing up claim, evidence, and reasoning
☐	**M-17**	Predator/prey	Predator and prey	Predator peaks come AFTER prey peaks
☐	**M-18**	Density-dependent vs. density-independent	Density-dependent vs. density-independent	Density-independent ≠ 'doesn't affect the population'; read dependence from data
☐	**M-19**	Habitat fragmentation	Habitat fragmentation	Fragmentation is about isolation and pattern, not just lost area
☐	**M-20**	Flow of energy: 10% rule calculation	Energy flow, metabolism & the 10% rule	Going up a level ÷10, going down ×10; count the steps
☐	**M-21**	CER: identify strong reasoning	Claim, evidence, reasoning	Evidence (data) vs. reasoning (science concept)
☐	**M-22**	Relative abundance	Relative abundance (with your class data)	Dividing by number of species, or leaving the species out of the total
☐	**M-23**	Overshoot, decline, recovery	Overshoot, decline & recovery	Overshoot leads to a crash (decline), not a smooth leveling off
☐	**M-24**	Organisms in communities: pioneer species & climax community	Who lives there: pioneers, climax community & more	Pioneers don't need soil; climax = large trees, not pioneers
☐	**M-25**	Reading graphs: points of interest	Reading and interpreting graphs	Compare the size of each change, not just which year is high